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(54) **ORGANIC LIGHT-EMITTING DEVICE AND APPARATUS INCLUDING THE SAME**

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CPC **H10K 85/654** (2023.02); **H10K 85/6572**
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See application file for complete search history.

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(57) **ABSTRACT**

An organic light-emitting device includes: a first electrode; a second electrode facing the first electrode; an organic layer disposed between the first electrode and the second electrode and includes an emission layer; and a first compound and a second compound; wherein the first compound includes a hole transport compound not including an electron transport moiety, and the second compound includes an electron transport compound including an electron transport moiety or a bipolar compound including an electron transport moiety, wherein the second compound includes deuterium.

16 Claims, 5 Drawing Sheets

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- (52) **U.S. Cl.**
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FIG. 1

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FIG. 2

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FIG. 3

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FIG. 4

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FIG. 5

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ORGANIC LIGHT-EMITTING DEVICE AND APPARATUS INCLUDING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority from and the benefit of Korean Patent Application No. 10-2019-0123357, filed on Oct. 4, 2019, which is hereby incorporated by reference for all purposes as if fully set forth herein.

BACKGROUND

Field

Exemplary implementations of the invention relate generally to an organic light-emitting device and, more specifically, to an apparatus including the same.

Discussion of the Background

Organic light-emitting devices are self emissive devices and, compared with a device of the related art, they have a wide viewing angle, a high contrast ratio, and a short response time, and show excellent characteristics in terms of luminance, driving voltage, and response speed.

In an organic light-emitting device, a first electrode is arranged on a substrate, and a hole transport region, an emission layer, an electron transport region, and a second electrode are sequentially formed on the first electrode. Holes provided from the first electrode may move toward the emission layer through the hole transport region, and electrons provided from the second electrode may move toward the emission layer through the electron transport region. The holes and the electrons, which are carriers, recombine in the emission layer to produce excitons. These excitons transition from an excited state to a ground state, thereby generating light. However, many existing organic light-emitting devices suffer from reduced lifespans during operation.

The above information disclosed in this Background section is only for understanding of the background of the inventive concepts, and, therefore, it may contain information that does not constitute prior art.

SUMMARY

Applicant discovered that by combining emission layers of organic light-emitting devices with different attributes unexpected synergistic improvements in efficiency and lifespan are obtained.

Organic emitting-light devices and apparatus including the same constructed according to the principles and exemplary implementations of the invention have high efficiency and long lifespan. For example, the emission layer of an organic light-emitting device composed of one or more layers of certain compounds that that satisfy certain of the equations disclosed herein, the emission layer has reduced resistance. In addition, the organic light-emitting device appropriately maintains ratios of holes and electrons based on the difference of the highest occupied molecular orbital HOMO energy levels of the hosts in the emission layer. For example, the organic light-emitting device may have first and second emission layers that satisfy certain of the disclosed equations such that the first emission layer has long lifespan, but relatively low efficiency, and the second emission layer has high efficiency, but relatively short lifespan,

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but the overall efficiency and lifespan of the organic light-emitting device are improved.

Additional features of the inventive concepts will be set forth in the description which follows, and in part will be apparent from the description, or may be learned by practice of the inventive concepts.

According to one aspect of the invention, an organic light-emitting device includes: a first electrode; a second electrode facing the first electrode; an organic layer disposed between the first electrode and the second electrode and includes an emission layer; and a first compound and a second compound; wherein the first compound includes a hole transport compound not including an electron transport moiety, and the second compound includes an electron transport compound including an electron transport moiety or a bipolar compound including an electron transport moiety, wherein the second compound includes deuterium.

The electron transport moiety may include at least one of a cyano group, a phosphine oxide group, a sulfone oxide group, a sulfonate group, and a π electron-depleted nitrogen-containing ring.

The first compound may be a compound of Formulae 301-1 to 301-5, or a compound of HT1-1, HT1-2, HT2-1, or HT2-2, as disclosed herein.

The first compound may be a fluorene-containing compound, a carbazole-containing compound, a diarylamine compound, a triarylamine compound, a dibenzofuran-containing compound, a dibenzothiophene-containing compound, or a dibenzosilole-containing compound.

The first compound may include deuterium.

The second compound may be a compound of Formulae 302-1 to 302-4, or a compound of ET2-1, ET2-2, ET4-1, or ET4-2, as disclosed herein.

The second compound may be a pyridine-containing compound, a pyrimidine-containing compound, a triazine-containing compound, a triazole-containing compound, an imidazole-containing compound, or an oxazine-containing compound.

The second compound may be a compound of Formula 303 or of Formulae 303-1 to 303-6, as disclosed herein.

The emission layer may include the first compound and the second compound.

The emission layer may include a first emission layer and a second emission layer, the first emission layer may include a first host and a second host, and the second emission layer may include a third host and a fourth host, wherein, the first host in the first emission layer and the third host in the second emission layer may each include the first compound, and the first host and the third host are same or different from each other, the second host in the first emission layer and the fourth host in the second emission layer may each include the second compound, the second compound included in the second host may include the electron transport compound, the second compound included in the fourth host may include the bipolar compound, the first host and the second host may each satisfy Equation 1 disclosed herein, and the third host and the fourth host may each satisfy Equation 2 disclosed herein.

The emission layer may emit green light having a maximum emission wavelength in a range of about 490 nm to about 560 nm.

The emission layer may further include a dopant, and the dopant may include a phosphorescent dopant or a fluorescent dopant.

The first electrode may be an anode, the second electrode may be a cathode, the organic layer may further include a hole transport region disposed between the first electrode

and the emission layer, and an electron transport region disposed between the emission layer and the second electrode; the hole transport region may include a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or any combination thereof, and the electron transport region may include a hole blocking layer, an electron transport layer, an electron injection layer, or any combination thereof.

The hole transport region may include an amine-based compound.

The electron transport region may include a metal-containing material.

An apparatus may include an organic light-emitting device, wherein the first electrode of the organic light-emitting device may be in electrical contact with one of a source electrode and a drain electrode of a thin-film transistor.

It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are intended to provide further explanation of the invention as claimed.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are included to provide a further understanding of the invention and are incorporated in and constitute a part of this specification, illustrate exemplary embodiments of the invention, and together with the description serve to explain the inventive concepts.

FIG. 1 is a schematic cross-sectional diagram of an exemplary embodiment of an organic light-emitting device constructed according to principles of the invention.

FIG. 2 is a schematic cross-sectional diagram of another exemplary embodiment of an organic light-emitting device constructed according to principles of the invention.

FIG. 3 is a schematic cross-sectional diagram of yet another exemplary embodiment of an organic light-emitting device constructed according to principles of the invention.

FIG. 4 is a schematic cross-sectional diagram of still another exemplary embodiment of an organic light-emitting device constructed according to principles of the invention.

FIG. 5 is a schematic diagram of an exemplary embodiment of an apparatus containing an organic light-emitting device constructed according to principles of the invention.

DETAILED DESCRIPTION

In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of various exemplary embodiments or implementations of the invention. As used herein "embodiments" and "implementations" are interchangeable words that are non-limiting examples of devices or methods employing one or more of the inventive concepts disclosed herein. It is apparent, however, that various exemplary embodiments may be practiced without these specific details or with one or more equivalent arrangements. In other instances, well-known structures and devices are shown in block diagram form in order to avoid unnecessarily obscuring various exemplary embodiments. Further, various exemplary embodiments may be different, but do not have to be exclusive. For example, specific shapes, configurations, and characteristics of an exemplary embodiment may be used or implemented in another exemplary embodiment without departing from the inventive concepts.

Unless otherwise specified, the illustrated exemplary embodiments are to be understood as providing exemplary

features of varying detail of some ways in which the inventive concepts may be implemented in practice. Therefore, unless otherwise specified, the features, components, modules, layers, films, panels, regions, and/or aspects, etc. (hereinafter individually or collectively referred to as "elements"), of the various embodiments may be otherwise combined, separated, interchanged, and/or rearranged without departing from the inventive concepts.

The use of cross-hatching and/or shading in the accompanying drawings is generally provided to clarify boundaries between adjacent elements. As such, neither the presence nor the absence of cross-hatching or shading conveys or indicates any preference or requirement for particular materials, material properties, dimensions, proportions, commonalities between illustrated elements, and/or any other characteristic, attribute, property, etc., of the elements, unless specified. Further, in the accompanying drawings, the size and relative sizes of elements may be exaggerated for clarity and/or descriptive purposes. When an exemplary embodiment may be implemented differently, a specific process order may be performed differently from the described order. For example, two consecutively described processes may be performed substantially at the same time or performed in an order opposite to the described order. Also, like reference numerals denote like elements.

When an element, such as a layer, is referred to as being "on," "connected to," or "coupled to" another element or layer, it may be directly on, connected to, or coupled to the other element or layer or intervening elements or layers may be present. When, however, an element or layer is referred to as being "directly on," "directly connected to," or "directly coupled to" another element or layer, there are no intervening elements or layers present. To this end, the term "connected" may refer to physical, electrical, and/or fluid connection, with or without intervening elements. Further, the D1-axis, the D2-axis, and the D3-axis are not limited to three axes of a rectangular coordinate system, such as the x, y, and z-axes, and may be interpreted in a broader sense. For example, the D1-axis, the D2-axis, and the D3-axis may be perpendicular to one another, or may represent different directions that are not perpendicular to one another. For the purposes of this disclosure, "at least one of X, Y, and Z" and "at least one selected from the group consisting of X, Y, and Z" may be construed as X only, Y only, Z only, or any combination of two or more of X, Y, and Z, such as, for instance, XYZ, XYY, YZ, and ZZ. As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items.

Although the terms "first," "second," etc. may be used herein to describe various types of elements, these elements should not be limited by these terms. These terms are used to distinguish one element from another element. Thus, a first element discussed below could be termed a second element without departing from the teachings of the disclosure.

Spatially relative terms, such as "beneath," "below," "under," "lower," "above," "upper," "over," "higher," "side" (e.g., as in "sidewall"), and the like, may be used herein for descriptive purposes, and, thereby, to describe one elements relationship to another element(s) as illustrated in the drawings. Spatially relative terms are intended to encompass different orientations of an apparatus in use, operation, and/or manufacture in addition to the orientation depicted in the drawings. For example, if the apparatus in the drawings is turned over, elements described as "below" or "beneath" other elements or features would then be oriented "above" the other elements or features. Thus, the exemplary term

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“below” can encompass both an orientation of above and below. Furthermore, the apparatus may be otherwise oriented (e.g., rotated 90 degrees or at other orientations), and, as such, the spatially relative descriptors used herein interpreted accordingly.

The terminology used herein is for the purpose of describing particular embodiments and is not intended to be limiting. As used herein, the singular forms, “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Moreover, the terms “comprises,” “comprising,” “includes,” and/or “including,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, components, and/or groups thereof, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. It is also noted that, as used herein, the terms “substantially,” “about,” and other similar terms, are used as terms of approximation and not as terms of degree, and, as such, are utilized to account for inherent deviations in measured, calculated, and/or provided values that would be recognized by one of ordinary skill in the art.

Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure is a part. Terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and should not be interpreted in an idealized or overly formal sense, unless expressly so defined herein.

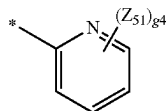
An aspect of the invention provides an organic light-emitting device including: a first electrode; a second electrode facing the first electrode; an organic layer disposed between the first electrode and the second electrode and including an emission layer; a first compound; and a second compound.

The first compound may be a hole transport compound not including an electron transport moiety, and the second compound may be an electron transport compound including an electron transport moiety or a bipolar compound including an electron transport moiety, wherein the second compound may include deuterium.

In one exemplary embodiment, the electron transport moiety may include at least one selected from a cyano group, a phosphine oxide group, a sulfone oxide group, a sulfonate group, and a π electron-depleted nitrogen-containing ring.

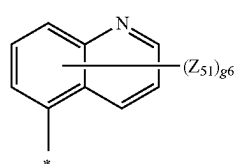
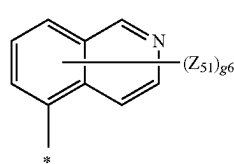
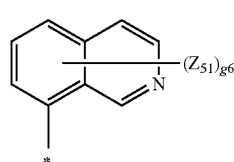
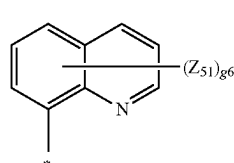
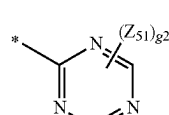
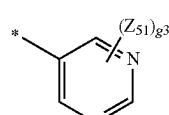
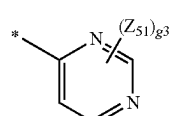
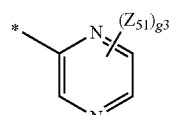
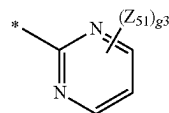
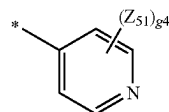
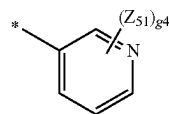
In one exemplary embodiment, the electron transport moiety may include at least one selected from:

- a cyano group, a phosphine oxide group, a sulfone oxide group, and a sulfonate group;
- a C_1 - C_{60} alkyl group, a phenyl group, a naphthyl group, a biphenyl group, and a terphenyl group, each substituted with at least one selected from a cyano group, a phosphine oxide group, a sulfone oxide group, and a sulfonate group; and
- a group represented by one of Formulae 7-1 to 7-59 below:



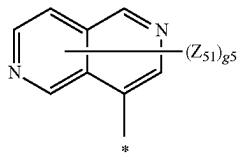
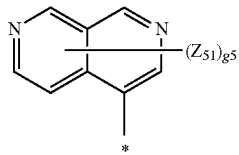
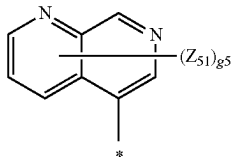
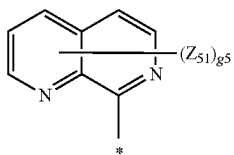
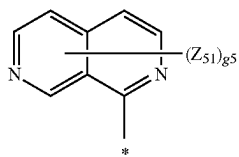
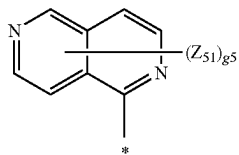
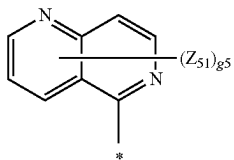
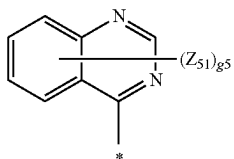
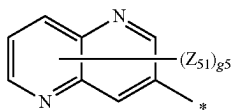
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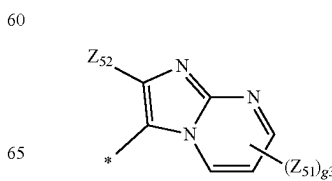
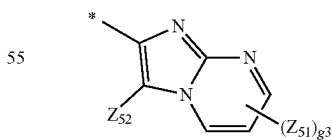
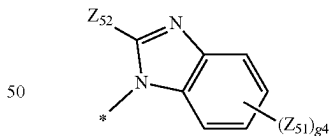
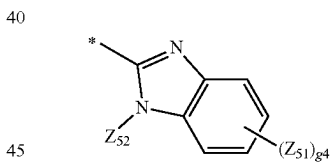
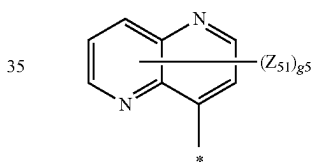
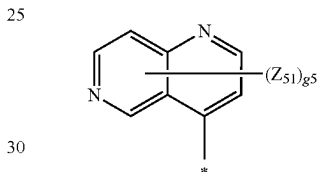
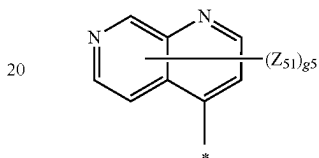
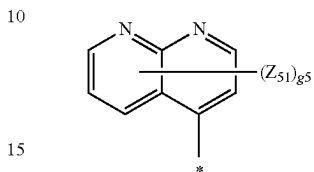
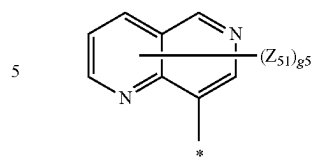


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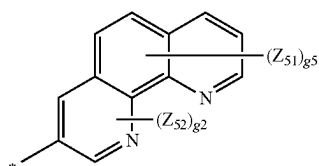
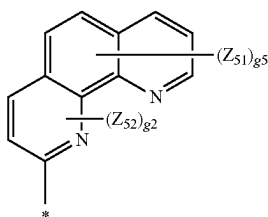
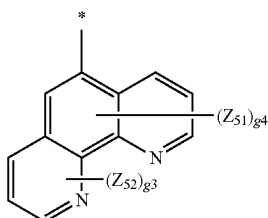
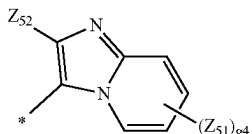
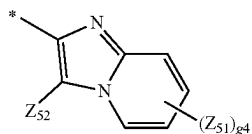
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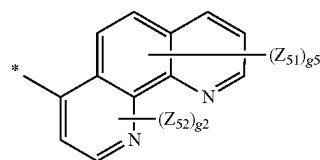


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In Formulae 7-1 to 7-59,

Z_{51} and Z_{52} may each independently be at least one selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, an imidazolyl group, a pyrazole group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, pyrimidinyl, a pyridazinyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a thiadiazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group;

7-56 15

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7-57 25

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 g_2 may be an integer from 0 to 2, g_3 may be an integer from 0 to 3 integer, g_4 may be an integer from 0 to 4 integer, g_5 may be an integer from 0 to 5 integer, g_6 may be an integer from 0 to 6, and

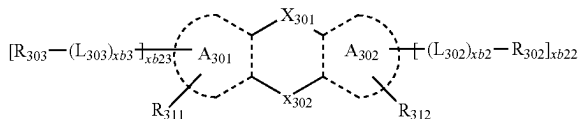
7-58 35

* and * each indicate a binding site to a neighboring site.

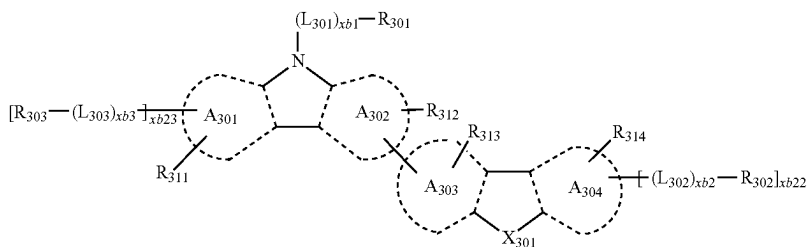
In one exemplary embodiment, the electron transport moiety may include at least one selected from a pyridine group, a pyrimidine group, a triazine group, a quinoline group, an isoquinoline group, and a quinazoline group.

40 In one exemplary embodiment, the first compound may be a compound represented by one of Formulae 301-1 to 301-5 below:

<Formula 301-1>



<Formula 301-2>

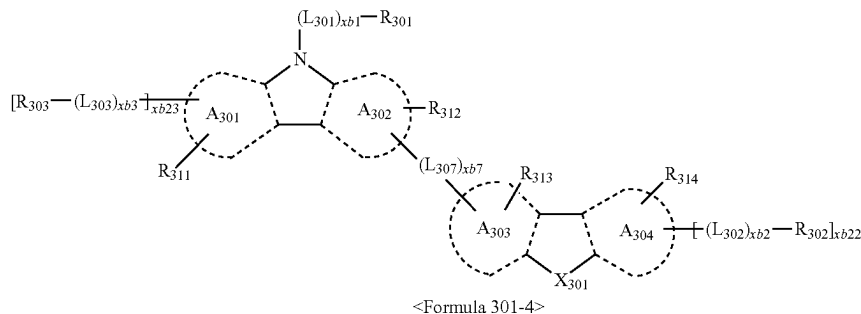


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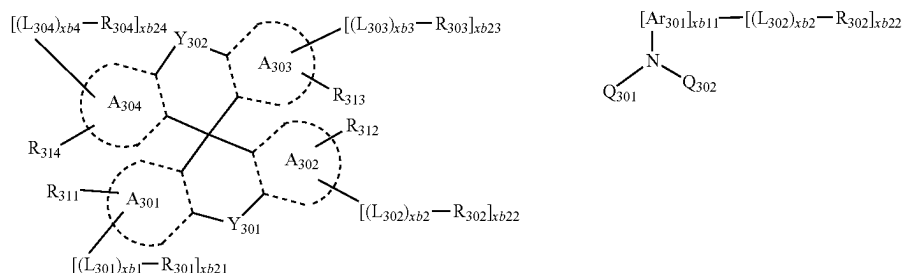
14

-continued

<Formula 301-3>



<Formula 301-5>



In Formulae 301-1 to 301-5,

A₃₀₁ to A₃₀₄ may each independently be selected from a benzene group, a naphthalene group, a phenanthrene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, an indene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, an indole group, a carbazole group, a benzocarbazole group, a dibenzocarbazole group, a furan group, a benzofuran group, a dibenzofuran group, a naphthofuran group, a benzonaphthofuran group, a dinaphthofuran group, a thiophene group, a benzothiophene group, a dibenzothiophene group, a naphthothiophene group, a benzonaphthothiophene group, and a dinaphthothiophene group,

X₃₀₁ may be O, S, N-[(L₃₀₅)_{xb5}-R₃₀₅], C[(L₃₀₅)_{xb5}-R₃₀₅], [(L₃₀₆)_{xb6}-R₃₀₆], or Si[(L₃₀₅)_{xb5}-R₃₀₅][(L₃₀₆)_{xb6}-R₃₀₆], Y₃₀₁, and Y₃₀₂ may each independently be a single bond, O, S, N-[(L₃₀₅)_{xb5}-R₃₀₅], C[(L₃₀₅)_{xb5}-R₃₀₅], [(L₃₀₆)_{xb6}-R₃₀₆], Si[(L₃₀₅)_{xb5}-R₃₀₅][(L₃₀₆)_{xb6}-R₃₀₆], or S(=O)₂,

xb22 and xb23 may each independently be 0, 1, or 2,

L₃₀₁ to L₃₀₇ may each independently be selected from a substituted or unsubstituted C₃-C₁₀ cycloalkylene group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkylene group, a substituted or unsubstituted C₃-C₁₀ cycloalkenylene group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenylene group, a substituted or unsubstituted C₆-C₆₀ arylene group, a substituted or unsubstituted C₁-C₆₀ heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group,

xb1 to xb6 may each independently be an integer from 0 to 5,

xb7 may be an integer from 1 to 5,

R₃₀₁ to R₃₀₆ may each independently be selected from deuterium, -F, -Cl, -Br, -I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a

hydrazino group, a hydrazono group, a substituted or unsubstituted C₁-C₆₀ alkyl group, a substituted or unsubstituted C₂-C₆₀ alkenyl group, a substituted or unsubstituted C₂-C₆₀ alkynyl group, a substituted or unsubstituted C₁-C₆₀ alkoxy group, a substituted or unsubstituted C₃-C₁₀ cycloalkyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkyl group, a substituted or unsubstituted C₃-C₁₀ cycloalkenyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenyl group, a substituted or unsubstituted C₆-C₆₀ aryl group, a substituted or unsubstituted C₆-C₆₀ aryloxy group, a substituted or unsubstituted C₆-C₆₀ arylthio group, a substituted or unsubstituted heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, -Si(Q₃₀₁)(Q₃₀₂)(Q₃₀₃), -N(Q₃₀₁)(Q₃₀₂), -B(Q₃₀₁)(Q₃₀₂), -C(=O)(Q₃₀₁), -S(=O)₂(Q₃₀₁), and -P(=O)(Q₃₀₁)(Q₃₀₂),

R₃₁₁ to R₃₁₄ may each independently be selected from hydrogen, deuterium, -F, -Cl, -Br, -I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group -Si(Q₃₀₁)(Q₃₀₂)(Q₃₀₃), -N(Q₃₀₁)(Q₃₀₂), -B(Q₃₀₁)(Q₃₀₂), -C(=O)(Q₃₀₁), -S(=O)₂(Q₃₀₁), and -P(=O)(Q₃₀₁)(Q₃₀₂), and

Q₃₀₁ to Q₃₀₃ may each independently be selected from hydrogen, deuterium, -F, -Cl, -Br, -I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a mon-

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ovalent non-aromatic condensed heteropolycyclic group, a biphenyl group, and a terphenyl group.

In one exemplary embodiment, the first compound may be selected from a fluorene-containing compound, a carbazole-containing compound, a diarylamine compound, a triarylamine compound, a dibenzofuran-containing compound, a dibenzothiophene-containing compound, and a dibenzosilole-containing compound.

In one exemplary embodiment, the first compound may include deuterium.

In one exemplary embodiment, in Formulae 301-1 to 301-5, at least one selected from R_{301} to R_{306} and R_{311} to R_{314} may be deuterium, a C_1 - C_{60} alkyl group substituted with deuterium, a C_3 - C_{10} cycloalkyl group substituted with deuterium, a C_6 - C_{60} aryl group substituted with deuterium, or a C_1 - C_{60} heteroaryl group substituted with deuterium.

In one exemplary embodiment, A_{301} to A_{304} in Formulae 301-1 to 301-5 may each independently be selected from a benzene group, an indene group, a naphthalene group, an anthracene group, a fluorene group, a phenanthrene group, and a carbazole group.

In one exemplary embodiment, A_{31} , A_{32} , and A_{34} in Formulae 301-1 to 301-5 may each independently be selected from a benzene group, a naphthalene group, and a phenanthrene group.

In one exemplary embodiment, A_{31} , A_{32} , and A_{34} in Formulae 301-1 to 301-5 may each independently be a benzene group or a naphthalene group.

In one exemplary embodiment, L_{301} to L_{307} in Formulae 301-1 to 301-5 may each independently be selected from:

a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthrenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylene group, a pentacenylene group, a thiophenylenylene group, a furanylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group; and

a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthrenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylene group, a pentacenylene group, a thiophenylenylene group, a furanylene group, a carbazolylenylene group, an indo-

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lylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthrenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, —Si(Q_{31})(Q_{32})(Q_{33}), —N(Q_{31})(Q_{32}), —B(Q_{31})(Q_{32}), —C(=O)(Q_{31}), —S(=O)₂(Q_{31}), and —P(=O)(Q_{31})(Q_{32}).

In one exemplary embodiment, R_{301} to R_{307} in Formulae 301-1 to 301-5 may each independently be selected from:

a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthrenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofura-

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nyl group, a dibenzothiophenyl group, a benzocarbazoyl group, a dibenzocarbazoyl group, and a dibenzosilolyl group; and

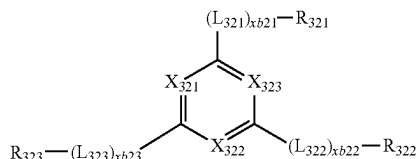
a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthrenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazoyl group, an indolyl group, an isoindolyl group, a benzofuran group, a benzothiophenyl group, a benzofuran group, a dibenzothiophenyl group, a benzocarbazoyl group, a dibenzocarbazoyl group, and a dibenzosilolyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthrenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazoyl group, an indolyl group, an isoindolyl group, a benzofuran group, a benzothiophenyl group, a dibenzofuran group, a dibenzothiophenyl group, a benzocarbazoyl group, a dibenzocarbazoyl group, a dibenzosilolyl group, —Si(Q₃₁)(Q₃₂)(Q₃₃), —N(Q₃₁)(Q₃₂), —B(Q₃₁)(Q₃₂), —C(=O)(Q₃₁), —S(=O)₂(Q₃₁), and —P(=O)(Q₃₁)(Q₃₂).

In one exemplary embodiment, Q₃₁ to Q₃₃ may each independently be selected from a C₁-C₁₀ alkyl group, a C₁-C₁₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

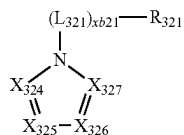
In one exemplary embodiment, the second compound may be an electron transport compound including an electron transport moiety.

In one or more exemplary embodiments, the second compound may be a compound represented by one of Formulae 302-1 to 302-4 below:

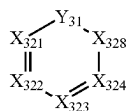
<Formula 302-1>



<Formula 302-2>



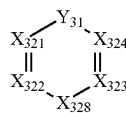
<Formula 302-3>



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-continued

<Formula 302-4>



In Formulae 302-1 to 302-4,

X₃₂₁ to X₃₂₇ may each independently be N or C[(L₃₂₄)_{xb24}-R₃₂₄],

Y₃₁ may be O or S,

X₃₂₈ may be N[(L₃₂₅)_{xb25}-R₃₂₅] or C[(L₃₂₅)_{xb25}-R₃₂₅] [(L₃₂₆)_{xb26}-R₃₂₆],

xb21 to xb26 may each independently be an integer from 0 to 5,

L₃₂₁ to L₃₂₆ may each independently be selected from a substituted or unsubstituted C₃-C₁₀ cycloalkylene group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkylene group, a substituted or unsubstituted C₃-C₁₀ cycloalkenylene group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenylene group, a substituted or unsubstituted C₆-C₆₀ arylene group, a substituted or unsubstituted C₁-C₆₀ heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group,

R₃₂₁ to R₃₂₆ may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C₁-C₆₀ alkyl group, a substituted or unsubstituted C₂-C₆₀ alkenyl group, a substituted or unsubstituted C₂-C₆₀ alkynyl group, a substituted or unsubstituted C₁-C₆₀ alkoxy group, a substituted or unsubstituted C₃-C₁₀ cycloalkyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkyl group, a substituted or unsubstituted C₃-C₁₀ cycloalkenyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenyl group, a substituted or unsubstituted C₆-C₆₀ aryl group, a substituted or unsubstituted C₆-C₆₀ aryloxy group, a substituted or unsubstituted C₆-C₆₀ arylthio group, a substituted or unsubstituted C₁-C₆₀ heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q₃₂₁)(Q₃₂₂)(Q₃₂₃), —N(Q₃₂₁)(Q₃₂₂), —B(Q₃₂₁)(Q₃₂₂), —C(=O)(Q₃₂₁), —S(=O)₂(Q₃₂₁), and —P(=O)(Q₃₂₁)(Q₃₂₂),

at least one selected from R₃₂₁ to R₃₂₆ may include deuterium,

two or more substituents selected from R₃₂₁ to R₃₂₆ may optionally be linked to each other to form a substituted or unsubstituted C₁-C₆₀ carbocyclic group or a substituted or unsubstituted C₁-C₆₀ heterocyclic group, and Q₃₂₁ to Q₃₂₃ may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, a monova-

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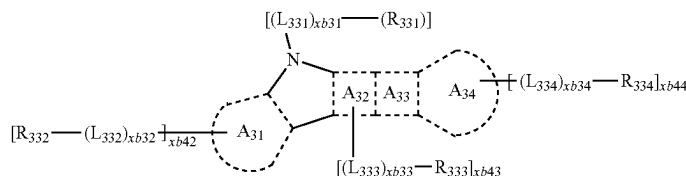
lent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a biphenyl group, and a terphenyl group.

In one exemplary embodiment, at least one selected from R_{321} to R_{326} in Formulae 302-1 to 302-4 may be deuterium, a C_1 - C_{60} alkyl group substituted with deuterium, a C_3 - C_{10} cycloalkyl group substituted with deuterium, a C_6 - C_{60} aryl group substituted with deuterium, or a C_1 - C_{60} heteroaryl group substituted with deuterium.

In one exemplary embodiment, the second compound may be a pyridine-containing compound, a pyrimidine-containing compound, a triazine-containing compound, a triazole-containing compound, an imidazole-containing compound, or an oxazine-containing compound.

In one exemplary embodiment, the second compound may be a bipolar compound including an electron transport moiety.

In one exemplary embodiment, the second compound may be a compound represented by Formula 303 below:



<Formula 303>



<Formula 303A>

In Formula 303,

A_{31} , A_{32} , and A_{34} may each independently be selected from a C_5 - C_{60} carbocyclic group and a C_1 - C_{30} heterocyclic group,

A_{33} may be a group represented by Formula 303A, X_{31} may be selected from $N[(L_{335})_{xb35}-(R_{335})]$, O, S, Se, $C[(L_{335})_{xb35}-(R_{335})][(L_{336})_{xb36}-(R_{336})]$, and $Si[(L_{335})_{xb35}-(R_{335})][(L_{336})_{xb36}-(R_{336})]$, $xb31$ to $xb36$ may each independently be an integer from 0 to 5,

L_{331} to L_{336} may each independently be selected from a substituted or unsubstituted C_3 - C_{10} cycloalkylene group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkylene group, a substituted or unsubstituted C_3 - C_{10} cycloalkenylene group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkenylene group, a substituted or unsubstituted C_6 - C_{60} arylene group, a substituted or unsubstituted C_1 - C_{60} heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group,

R_{331} to R_{336} may each independently be selected from hydrogen, deuterium, $-F$, $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C_1 - C_{60} alkyl group, a substituted or unsubstituted C_2 - C_{60} alkenyl group, a substituted or

unsubstituted C_2 - C_{60} alkynyl group, a substituted or unsubstituted C_1 - C_{60} alkoxy group, a substituted or unsubstituted C_3 - C_{10} cycloalkyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3 - C_{10} cycloalkenyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6 - C_{60} aryl group, a substituted or unsubstituted C_6 - C_{60} aryloxy group, a substituted or unsubstituted C_6 - C_{60} arylthio group, a substituted or unsubstituted C_1 - C_{60} heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, $-Si(Q_{331})(Q_{332})(Q_{333})$, $-N(Q_{331})(Q_{332})$, $-B(Q_{331})(Q_{332})$, $-C(=O)(Q_{331})$, $-S(=O)_2(Q_{331})$, and $-P(=O)(Q_{331})(Q_{332})$,

at least one selected from L_{331} to L_{336} and R_{331} to R_{336} may include an electron transport moiety,

at least one selected from R_{331} to R_{336} may include deuterium, $xb42$ to $xb44$ may each independently be an integer from 0 to 10, and

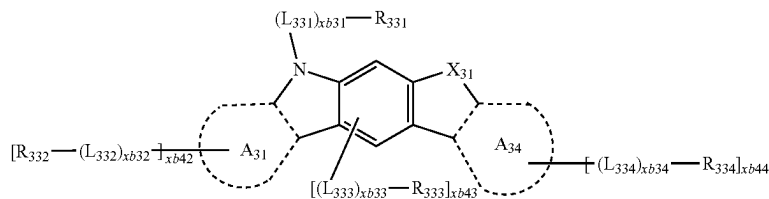
Q_{331} to Q_{333} may each independently be selected from hydrogen, deuterium, $-F$, $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{60} alkyl group, a C_2 - C_{60} alkenyl group, a C_2 - C_{60} alkynyl group, a C_1 - C_{60} alkoxy group, a C_3 - C_{10} cycloalkyl group, a C_1 - C_{10} heterocycloalkyl group, a C_3 - C_{10} cycloalkenyl group, a C_1 - C_{10} heterocycloalkenyl group, a C_6 - C_{60} aryl group, a C_6 - C_{60} aryloxy group, a C_6 - C_{60} arylthio group, a C_1 - C_{60} heteroaryl group, a C_1 - C_{60} heteroaryloxy group, a C_1 - C_{60} heteroarylthio group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a biphenyl group, and a terphenyl group.

In one exemplary embodiment, at least one selected from R_{331} to R_{336} in Formula 303 may be deuterium, a C_1 - C_{60} alkyl group substituted with deuterium, a C_3 - C_{10} cycloalkyl group substituted with deuterium, a C_6 - C_{60} aryl group, the substituted with deuterium, or a C_1 - C_{60} heteroaryl group substituted with deuterium.

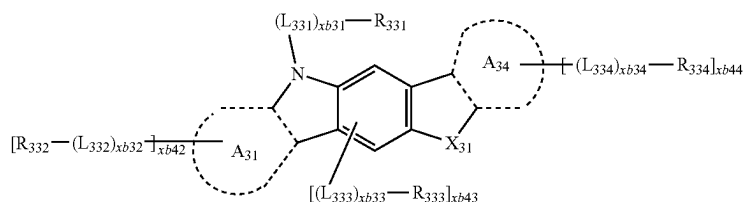
In one exemplary embodiment, the second compound may be a compound represented by one of Formulae 303-1 to 303-6 below:

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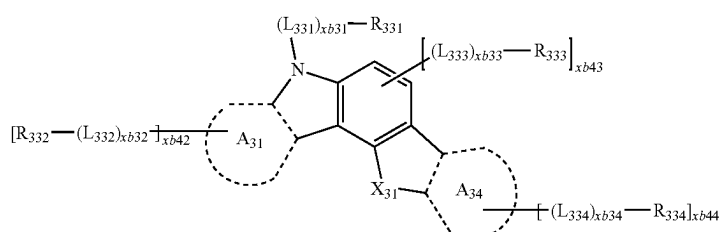
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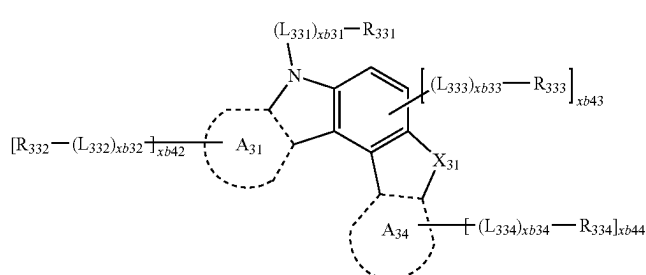
<Formula 303-1>



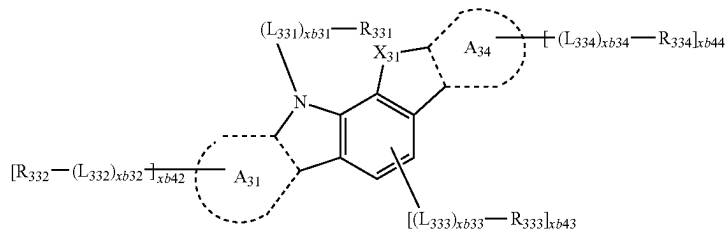
<Formula 303-2>



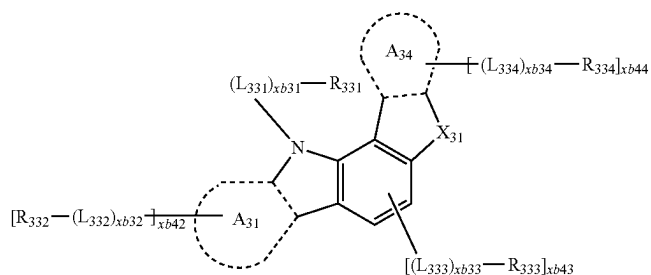
<Formula 303-3>



<Formula 303-4>



<Formula 303-5>



<Formula 303-6>

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In Formulae 303-1 to 303-6,

A_{31} , A_{34} , X_{31} , xb_{31} to xb_{34} , L_{331} to L_{336} , R_{331} to R_{336} , xb_{42} , and xb_{44} may each be understood by referring to the corresponding descriptions thereof provided above, and xb_{43} may be 1 or 2.

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In one exemplary embodiment, A_{31} , A_{32} , and A_{34} in Formula 303 may each independently be selected from a benzene group, an indene group, a naphthalene group, an anthracene group, a fluorene group, a phenanthrene group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, and a carbazole group.

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In one exemplary embodiment, A_{31} , A_{32} , and A_{34} in Formula 303 may each independently be selected from a benzene group, a naphthalene group, a phenanthrene group, a pyridine group, a pyrimidine group, a pyrazine group, and a pyridazine group.

In one exemplary embodiment, A_{31} , A_{32} , and A_{34} in Formula 303 may each independently be a benzene group, a naphthalene group, or a phenanthrene group.

In one exemplary embodiment, A_{31} , A_{32} , and A_{34} in Formula 303 may each independently be a benzene group or a naphthalene group.

In one exemplary embodiment, L_{321} to L_{326} and L_{331} to L_{336} in Formulae 302-1 to 302-4 and 303 may each independently be selected from:

a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthrenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group; and

a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylenylene group, a fluoranthrenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinox-

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alinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthrenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenylenylene group, a pentacenylenylene group, a thiophenyl group, a furanyl group, a carbazoyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazoyl group, a dibenzocarbazoyl group, a dibenzosiloyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinylenylene group, a quinazolinyl group, a cinnolinylenylene group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinylenylene group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazoyl group, —Si(Q_{31})(Q_{32})(Q_{33}), —N(Q_{31})(Q_{32}), —B(Q_{31})(Q_{32}), —C(=O)(Q_{31}), —S(=O)₂(Q_{31}), and —P(=O)(Q_{31})(Q_{32}).

In one exemplary embodiment, R_{321} to R_{326} and R_{331} to R_{336} in Formulae 302-1 to 302-4 and 303 may each independently be selected from:

a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenylenylene group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthrenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenylenylene group, a pentacenylenylene group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazoyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazoyl group, a dibenzocarbazoyl group, a dibenzosiloyl group, and a pyridinyl group; and

a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenylenylene group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a

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spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluo-
 ranthenyl group, a triphenylenyl group, a pyrenyl
 group, a chrysenyl group, a naphthacenyl group, a
 picenyl group, a perylenyl group, a pentaphenyl group,
 a hexacenyl group, a pentacenyl group, a rubicenyl
 group, a coronenyl group, an ovalenyl group, a thio-
 phenyl group, a furanyl group, a carbazolyl group, an
 indolyl group, an isoindolyl group, a benzofuranyl
 group, a benzothiophenyl group, a dibenzofuranyl
 group, a dibenzothiophenyl group, a benzocarbazolyl
 group, a dibenzocarbazolyl group, a dibenzosilolyl
 group, and a pyridinyl group, each substituted with at
 least one selected from deuterium, —F, —Cl, —Br,
 —I, a hydroxyl group, a cyano group, a nitro group, an
 amidino group, a hydrazino group, a hydrazono group,
 a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a cyclo-
 pentyl group, a cyclohexyl group, a cycloheptyl group,
 a cyclopentenyl group, a cyclohexenyl group, a phenyl
 group, a biphenyl group, a terphenyl group, a phenyl
 group substituted with a C₁-C₁₀ alkyl group, a phenyl
 group substituted with —F, a pentalenyl group, an
 indenyl group, a naphthyl group, an azulenyl group,
 a heptalenyl group, an indacenyl group, an acenaphthyl
 group, a fluorenyl group, a spiro-bifluorenyl group, a
 benzofluorenyl group, a dibenzofluorenyl group, a phe-
 nalenyl group, a phenanthrenyl group, an anthracenyl
 group, a fluoranthenyl group, a triphenylenyl group, a
 pyrenyl group, a chrysenyl group, a naphthacenyl
 group, a picenyl group, a perylenyl group, a pentaphe-
 nyl group, a hexacenyl group, a pentacenyl group, a
 rubicenyl group, a coronenyl group, an ovalenyl group,
 a thiophenyl group, a furanyl group, a carbazolyl
 group, an indolyl group, an isoindolyl group, a benzo-
 furanyl group, a benzothiophenyl group, a dibenzofura-
 nyl group, a dibenzothiophenyl group, a benzocarpa-
 zolyl group, a dibenzocarbazolyl group, a
 dibenzosilolyl group, a pyridinyl group, —Si(Q₃₁)
 (Q₃₂)(Q₃₃), and —N(Q₃₁)(Q₃₂).

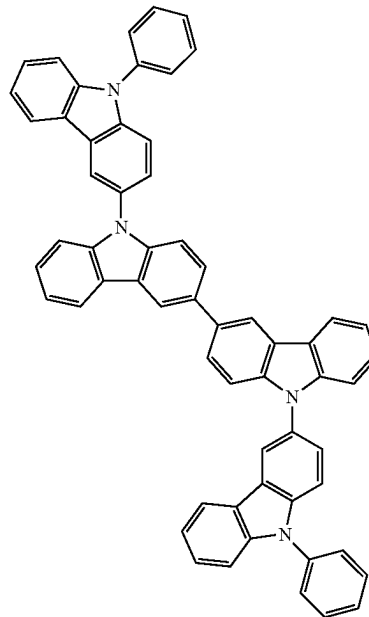
In one exemplary embodiment, Q₃₁ to Q₃₃ may each
 independently be selected from a C₁-C₁₀ alkyl group, a
 C₁-C₁₀ alkoxy group, a phenyl group, a biphenyl group, a
 terphenyl group, and a naphthyl group.

In one exemplary embodiment, Q₃₀₁ to Q₃₀₃, Q₃₂₁ to
 Q₃₂₃, and Q₃₃₁ to Q₃₃₃ may each independently be selected
 from a C₁-C₁₀ alkyl group, a C₁-C₁₀ alkoxy group, a phenyl
 group, a biphenyl group, a terphenyl group, a pentalenyl
 group, an indenyl group, a naphthyl group, an azulenyl
 group, a heptalenyl group, an indacenyl group, an acenaph-
 thyl group, a fluorenyl group, a spiro-bifluorenyl group, a
 benzofluorenyl group, a dibenzofluorenyl group, a phenale-
 nyl group, a phenanthrenyl group, an anthracenyl group, a
 fluoranthenyl group, a triphenylenyl group, a pyrenyl group,
 a chrysenyl group, a naphthacenyl group, a picenyl group, a
 perylenyl group, a pentaphenyl group, a hexacenyl group, a
 pentacenyl group, a rubicenyl group, a coronenyl group, an
 ovalenyl group, a thiophenyl group, a furanyl group, a
 carbazolyl group, an indolyl group, an isoindolyl group, a
 benzofuranyl group, a benzothiophenyl group, a dibenzo-
 furanyl group, a dibenzothiophenyl group, a benzocarpa-
 zolyl group, a dibenzocarbazolyl group, a dibenzosilolyl
 group, and a pyridinyl group.

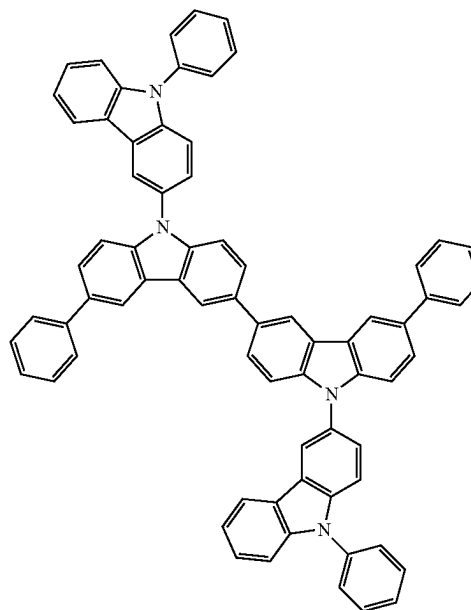
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In one exemplary embodiment, the first compound may
 be selected from Compounds HT1-1, HT1-2, HT2-1, and
 HT2-2 below:

<HT1-1>

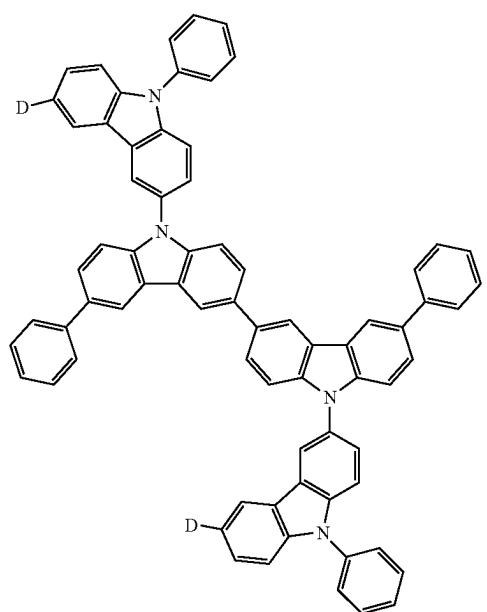
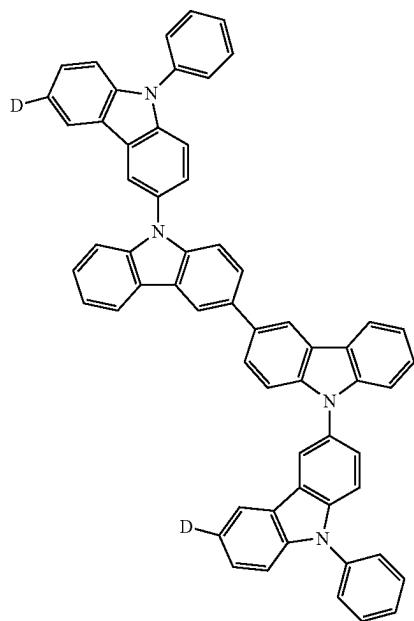


<HT1-2>



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-continued



In one exemplary embodiment, the first compound may be selected from Compounds HT2-1 and HT2-2 below:

In one exemplary embodiment, the second compound may be selected from Compounds ET2-1, ET2-2, ET4-1, and ET4-2 below:

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<HT2-1>

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<HT2-2>

<ET2-1>

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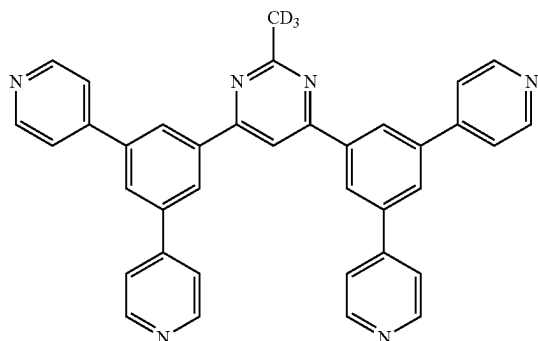
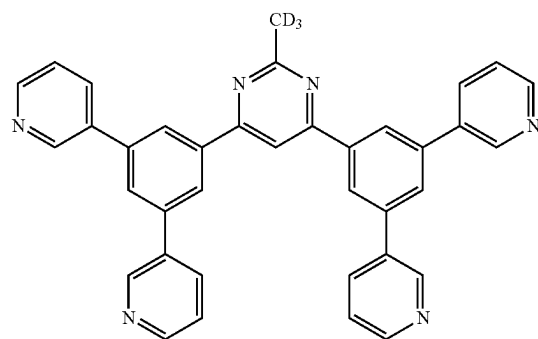
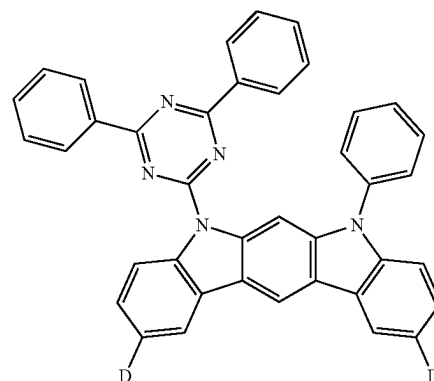
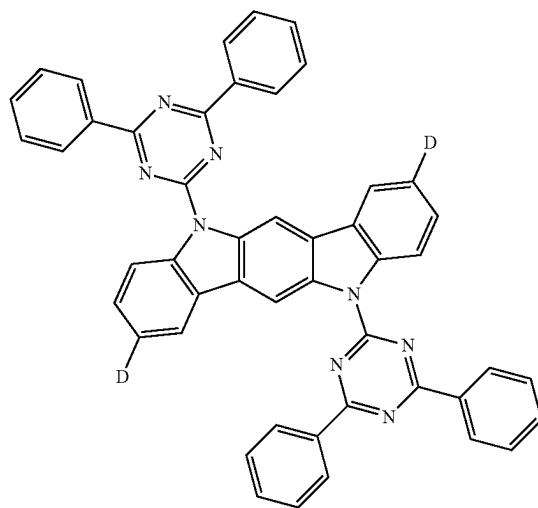
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<ET2-2>

<ET4-1>

<ET4-2>



The organic light-emitting device may include the first compound and the second compound. The first compound may be a hole transport compound not including an electron transport moiety, and the second compound may be an electron transport compound including an electron transport moiety or a bipolar compound including an electron transport moiety, wherein a combination of the first compound and the second compound may control the distribution of holes and electrons in the organic light-emitting device.

The second compound may include an electron transport moiety and deuterium. Deuterium is an isotope of hydrogen, and a deuterium-carbon bond thereof is more stable than a hydrogen-carbon bond, and thus, the second compound may have high molecular stability, and more particularly, may have excellent electrical stability.

Therefore, according to one exemplary embodiment, when the emission layer of the organic light-emitting device is formed by mixing or co-depositing the first compound and the second compound, injection characteristics of holes and electrons are improved, and charge balance in the emission layer is also improved, thereby increasing both efficiency and lifespan.

In one exemplary embodiment, when the first compound includes deuterium, the molecular stability, especially the electrical stability, of the first compound may be improved, and accordingly, the organic light-emitting device including the first compound and the second compound may have further improved efficiency and further improved lifespan.

In one exemplary embodiment, the first electrode may be an anode.

In one exemplary embodiment, the second electrode may be a cathode.

In one exemplary embodiment, the organic light-emitting device may further include a hole transport region between the first electrode and the emission layer and an electron transport region between the emission layer and the second electrode.

In one exemplary embodiment, the hole transport region may include at least one selected from a hole injection layer, a hole transport layer, a buffer layer, an emission auxiliary layer, and an electron blocking layer.

In one exemplary embodiment, the electron transport region may include at least one selected from a hole blocking layer, an electron transport layer, and an electron injection layer.

In one exemplary embodiment, the hole transport region may include an amine-based compound.

In one exemplary embodiment, the first compound and the second compound may be included in the emission layer.

In one exemplary embodiment, based on weights of the first compound and the second compound, an amount of the first compound in the emission layer may be greater than that of the second compound.

In one exemplary embodiment, the emission layer may emit green light having the maximum emission wavelength in a range of about 490 nm to about 560 nm.

In one exemplary embodiment, the emission layer may further include a dopant, and the dopant may include a phosphorescent dopant or a fluorescent dopant.

In one exemplary embodiment, the emission layer may further include a dopant, and the dopant may include a phosphorescent dopant.

In one exemplary embodiment, the electron transport region may include a metal-containing material.

In one exemplary embodiment, the first emission layer may further include a first dopant, and the second emission layer may further include a second dopant. In one exemplary

embodiment, the first dopant and the second dopant may be identical to or different from each other. The first dopant and the second dopant may each be understood by referring to the descriptions provided in connection with the dopant.

Another aspect of the invention provides an apparatus including: the organic light-emitting device; and a thin-film transistor, wherein the first electrode of the organic light-emitting device is in electrical contact with one of a source electrode or a drain electrode of the thin-film transistor.

The term "organic layer" as used herein refers to a single layer and/or all layers between the first electrode and the second electrode of the organic light-emitting device. A material included in "the organic layer" is not limited to an organic material.

15 Description of FIG. 1

FIG. 1 is a schematic view of an organic light-emitting device 10 according to one exemplary embodiment. The organic light-emitting device 10 includes a first electrode 110, an organic layer 150, and a second electrode 190.

Hereinafter, the structure of the organic light-emitting device 10 according to an embodiment and a method of manufacturing the organic light-emitting device 10 will be described in connection with FIG. 1.

First Electrode 110

Referring to FIG. 1, a substrate may be additionally located under the first electrode 110 or above the second electrode 190. The substrate may be a glass substrate or a plastic substrate, each having excellent mechanical strength, thermal stability, transparency, surface smoothness, ease of handling, and water resistance.

The first electrode 110 may be, for example, formed by depositing or sputtering a material for forming the first electrode 110 on the substrate. When the first electrode 110 is an anode, the material for a first electrode 110 may be selected from materials with a high work function to facilitate hole injection.

The first electrode 110 may be a reflective electrode, a semi-transmissive electrode, or a transmissive electrode. When the first electrode 110 is a transmissive electrode, the material for forming the first electrode 110 may be selected from indium tin oxide (ITO), indium zinc oxide (IZO), tin oxide (SnO₂), zinc oxide (ZnO), and any combinations thereof, but exemplary embodiments of the invention are not limited thereto. In one or more exemplary embodiments, when the first electrode 110 is a semi-transmissive electrode or a reflective electrode, the material for forming the first electrode 110 may be selected from magnesium (Mg), silver (Ag), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), and any combinations thereof, but exemplary embodiments of the invention are not limited thereto.

The first electrode 110 may have a single-layered structure, or a multi-layered structure including two or more layers. For example, the first electrode 110 may have a three-layered structure of ITO/Ag/ITO, but the structure of the first electrode 110 is not limited thereto.

Organic Layer 150

The organic layer 150 is disposed on the first electrode 110. The organic layer 150 may include an emission layer.

The organic layer 150 may further include a hole transport region between the first electrode 110 and the emission layer and an electron transport region between the emission layer and the second electrode 190.

Hole Transport Region in Organic Layer 150

The hole transport region may have i) a single-layered structure including a single layer including a single material,

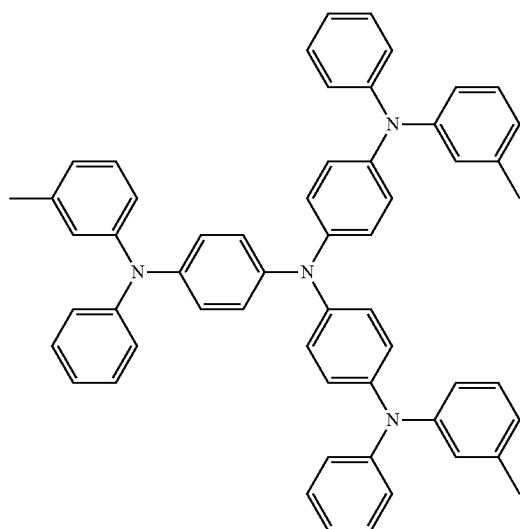
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ii) a single-layered structure including a single layer including a plurality of different materials, or iii) a multi-layered structure having a plurality of layers including a plurality of different materials.

The hole transport region may include at least one layer selected from a hole injection layer, a hole transport layer, an emission auxiliary layer, and an electron blocking layer.

For example, the hole transport region may have a single-layered structure including a single layer including a plurality of different materials, or a multi-layered structure having a hole injection layer/hole transport layer structure, a hole injection layer/hole transport layer/emission auxiliary layer structure, a hole injection layer/emission auxiliary layer structure, a hole transport layer/emission auxiliary layer structure, or a hole injection layer/hole transport layer/electron blocking layer structure, wherein for each structure, constituting layers are sequentially stacked from the first electrode 110 in this stated order, but the structure of the hole transport region is not limited thereto.

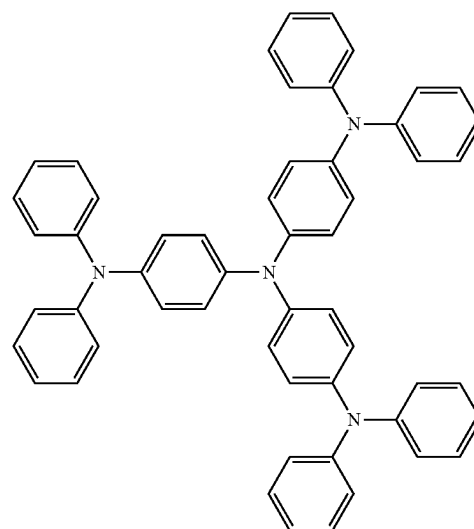
In one exemplary embodiment, the hole transport region may include at least one selected from 4,4',4''-Tris[phenyl (m-tolyl)amino]triphenylamine (m-MTDATA), 1-N,1-N-bis[4-(diphenylamino)phenyl]-4-N,4-N-diphenylbenzene-1,4-diamine (TDATA), 4,4',4''-Tris[2-naphthyl(phenyl)amino]triphenylamine (2-TNATA), N,N'-Di(1-naphthyl)-N,N'-diphenyl-(1,1'-biphenyl)-4,4'-diamine (NPB or NPD), N4,N4'-Di(naphthalen-2-yl)-N4,N4'-diphenyl-[1,1'-biphenyl]-4,4'-diamine (β -NPB), N,N'-Bis(3-methylphenyl)-N,N'-diphenylbenzidine (TPD), 4(N,N'-Bis(3-methylphenyl)-N,N'-diphenyl-9,9-spirobifluorene-2,7-diamine (spiro-TPD), 2,7-Bis[N-(1-naphthyl)anilino]-9,9'-spirobifluorene (spiro-NPB), 2,2'-Dimethyl-N,N'-di-[(1-naphthyl)-N,N'-diphenyl]-1,1'-biphenyl-4,4'-diamine (methylated-NPB), 4-(1-{4-[bis(4-methylphenyl)amino]phenyl}cyclohexyl)-N,N-bis(4-methylphenyl)aniline (TAPC), N,N,N',N'-Tetrakis(3-methylphenyl)-3,3'-dimethylbenzidine (HMTPD), 4,4',4''-tris(N-carbazolyl)triphenylamine (TCTA), polyaniline/dodecylbenzenesulfonic acid (PANI/DBSA), poly(3,4-ethylenedioxythiophene)/poly(4-styrenesulfonate) (PEDOT/PSS), polyaniline/camphor sulfonic acid (PANI/CSA), polyaniline/poly(4-styrenesulfonate) (PANI/PSS), a compound represented by Formula 201 below, and a compound represented by Formula 202 below:



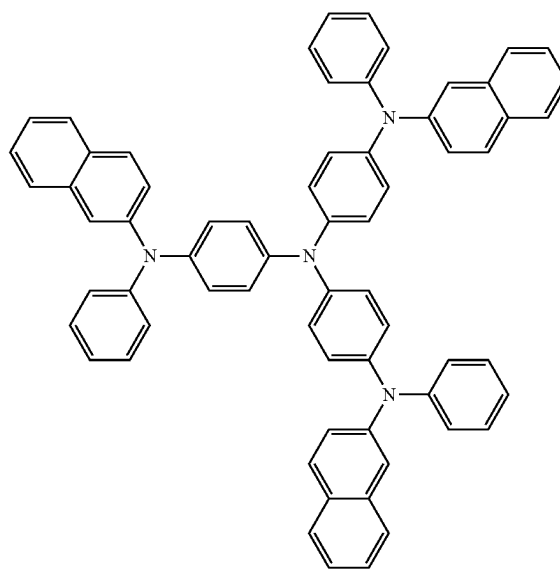
m-MTDATA

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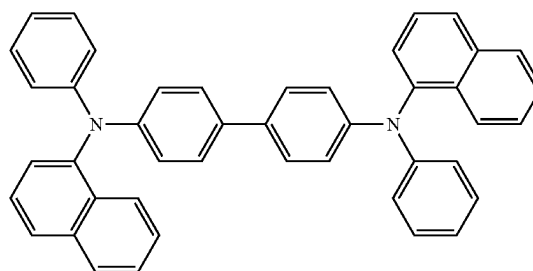
-continued



TDATA



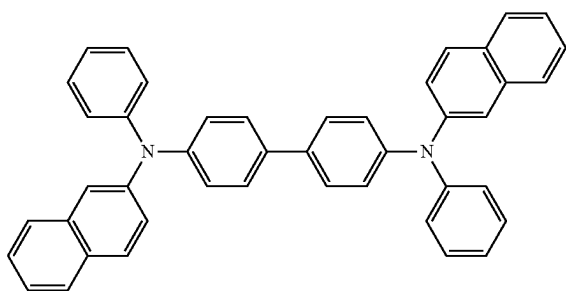
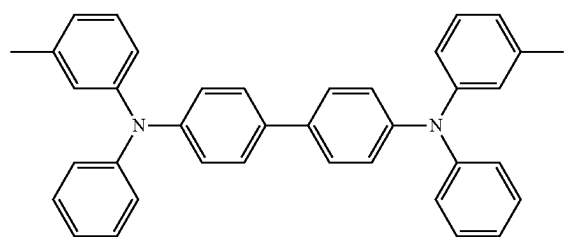
2-TNATA



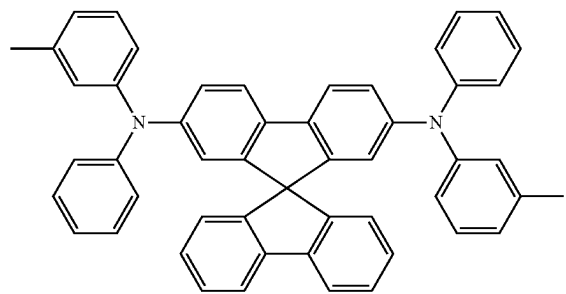
NPB

33

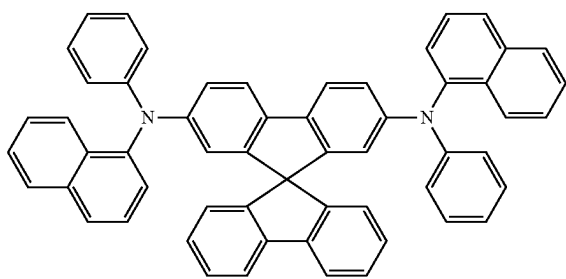
-continued

 β -NPB

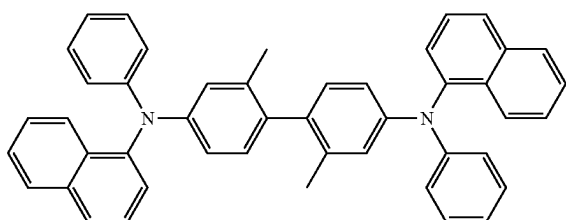
TPD



Spiro-TPD



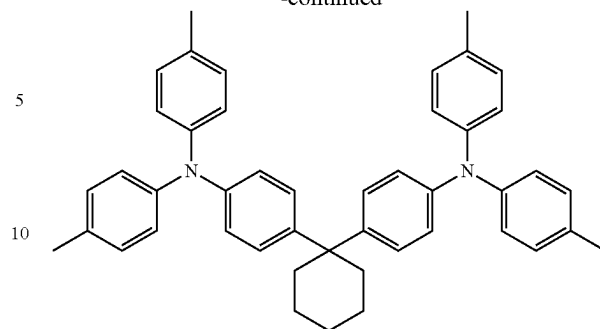
Spiro-NPB



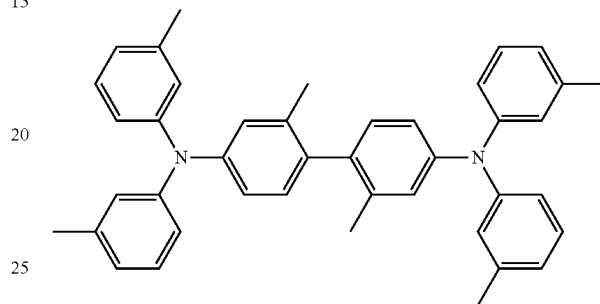
methylated NPB

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-continued

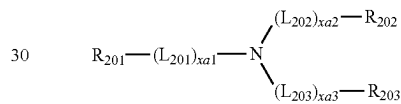


TAPC

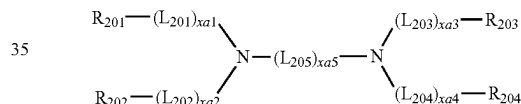


HMTPD

<Formula 201>



<Formula 202>



In Formulae 201 and 202,

L_{201} to L_{204} may each independently be selected from a substituted or unsubstituted C_3 - C_{10} cycloalkylene group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkylene group, a substituted or unsubstituted C_3 - C_{10} cycloalkenylene group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkenylene group, a substituted or unsubstituted C_6 - C_{60} arylene group, a substituted or unsubstituted C_1 - C_{60} heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group,

L_{205} may be selected from $^*\text{—O—}^*$, $^*\text{—S—}^*$, $^*\text{—N—}^*$ (Q_{201})- * , a substituted or unsubstituted C_1 - C_{20} alkylene group, a substituted or unsubstituted C_2 - C_{20} alkenylene group, a substituted or unsubstituted C_3 - C_{10} cycloalkylene group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkylene group, a substituted or unsubstituted C_3 - C_{10} cycloalkenylene group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkenylene group, a substituted or unsubstituted C_6 - C_{60} arylene group, a substituted or unsubstituted C_1 - C_{60} heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group,

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xa1 to xa4 may each independently be an integer from 0 to 3,

xa5 may be an integer from 1 to 10,

R₂₀₁ to R₂₀₄ and Q₂₀₁ may each independently be selected from a substituted or unsubstituted C₃-C₁₀ cycloalkyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkyl group, a substituted or unsubstituted C₃-C₁₀ cycloalkenyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenyl group, a substituted or unsubstituted C₆-C₆₀ aryl group, a substituted or unsubstituted C₆-C₆₀ aryloxy group, a substituted or unsubstituted C₆-C₆₀ arylthio group, a substituted or unsubstituted C₁-C₆₀ heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group.

In one exemplary embodiment, in Formula 202, R₂₀₁ and R₂₀₂ may optionally be linked via a single bond, a dimethylmethylene group, or a diphenyl-methylene group, and R₂₀₃ and R₂₀₄ may optionally be linked via a single bond, a dimethyl-methylene group, or a diphenyl-methylene group.

In one exemplary embodiment, in Formulae 201 and 202,

L₂₀₁ to L₂₀₅ may each independently be selected from:

a phenylene group, a pentalenylene group, an indenylene group, a naphthylene group, an azulenylene group, a heptalenylene group, an indacenylene group, an acenaphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenalenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthrenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a naphthacenylenylene group, a picenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylene group, a pentacenylene group, a rubicenylene group, a coronenylene group, an ovalenylene group, a thiophenylenylene group, a furanylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, and a pyridinylenylene group; and

a phenylene group, a pentalenylene group, an indenylene group, a naphthylene group, an azulenylene group, a heptalenylene group, an indacenylene group, an acenaphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenalenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthrenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a naphthacenylenylene group, a picenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylene group, a pentacenylene group, a rubicenylene group, a coronenylene group, an ovalenylene group, a thiophenylenylene group, a furanylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, and a pyridinylenylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl

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group, a C₁-C₂₀ alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C₁-C₁₀ alkyl group, a phenyl group substituted with —F, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthrenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, —Si(Q₃₁)(Q₃₂)(Q₃₃) and —N(Q₃₁)(Q₃₂), wherein

Q₃₁ to Q₃₃ may each independently be selected from a C₁-C₁₀ alkyl group, a C₁-C₁₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

In one or more exemplary embodiments, xa1 to xa4 may each independently be 0, 1, or 2.

In one or more exemplary embodiments, xa5 may be 1, 2, 3, or 4.

In one or more exemplary embodiments, R₂₀₁ to R₂₀₄ and Q₂₀₁ may each independently be selected from a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthrenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group; and

a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthrenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group, each substituted with at

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least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C₁-C₁₀ alkyl group, a phenyl group substituted with —F, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, —Si(Q₃₁)(Q₃₂)(Q₃₃) and —N(Q₃₁)(Q₃₂); and

Q₃₁ to Q₃₃ are the same as described above.

In one or more exemplary embodiments, at least one selected from R₂₀₁ to R₂₀₃ in Formula 201 may each independently be selected from:

a fluorenyl group, a spiro-bifluorenyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group,

a fluorenyl group, a spiro-bifluorenyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C₁-C₁₀ alkyl group, a phenyl group substituted with —F, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group, but exemplary embodiments of the invention are not limited thereto.

In one or more exemplary embodiments, in Formula 202, i) R₂₀₁ and R₂₀₂ may be linked via a single bond, and/or ii) R₂₀₃ and R₂₀₄ may be linked via a single bond.

In one or more exemplary embodiments, R₂₀₁ to R₂₀₄ in Formula 202 may be selected from;

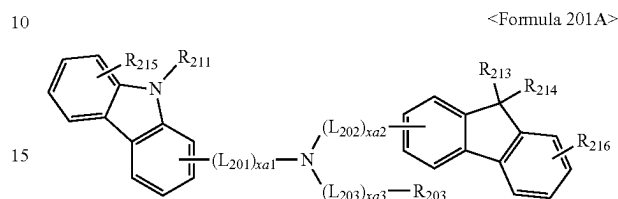
a carbazolyl group; and

a carbazolyl group substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C₁-C₁₀ alkyl group, a phenyl group substituted with —F, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a carbazolyl group, a diben-

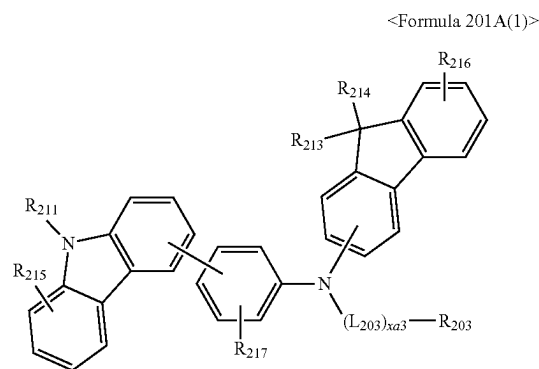
38

zofuranyl group, and a dibenzothiophenyl group, but exemplary embodiments of the invention are not limited thereto.

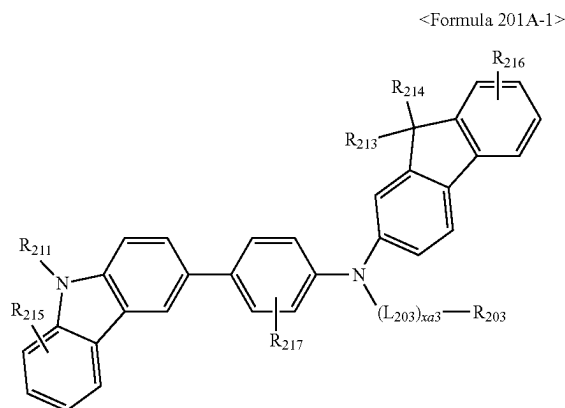
In one or more exemplary embodiments, the compound represented by Formula 201 may be represented by Formula 201A below:



In one or more exemplary embodiments, the compound represented by Formula 201 may be represented by Formula 201A(1) below, but exemplary embodiments of the invention are not limited thereto:

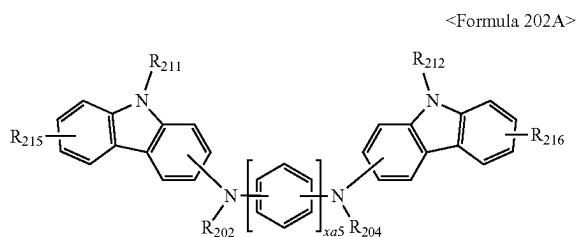


In one or more embodiments, the compound represented by Formula 201 may be represented by Formula 201A-1 below, but exemplary embodiments of the invention are not limited thereto:

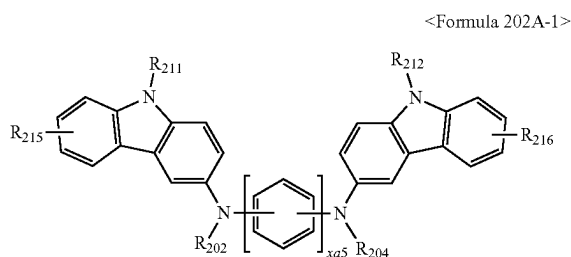


In one or more exemplary embodiments, the compound represented by Formula 202 may be represented by Formula 202A below:

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In one or more exemplary embodiments, the compound represented by Formula 202 may be represented by Formula 202A-1 below:



In Formulae 201A, 201A(1), 201A-1, 202A, and 202A-1, L_{201} to L_{203} , $xa1$ to $xa3$, $xa5$, and R_{202} to R_{204} may each be understood by referring to the corresponding descriptions thereof provided above,

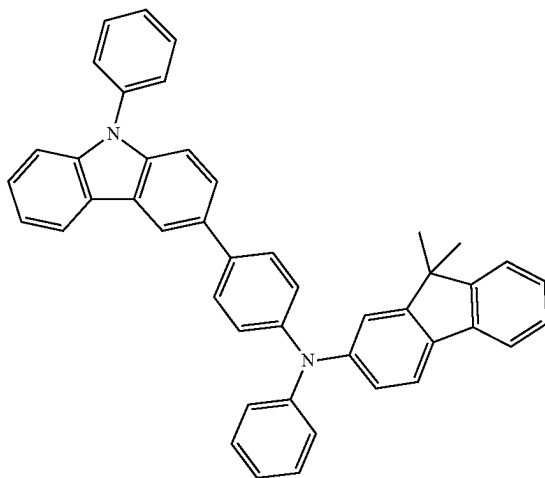
40

R_{211} and R_{212} may each be understood by referring to the description presented in connection with R_{203} , and

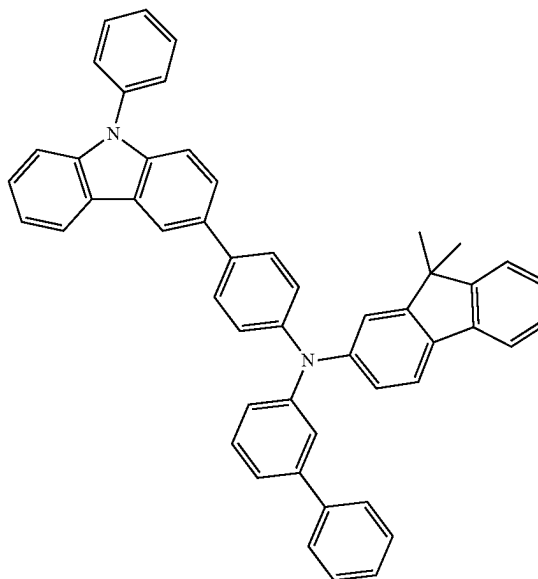
R_{213} to R_{217} may each independently be selected from hydrogen, deuterium, $-F$, $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C_1 - C_{10} alkyl group, a phenyl group substituted with $-F$, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group.

The hole transport region may include at least one compound selected from Compounds HT1 to HT39 below, but compounds to be included in the hole transport region are not limited thereto:

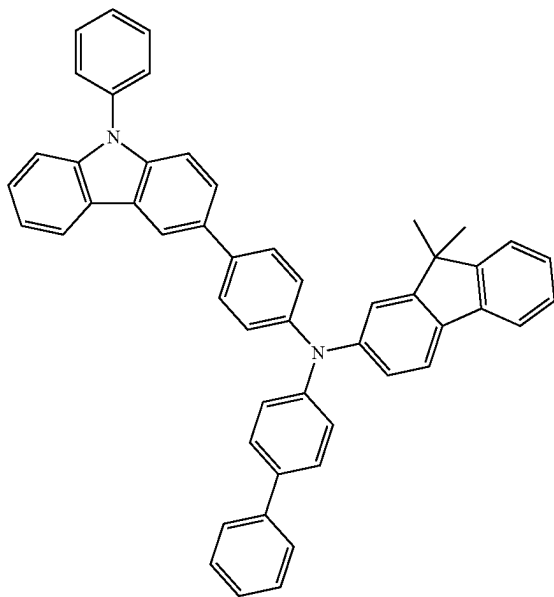
HT1



HT2

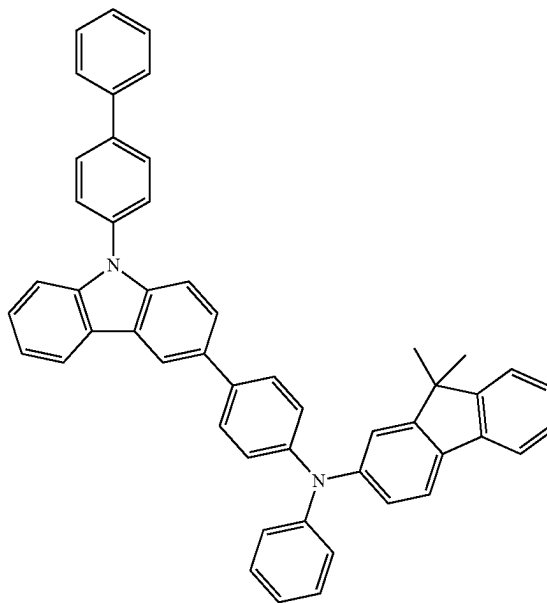


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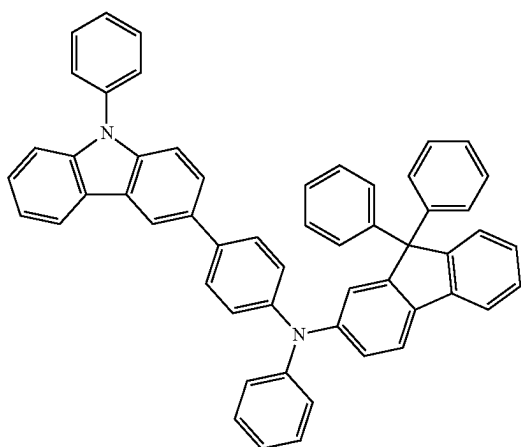
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HT3

42

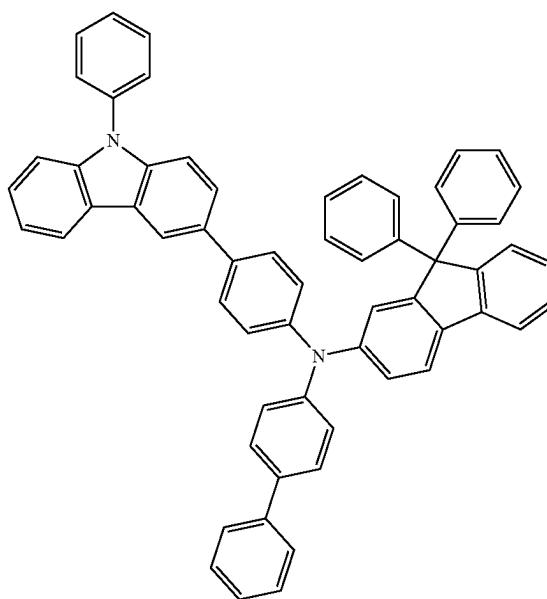
HT4



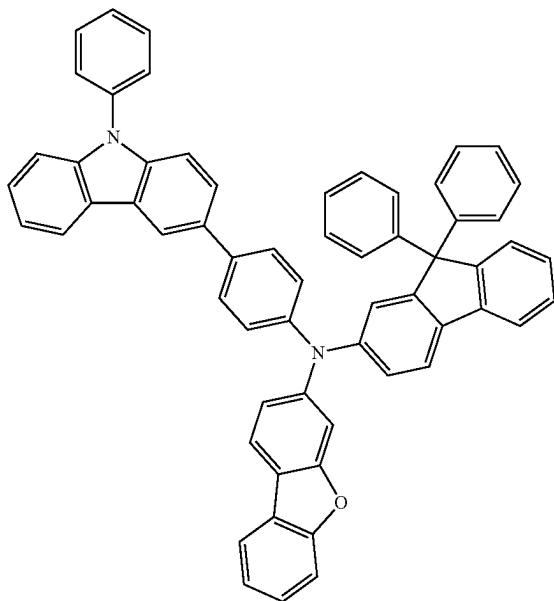
HT5



HT6

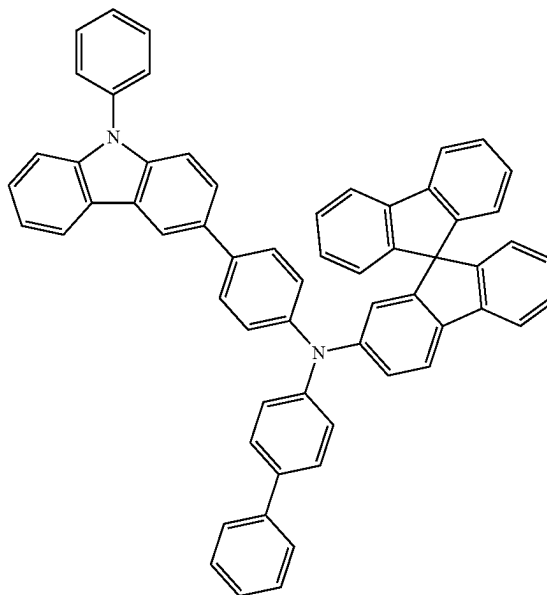


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-continued
HT7

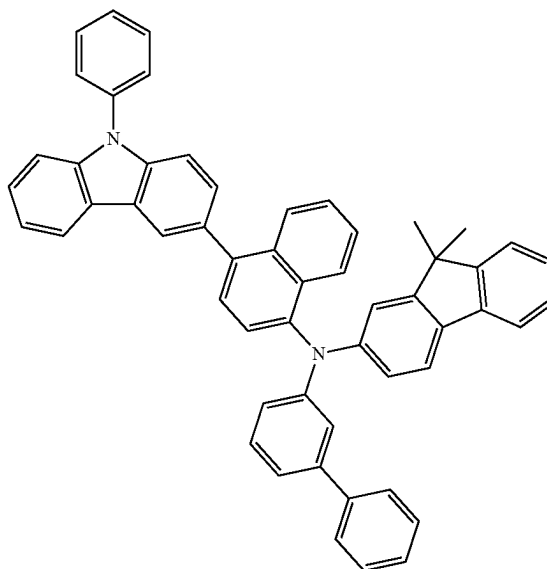
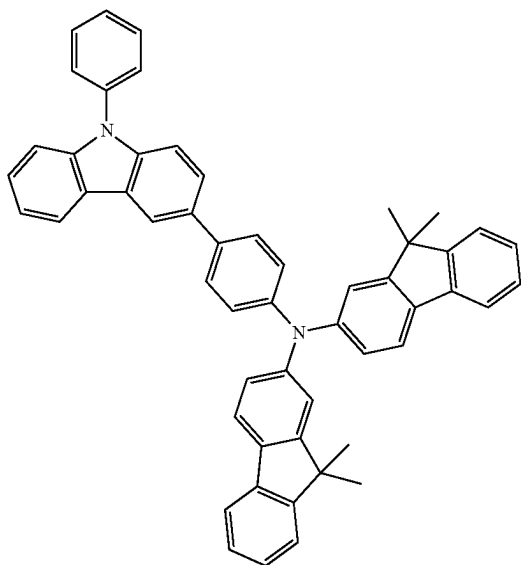
44

HT8

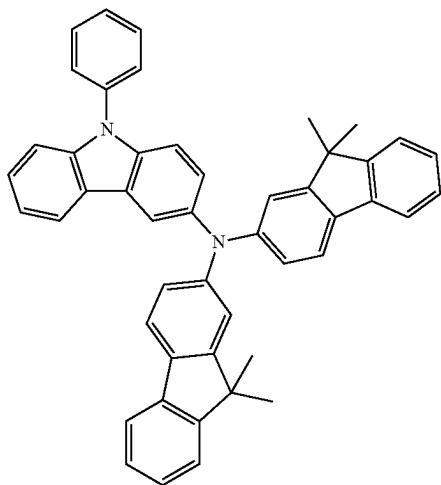


HT9

HT10

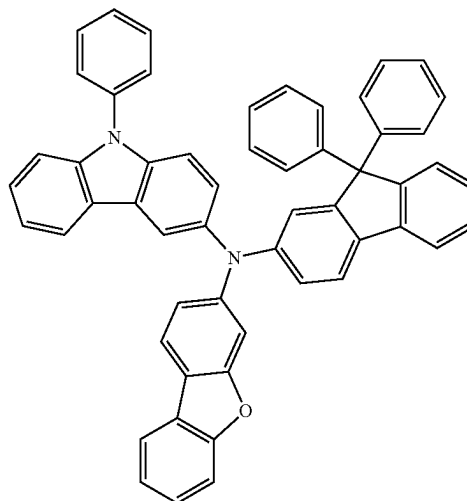


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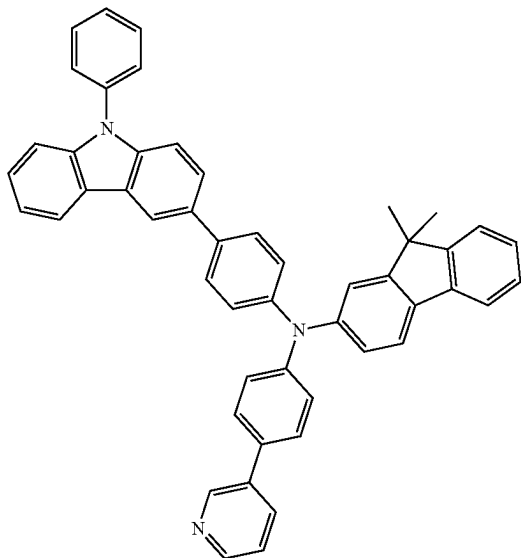
-continued
HT11

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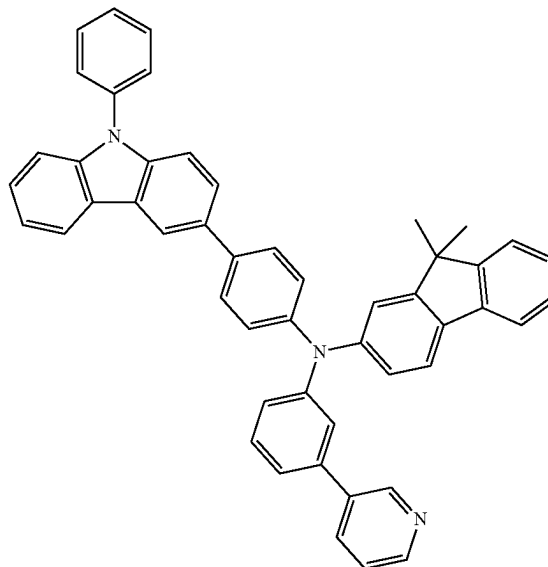


HT12

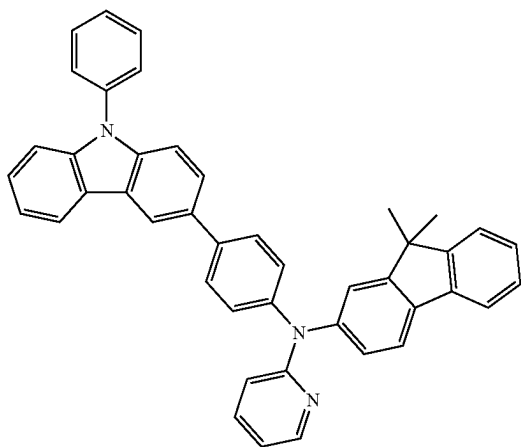
HT13



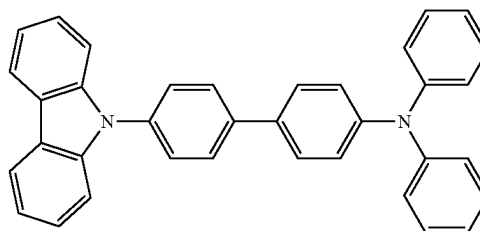
HT14



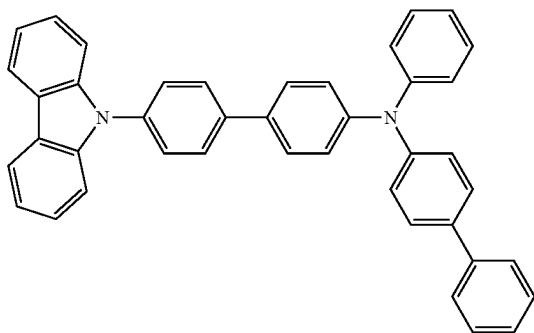
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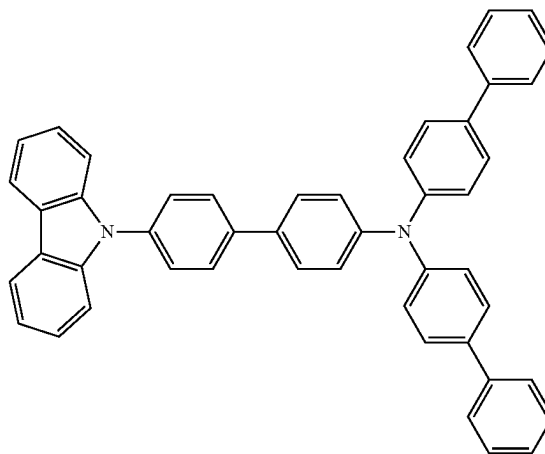


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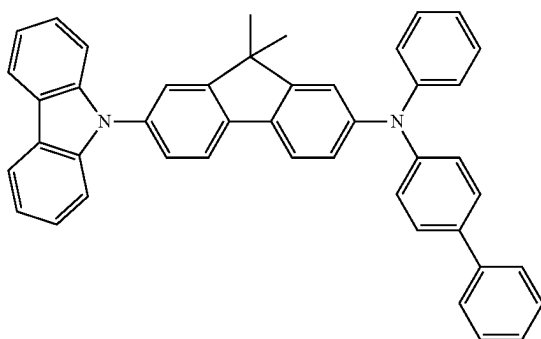
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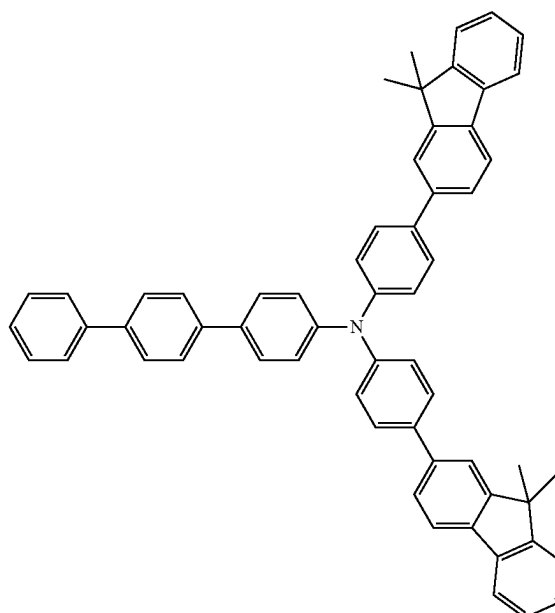
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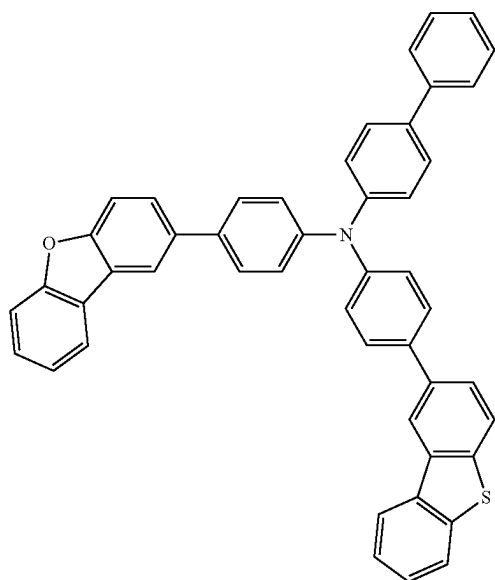
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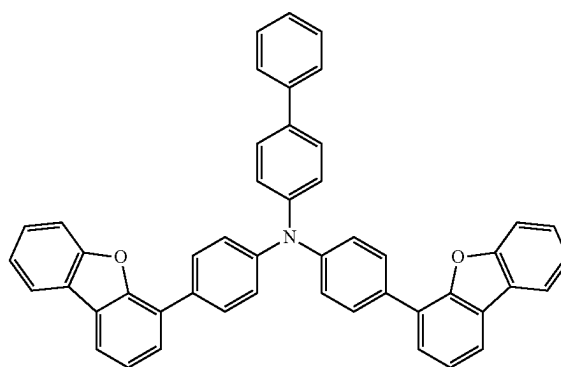


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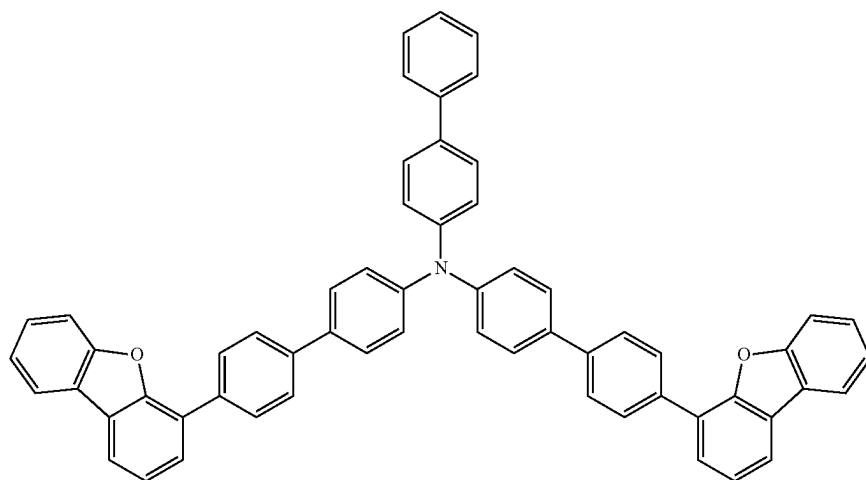
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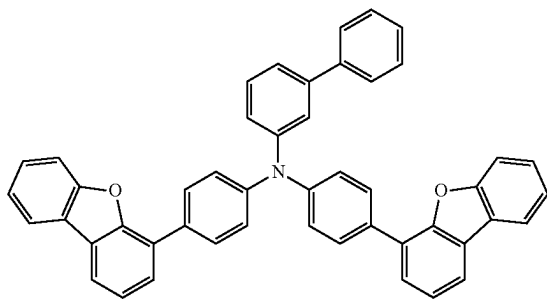
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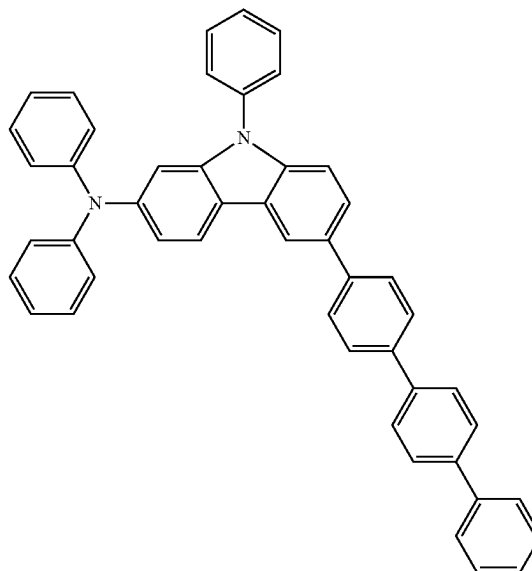


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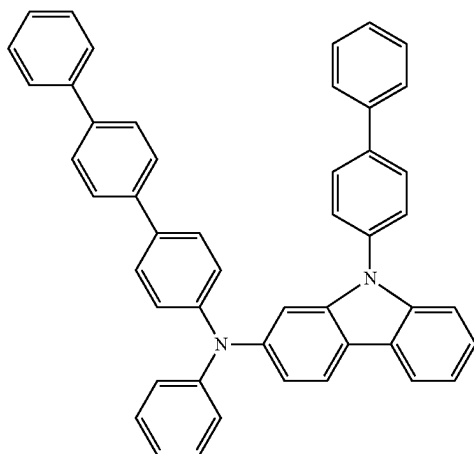
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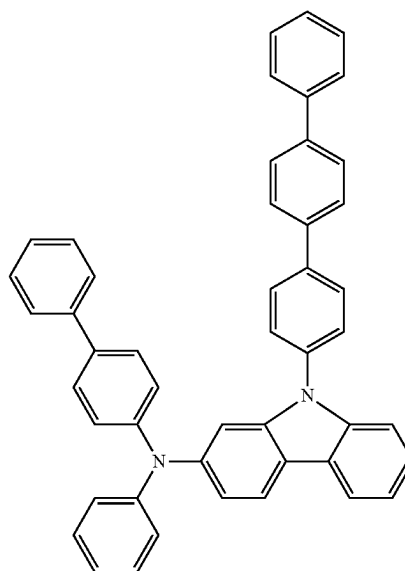
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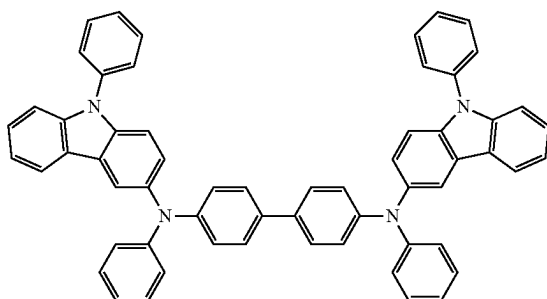
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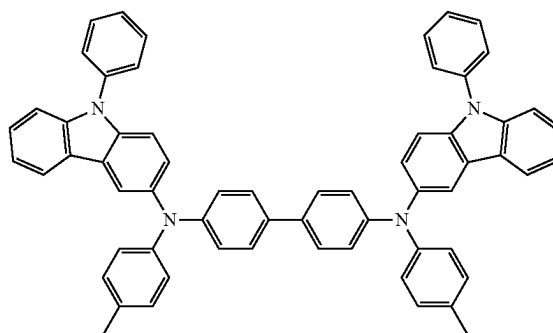
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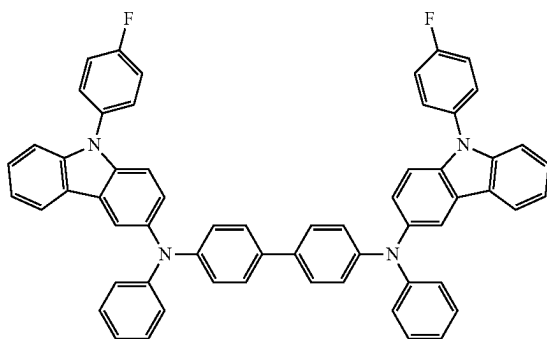
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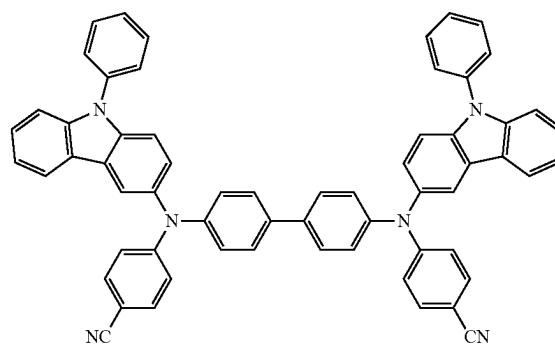


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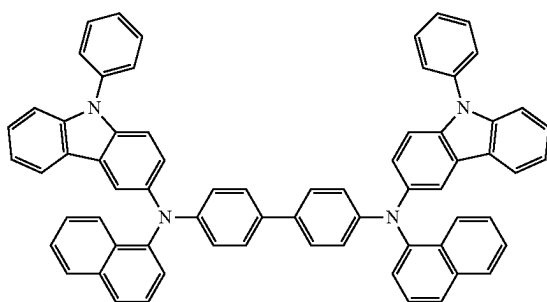
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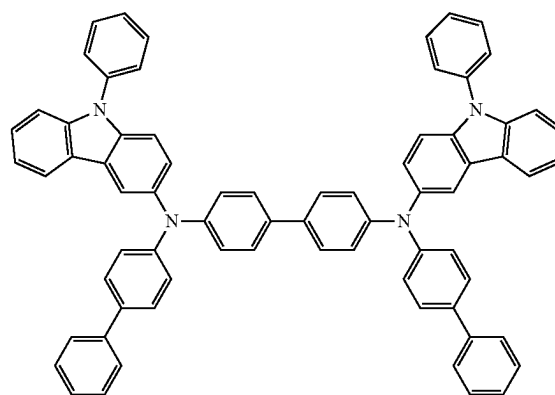
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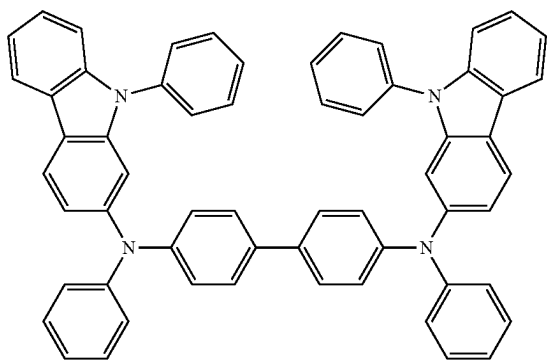
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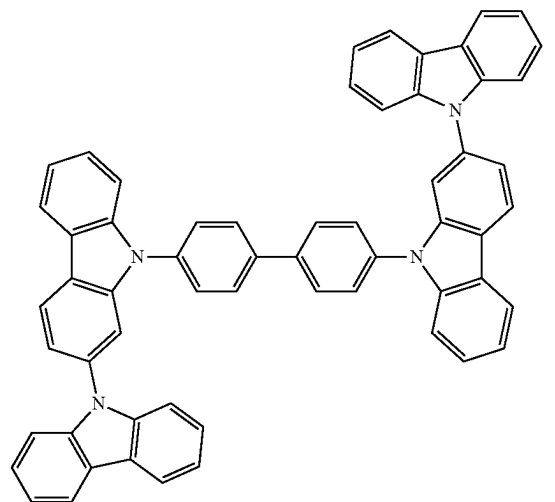
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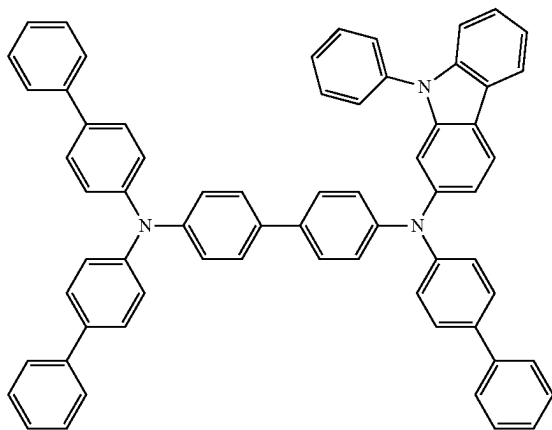
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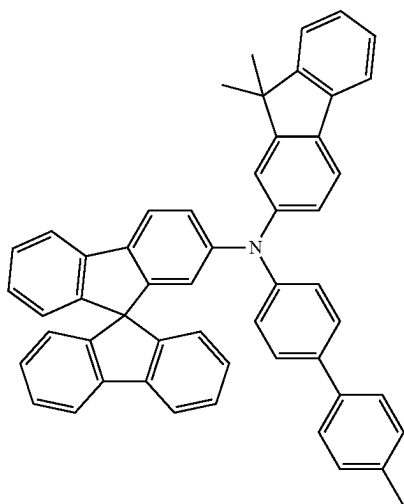
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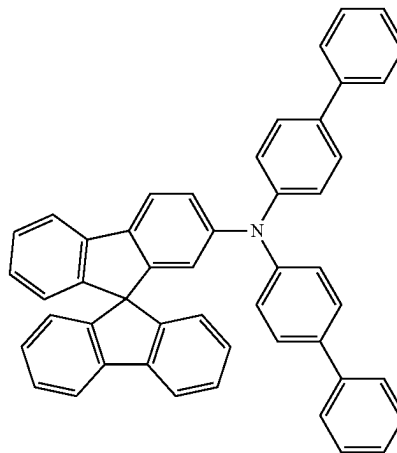
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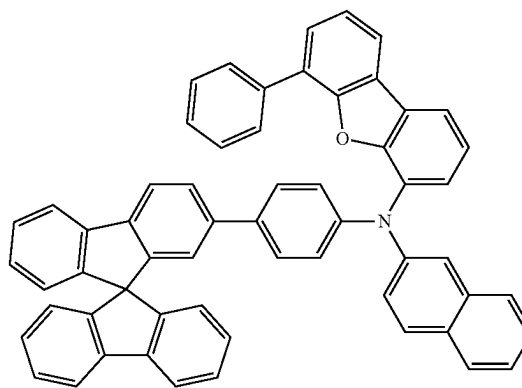


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HT37



HT39



The thickness of the hole transport region may be in a range of about 100 Å to about 10,000 Å, for example, about 100 Å to about 1,000 Å. When the hole transport region includes at least one of a hole injection layer and a hole transport layer, the thickness of the hole injection layer may be in a range of about 100 Å to about 9,000 Å, for example, about 100 Å to about 1,000 Å, and the thickness of the hole transport layer may be in a range of about 50 Å to about 2,000 Å, for example about 100 Å to about 1,500 Å. When the thicknesses of the hole transport region, the hole injection layer and the hole transport layer are within these ranges, satisfactory hole transporting characteristics may be obtained without a substantial increase in driving voltage.

The emission auxiliary layer may increase light-emission efficiency by compensating for the optical resonance distance according to the wavelength of light emitted by an emission layer, and the electron blocking layer may block the flow of electrons from an electron transport region. The emission auxiliary layer and the electron blocking layer may include the materials as described above.

p-Dopant

The hole transport region may further include, in addition to these materials, a charge-generation material for the improvement of conductive properties. The charge-gener-

ation material may be homogeneously or non-homogeneously dispersed in the hole transport region.

The charge-generation material may be, for example, a p-dopant.

In one exemplary embodiment, a lowest unoccupied molecular orbital (LUMO) of the p-dopant may be -3.5 eV or less.

The p-dopant may include at least one selected from a quinone derivative, a metal oxide, and a cyano group-containing compound, but exemplary embodiments of the invention are not limited thereto.

For example, the p-dopant may include at least one selected from

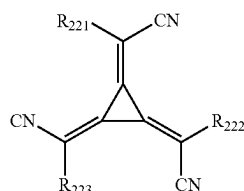
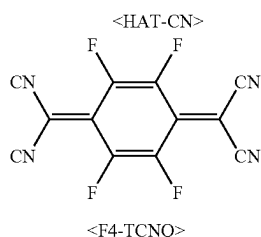
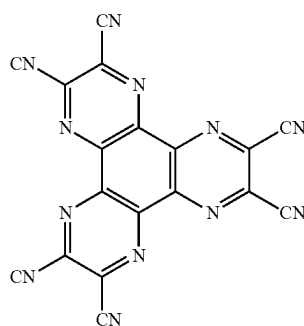
a quinone derivative, such as tetracyanoquinodimethane (TCNQ) and F4-2,3,5,6-Tetrafluoro-7,7,8,8-tetracyanoquinodimethane (TCNQ);

a metal oxide, such as tungsten oxide or molybdenum oxide;

1,4,5,8,9,12-hexaazatriphenylene-hexacarbonitrile (HAT-CN); and

a compound represented by Formula 221 below, but exemplary embodiments of the invention are not limited thereto:

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<Formula 221>

In Formula 221,

R₂₂₁ to R₂₂₃ may each independently be selected from a substituted or unsubstituted C₃-C₁₀ cycloalkyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkyl group, a substituted or unsubstituted C₃-C₁₀ cycloalkenyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenyl group, a substituted or unsubstituted C₆-C₆₀ aryl group, a substituted or unsubstituted C₁-C₆₀ heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, wherein at least one selected from R₂₂₁ to R₂₂₃ may have at least one substituent selected from a cyano group, —F, —Cl, —Br, —I, a C₁-C₂₀ alkyl group substituted with —F, a C₁-C₂₀ alkyl group substituted with —Cl, a C₁-C₂₀ alkyl group substituted with —Br, and a C₁-C₂₀ alkyl group substituted with —I.

Emission Layer in Organic Layer 150

When the organic light-emitting device 10 is a full-color organic light-emitting device, the emission layer may be patterned into a red emission layer, a green emission layer, or a blue emission layer, according to a representative sub-pixel. In one or more exemplary embodiments, the emission layer may have a stacked structure of two or more layers selected from a red emission layer, a green emission layer, and a blue emission layer, in which the two or more layers contact each other or are separated from each other. In one or more exemplary embodiments, the emission layer may include two or more materials selected from a red light-emitting material, a green light-emitting material, and a blue light-emitting material, in which the two or more materials are mixed with each other in a single layer to emit white light.

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The emission layer may include at least one host and a dopant. The dopant may include at least one selected from a phosphorescent dopant and a fluorescent dopant.

In the emission layer, an amount of the dopant may be in a range of about 0.01 parts by weight to about 15 parts by weight based on 100 parts by weight of the host, but exemplary embodiments of the invention are not limited thereto.

The thickness of the emission layer may be in a range of about 100 Å to about 1,000 Å, for example, about 200 Å to about 600 Å. When the thickness of the emission layer is within this range, excellent light-emission characteristics may be obtained without a substantial increase in driving voltage.

At Least One Host in Emission Layer

In one exemplary embodiment, the emission layer may include a first emission layer and a second emission layer, wherein the first emission layer includes a first host and a second host, and the second emission layer includes a third host and a fourth host.

In one exemplary embodiment, in the first emission layer and the first host may be formed from the first compound, and the second host may be formed from the second compound. For example, in the first emission layer, the first host may be a hole transport compound not including the electron transport moiety, and the second host may be an electron transport compound including an electron transport moiety. In one or more exemplary embodiments, in the first emission layer, the first host may be a hole transport compound not including an electron transport moiety, and the second host may be a bipolar compound including an electron transport moiety.

In one exemplary embodiment, in the second emission layer and the third host may be formed from the first compound, and the fourth host may be formed from the second compound. For example, in the second emission layer and the third host may be formed from a hole transport compound not including an electron transport moiety, and the fourth host may be formed from an electron transport compound including an electron transport moiety. In one or more exemplary embodiments, in the second emission layer and the third host may be formed from a hole transport compound not including an electron transport moiety, and the fourth host may be formed from a bipolar compound including an electron transport moiety.

In one exemplary embodiment, the first host in the first emission layer and the third host in the second emission layer may each be understood by referring to the descriptions provided herein of the first compound.

In one exemplary embodiment, the second host in the first emission layer and the fourth host in the second emission layer may each be understood by referring to the descriptions provided herein of the second compound.

In one exemplary embodiment, the first host and the third host may be identical to or different from each other. In one exemplary embodiment, the second host and the fourth host may be identical to or different from each other.

In one exemplary embodiment, the first host and the third host may be different from each other, and the second host and the fourth host may be different from each other.

In one exemplary embodiment, the first host in the first emission layer and the third host in the second emission layer may each be selected from the first compound, and may be identical to or different from each other.

In one exemplary embodiment, the second host in the first emission layer and the fourth host in the second emission

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layer may each be selected from the second compound, and may be identical to or different from each other.

In one exemplary embodiment, the first host in the first emission layer and the third host in the second emission layer may each include the first compound, and may be identical to or different from each other,

The second host in the first emission layer and the fourth host in the second emission layer may each include the second compound, the second compound in the second host may be the electron transport compound, and the second compound in the fourth host may be the bipolar compound.

Referring now to the equations below, the first host and the second host may each satisfy Equation 1 below, and the third host and the fourth host may each satisfy Equation 2 below:

$$E_{HOMO1}-E_{HOMO2}\leq 0.2\text{ eV} \quad \text{<Equation 1>}$$

$$E_{HOMO3}-E_{HOMO4}\geq 0.3\text{ eV} \quad \text{<Equation 2>}$$

In Equations 1 and 2,

E_{HOMO1} indicates a highest occupied molecular orbital (HOMO) of the first host,

E_{HOMO2} indicates a HOMO energy level of the second host,

E_{HOMO3} indicates a HOMO energy level of the third host, and

E_{HOMO4} indicates a HOMO energy level of the fourth host.

When the organic light-emitting device satisfies Equations 1 and 2 the emission layer has reduced resistance. In addition, the organic light-emitting device satisfying Equations 1 and 2 appropriately maintains ratios of holes and electrons based on the difference of the HOMO energy levels of the hosts in the emission layer. That is, in the organic light-emitting device satisfying Equations 1 and 2, the first emission layer, which has long lifespan, but relatively low efficiency, and the second emission layer, which has high efficiency, but relatively short lifespan, are appropriately used, thereby improving efficiency and lifespan of the organic light-emitting device at the same time.

In detail, when the first host and the second host in the first emission layer satisfy Equation 1, the first emission layer may have improved hole transport characteristics. Since the first emission layer has excellent hole transport characteristics, the hole transport from the first electrode toward the first emission layer may be smoothly performed. In further detail, the first host and the second host in the first emission layer may further satisfy Equation 1-1 below:

$$0\text{ eV}\leq E_{HOMO1}-E_{HOMO2}\leq 0.2\text{ eV} \quad \text{<Equation 1-1>}$$

In addition, when the third host and the fourth host in the second emission layer satisfy Equation 1, the second emission layer may perform less hole transport than the first emission layer. Due to low performance of the hole transport in the second emission, holes in the emission layer do not form excitons, and accordingly, the movement of the excitons toward the second electrode may be presented, thereby increasing emission efficiency of the organic light-emitting device. In further detail, the third host and the fourth host in the second emission layer may further satisfy Equation 2-1 below:

$$0.3\text{ eV}\leq E_{HOMO3}-E_{HOMO4}\leq 0.8\text{ eV} \quad \text{<Equation 2-1>}$$

The at least one host may include the first compound and the second compound.

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The at least one host may further include a compound represented by Formula 301 below:



In Formula 301,

A_{301} may be a substituted or unsubstituted $\text{C}_5\text{-C}_{60}$ carbocyclic group or a substituted or unsubstituted $\text{C}_1\text{-C}_{60}$ heterocyclic group,

$xb11$ may be 1, 2, or 3,

L_{301} may be selected from a substituted or unsubstituted $\text{C}_3\text{-C}_{10}$ cycloalkylene group, a substituted or unsubstituted $\text{C}_1\text{-C}_{10}$ heterocycloalkylene group, a substituted or unsubstituted $\text{C}_3\text{-C}_{10}$ cycloalkenylene group, a substituted or unsubstituted $\text{C}_1\text{-C}_{10}$ heterocycloalkenylene group, a substituted or unsubstituted $\text{C}_6\text{-C}_{60}$ arylene group, a substituted or unsubstituted $\text{C}_1\text{-C}_{60}$ heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group,

$xb1$ may be an integer from 0 to 5,

R_{301} may be selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted $\text{C}_1\text{-C}_{60}$ alkyl group, a substituted or unsubstituted $\text{C}_2\text{-C}_{60}$ alkenyl group, a substituted or unsubstituted $\text{C}_2\text{-C}_{60}$ alkynyl group, a substituted or unsubstituted $\text{C}_1\text{-C}_{60}$ alkoxy group, a substituted or unsubstituted $\text{C}_3\text{-C}_{10}$ cycloalkyl group, a substituted or unsubstituted $\text{C}_1\text{-C}_{10}$ heterocycloalkyl group, a substituted or unsubstituted $\text{C}_3\text{-C}_{10}$ cycloalkenyl group, a substituted or unsubstituted $\text{C}_1\text{-C}_{10}$ heterocycloalkenyl group, a substituted or unsubstituted $\text{C}_6\text{-C}_{60}$ aryl group, a substituted or unsubstituted $\text{C}_6\text{-C}_{60}$ aryloxy group, a substituted or unsubstituted $\text{C}_6\text{-C}_{60}$ arylthio group, a substituted or unsubstituted $\text{C}_1\text{-C}_{60}$ heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, —Si(Q_{301})(Q_{302})(Q_{303}), —N(Q_{301})(Q_{302}), —B(Q_{301})(Q_{302}), —C(=O)(Q_{301}), —S(=O)₂(Q_{301}), and —P(=O)(Q_{301})(Q_{302}),

$xb21$ may be an integer from 1 to 5, and

Q_{301} to Q_{303} may each independently be selected from a $\text{C}_1\text{-C}_{10}$ alkyl group, a $\text{C}_1\text{-C}_{10}$ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group, but exemplary embodiments of the invention are not limited thereto.

In one exemplary embodiment, Ar_{301} in Formula 301 may be selected from:

a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, a dibenzofuran group, and a dibenzothiophene group; and

a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, a dibenzofuran group, and a dibenzothiophene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br,

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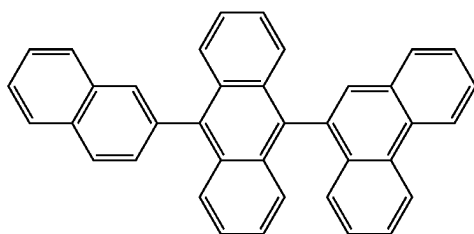
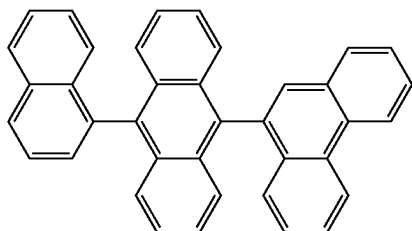
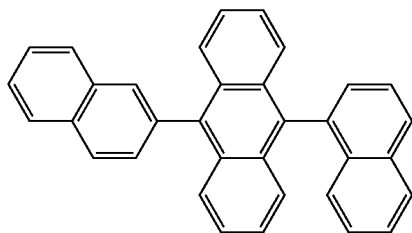
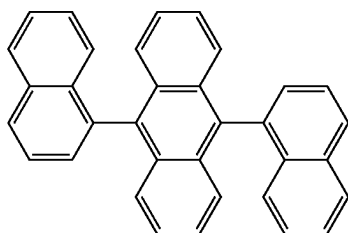
—I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, —Si(Q₃₁)(Q₃₂)(Q₃₃), —N(Q₃₁)(Q₃₂), —B(Q₃₁)(Q₃₂), —C(=O)(Q₃₁), —S(=O)₂(Q₃₁), and —P(=O)(Q₃₁)(Q₃₂), and

Q₃₁ to Q₃₃ may each independently be selected from a C₁-C₁₀ alkyl group, a C₁-C₁₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group, but exemplary embodiments of the invention are not limited thereto.

When xb11 in Formula 301 is 2 or more, two or more Ar₃₀₁(s) may be linked to each other via a single bond.

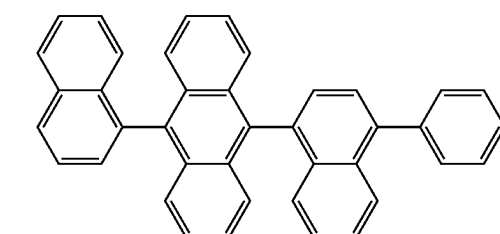
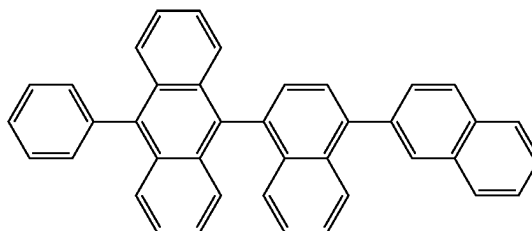
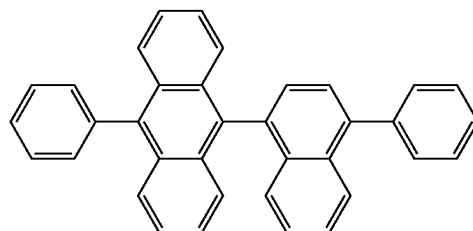
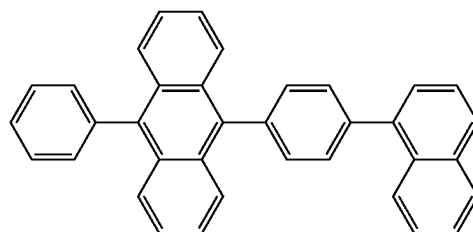
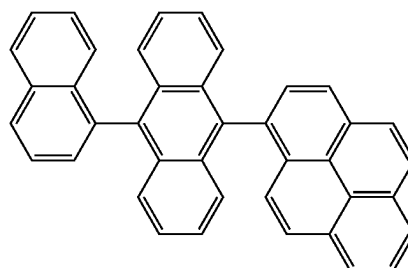
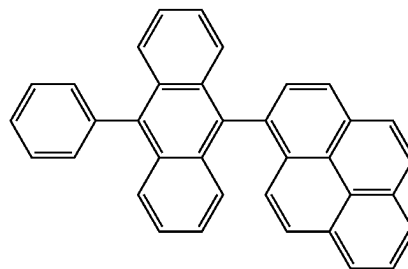
In one or more exemplary embodiments, the host may include an alkaline earth metal complex. For example, the host may include a Be complex (for example, Compound H55 below), a Mg complex, and a Zn complex.

The at least one host may further include at least one selected from 9,10-di(2-naphthyl)anthracene (ADN), 2-methyl-9,10-bis(naphthalen-2-yl)anthracene (MADN), 9,10-di(2-naphthyl)-2-t-butyl-anthracene (TBADN), 4,4'-bis(N-carbazolyl)-1,1'-biphenyl (CBP), 1,3-di-9-carbazolyl-benzene (mCP), 1,3,5-tri(carbazol-9-yl)benzene (TCP), and Compounds H1 to H55 below, but exemplary embodiments of the invention are not limited thereto:



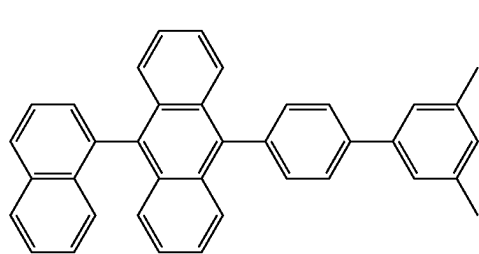
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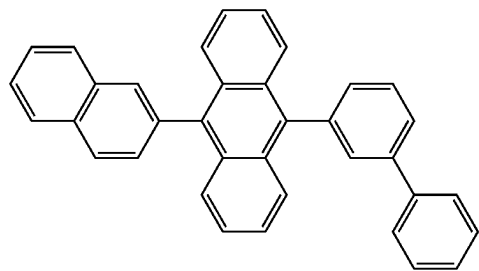
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H11

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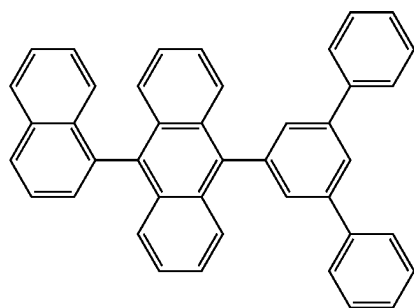
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H12

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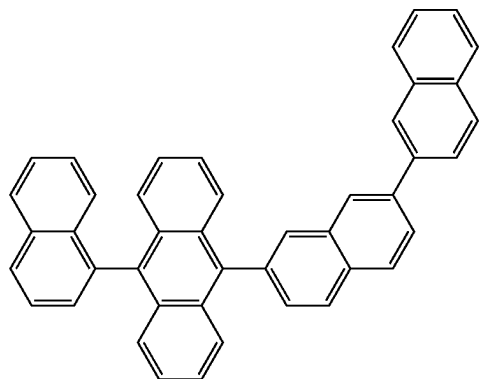


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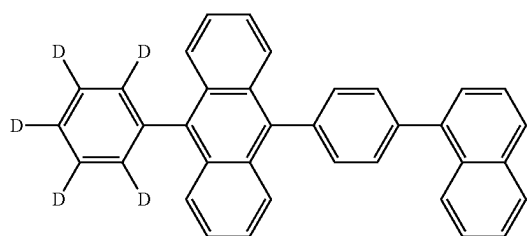


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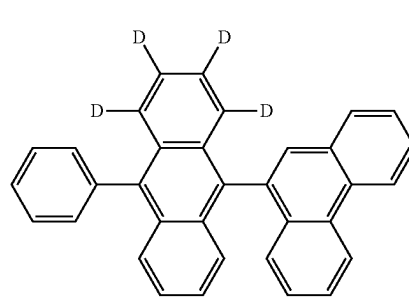
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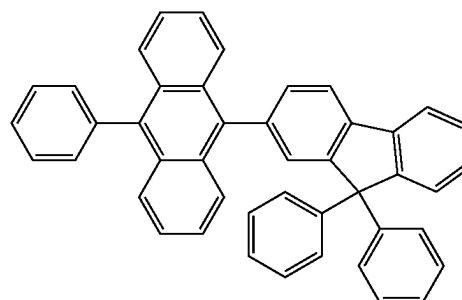
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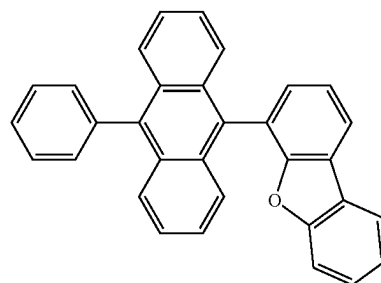
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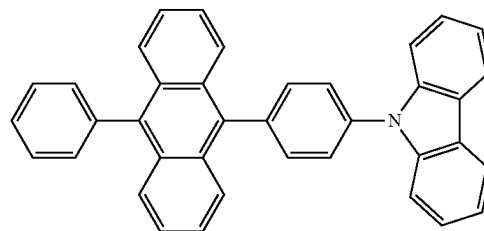
H16



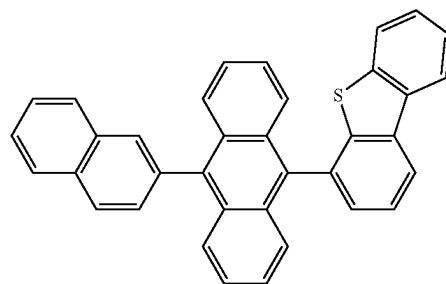
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H18



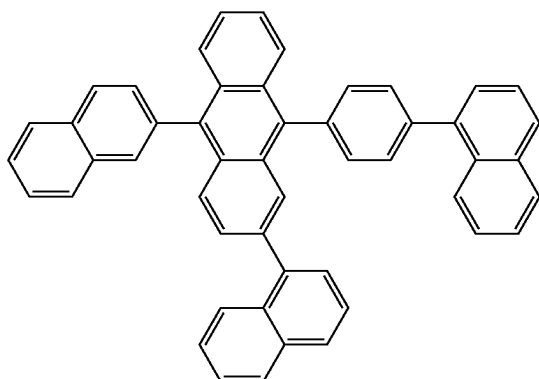
H19



H20

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H21

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H22

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H23

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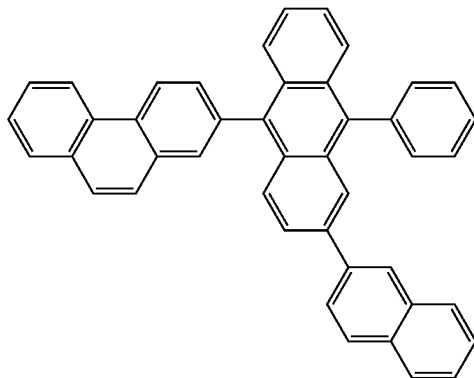
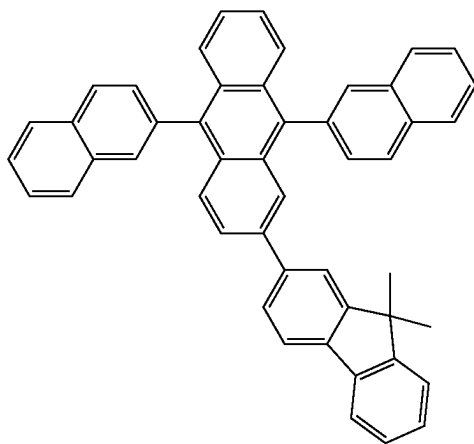
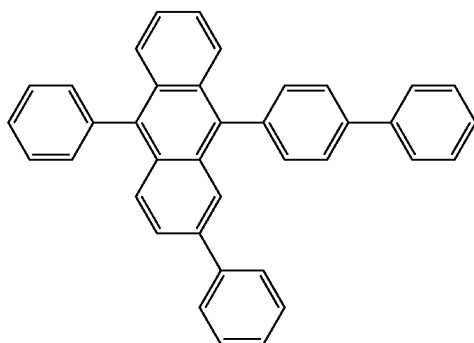
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H24

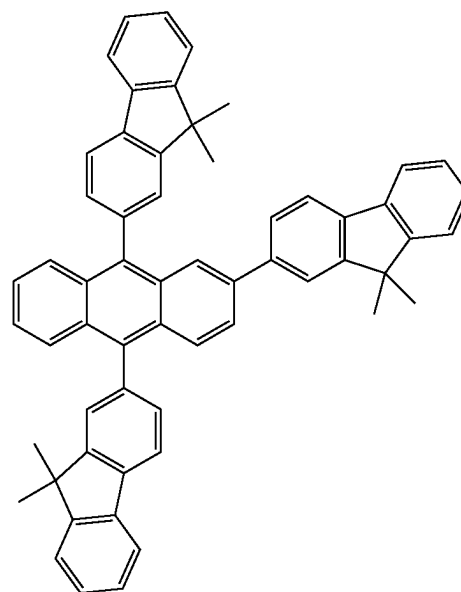
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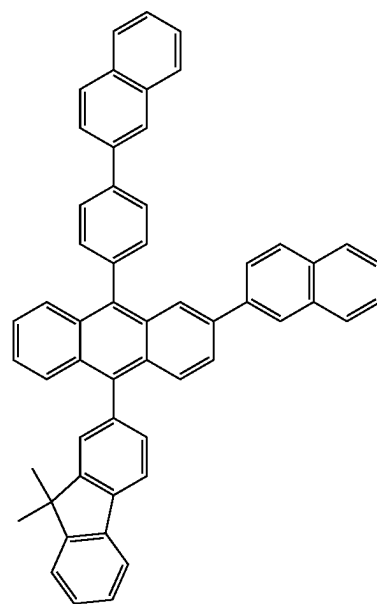
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**66**

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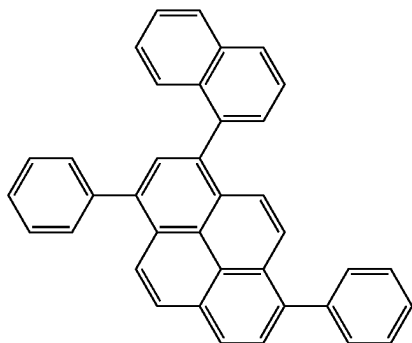
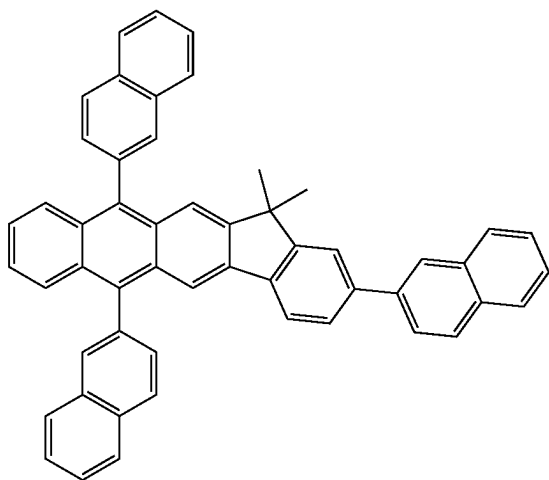
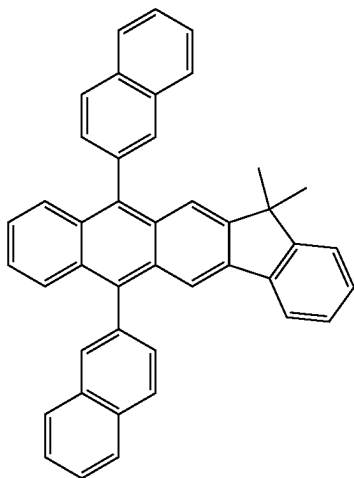
H25



H26

67

-continued

**68**

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H27

H30

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H28

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H29

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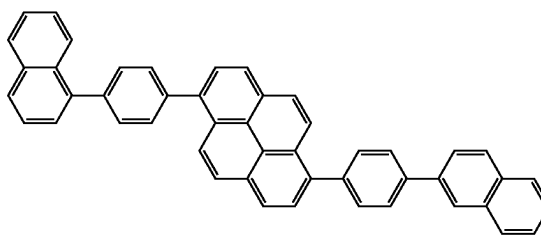
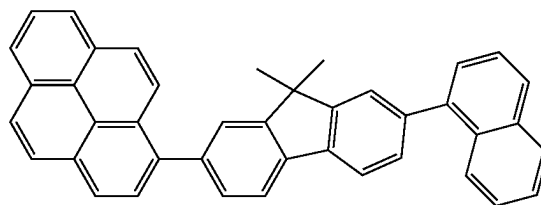
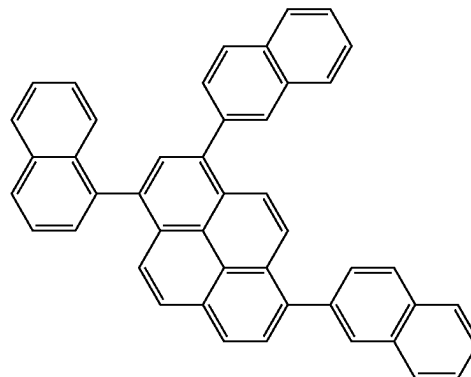
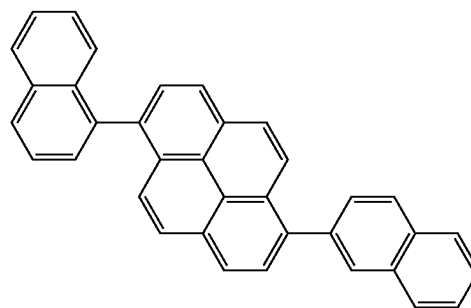
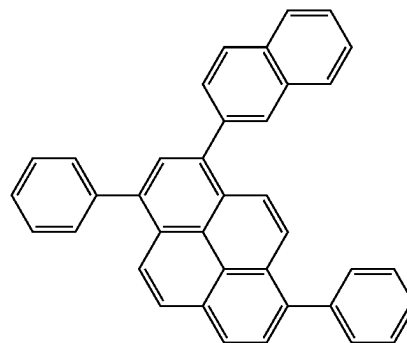
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H31

H32

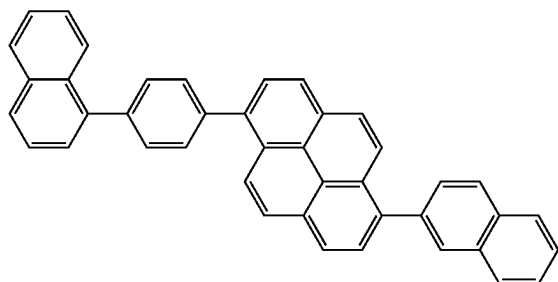
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H34



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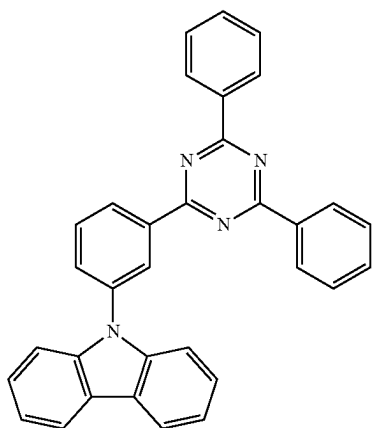


H35

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H36 15

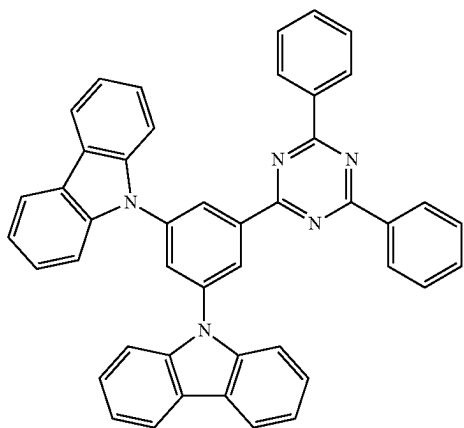


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H37 35

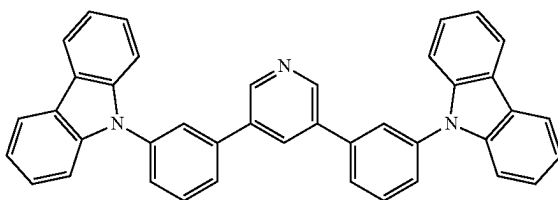


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H38

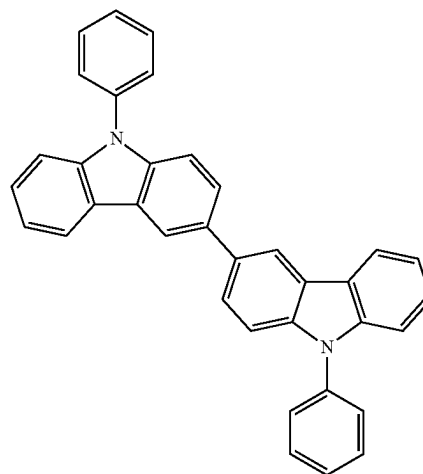


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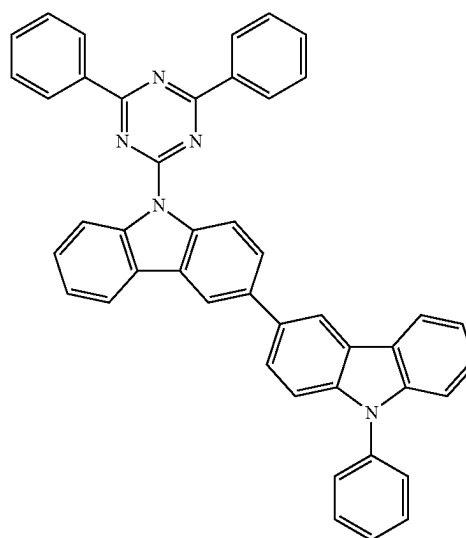
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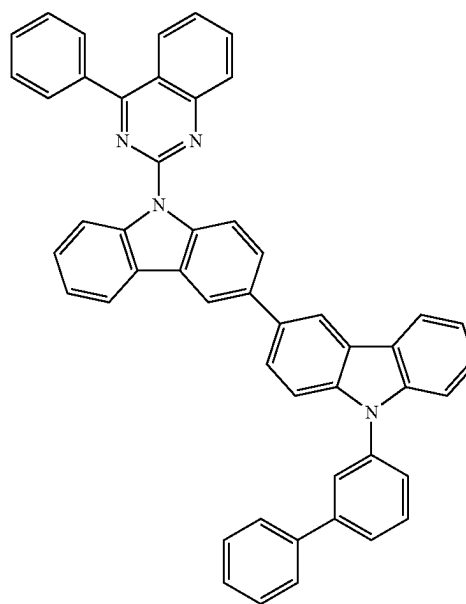
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H39



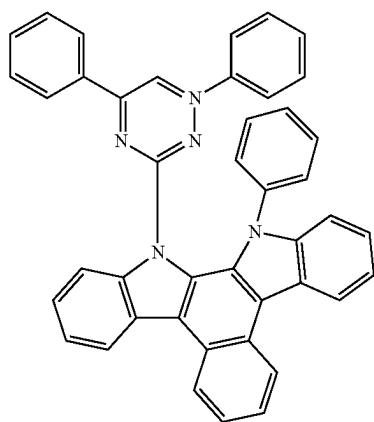
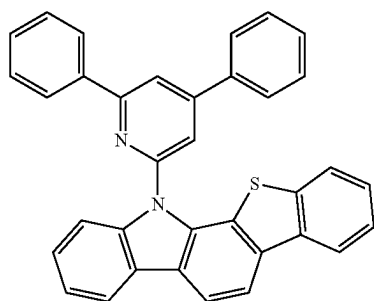
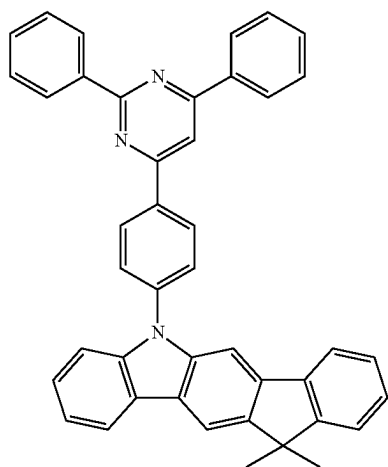
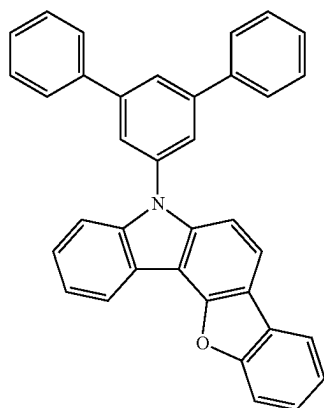
H40



H41

71

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**72**

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H42

H46

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H43

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H47

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H48

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H44

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H49

H45

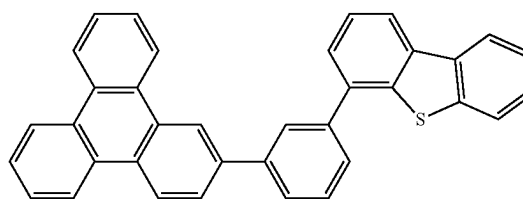
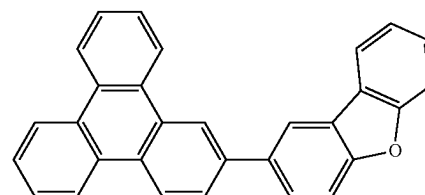
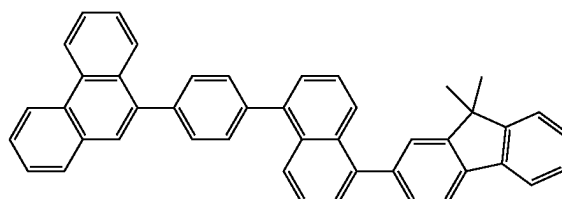
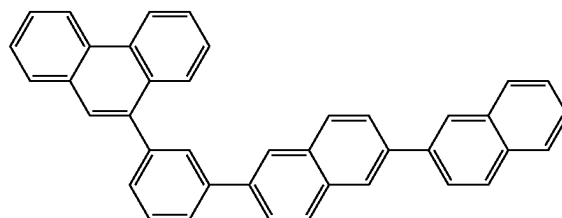
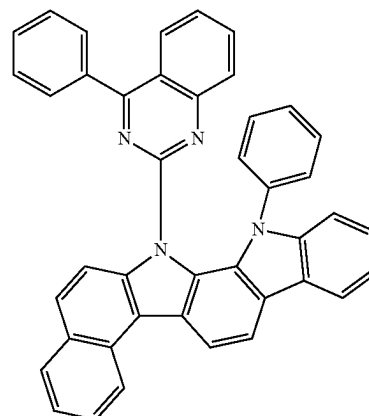
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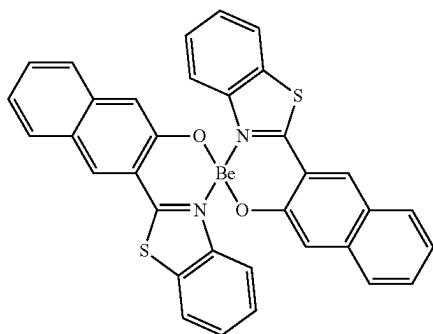
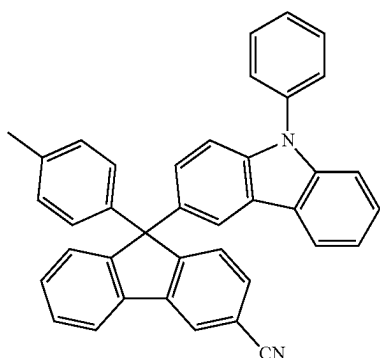
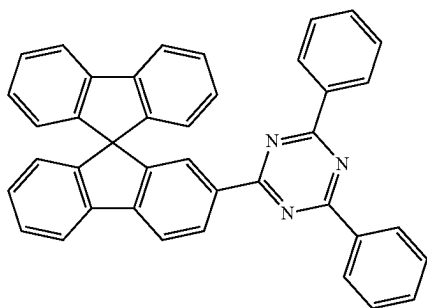
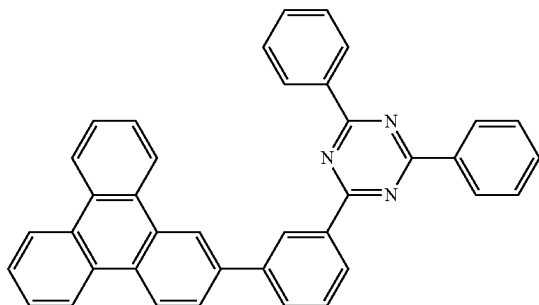
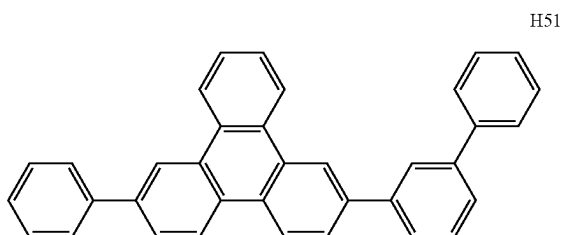
H50

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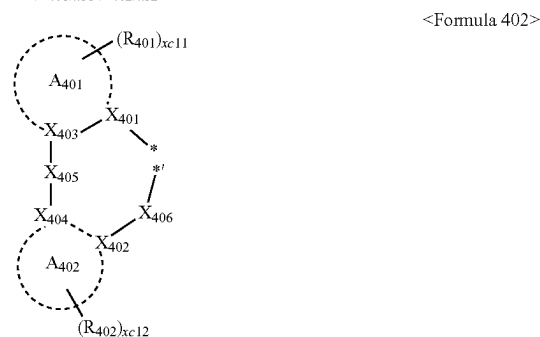
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Phosphorescent Dopant Included in Emission Layer in Organic Layer 150

The phosphorescent dopant may include an organometallic complex represented by Formula 401 below:



In Formulae 401 and 402,

M may be selected from iridium (Ir), platinum (Pt), palladium (Pd), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), rhodium (Rh), and thulium (Tm),

L_{401} may be a ligand represented by Formula 402, and $xc1$ may be 1, 2, or 3, wherein when $xc1$ is two or more, two or more L_{401} (s) may be identical to or different from each other,

L_{402} may be an organic ligand, and $xc2$ may be an integer from 0 to 4, wherein when $xc2$ may be two or more, two or more L_{402} (s) may be identical to or different from each other,

X_{401} to X_{404} may each independently be nitrogen or carbon,

X_{401} and X_{403} may be linked via a single bond or a double bond, and X_{402} and X_{404} may be linked via a single bond or a double bond,

A_{401} and A_{402} may each independently be a C_5 - C_{60} carbocyclic group or a C_1 - C_{60} heterocyclic group,

X_{405} may be a single bond, $*-O-*$, $*-S-*$, $*-C(=O)-*$, $*-N(Q_{411})-*$, $*-C(Q_{411})(Q_{412})-*$, $*-C(Q_{411})=C(Q_{412})-*$, $*-C(Q_{411})=*$, or $*-C=*$, wherein Q_{411} and Q_{412} may each independently be hydrogen, deuterium, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, or a naphthyl group,

X_{406} may be a single bond, O, or S,

R_{401} and R_{402} may each independently be selected from hydrogen, deuterium, $-F$, $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C_1 - C_{20} alkyl group, a substituted or unsubstituted C_1 - C_{20} alkoxy group, a substituted or unsubstituted C_3 - C_{10} cycloalkyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3 - C_{10} cycloalkenyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6 - C_{60} aryl group, a substituted or unsubstituted C_6 - C_{60} aryloxy group, a substituted or unsubstituted C_6 - C_{60} arylthio group, a substituted or unsubstituted C_1 - C_{60} heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted monovalent non-aromatic condensed het-

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eropolycyclic group, $-\text{Si}(\text{Q}_{401})(\text{Q}_{402})(\text{Q}_{403})$, $-\text{N}(\text{Q}_{401})(\text{Q}_{402})$, $-\text{B}(\text{Q}_{401})(\text{Q}_{402})$, $-\text{C}(=\text{O})(\text{Q}_{401})$, $-\text{S}(=\text{O})_2(\text{Q}_{401})$, and $-\text{P}(=\text{O})(\text{Q}_{401})(\text{Q}_{402})$, and Q_{401} to Q_{403} may each independently be selected from a C_1 - C_{10} alkyl group, a C_1 - C_{10} alkoxy group, a C_6 - C_{20} aryl group, and a C_1 - C_{20} heteroaryl group, xc11 and xc12 may each independently be an integer from 0 to 10, * and *' in Formula 402 each indicate a binding site to M in Formula 401.

In one exemplary embodiment, A_{401} and A_{402} in Formula 402 may each independently be selected from a benzene group, a naphthalene group, a fluorene group, a spiro-bifluorene group, an indene group, a pyrrole group, a thiophene group, a furan group, an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine group, a pyrazine group, a pyrimidine group, a pyridazine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a quinoxaline group, a quinazoline group, a carbazole group, a benzimidazole group, a benzofuran group, a benzothiophene group, an isobenzothiophene group, a benzoxazole group, an isobenzoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a dibenzofuran group, and a dibenzothiophene group.

In one or more exemplary embodiments, in Formula 402, i) X_{401} may be nitrogen and X_{402} may be carbon, or ii) X_{401} and X_{402} may each be nitrogen at the same time.

In one or more exemplary embodiments, R_{401} and R_{402} in Formula 402 may each independently be selected from:

hydrogen, deuterium, $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, and a C_1 - C_{20} alkoxy group;

a C_1 - C_{20} alkyl group and a C_1 - C_{20} alkoxy group, each substituted with at least one selected from deuterium, $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a phenyl group, a naphthyl group, a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, and a norbornenyl group; a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group;

a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group, each substituted with at least one selected from deuterium, $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a cyclopentyl group, a cyclohexyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a pyridinyl group, a pyrazinyl

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group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a carbazolyl group, a dibenzofuranyl group, and a dibenzothiophenyl group; and

$-\text{Si}(\text{Q}_{401})(\text{Q}_{402})(\text{Q}_{403})$, $-\text{N}(\text{Q}_{401})(\text{Q}_{402})$, $-\text{B}(\text{Q}_{401})(\text{Q}_{402})$, $-\text{C}(=\text{O})(\text{Q}_{401})$, $-\text{S}(=\text{O})_2(\text{Q}_{401})$, and $-\text{P}(=\text{O})(\text{Q}_{401})(\text{Q}_{402})$, and

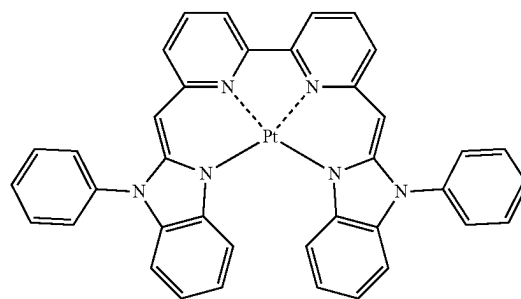
Q_{401} to Q_{403} may each independently be selected from a C_1 - C_{10} alkyl group, a C_1 - C_{10} alkoxy group, a phenyl group, a biphenyl group, and a naphthyl group, but exemplary embodiments of the invention are not limited thereto.

In one or more exemplary embodiments, when xc1 in Formula 401 is two or more, two $\text{A}_{401}(\text{s})$ in two or more $\text{L}_{401}(\text{s})$ may optionally be linked to each other via X_{407} , which is a linking group, two $\text{A}_{402}(\text{s})$ may optionally be linked to each other via X_{408} , which is a linking group (see Compounds PD1 to PD4 and PD7 below). X_{407} and X_{408} may each independently be a single bond, $^*\text{—O—}^*$, $^*\text{—S—}^*$, $^*\text{—C}(=\text{O})\text{—}^*$, $^*\text{—N}(\text{Q}_{413})\text{—}^*$, $^*\text{—C}(\text{Q}_{413})(\text{Q}_{414})\text{—}^*$ or $^*\text{—C}(\text{Q}_{413})=\text{C}(\text{Q}_{414})\text{—}^*$ (where Q_{413} and Q_{414} may each independently be hydrogen, deuterium, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, or a naphthyl group), but exemplary embodiments of the invention are not limited thereto.

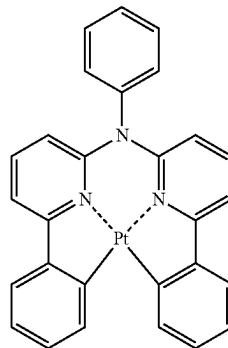
L_{402} in Formula 401 may be a monovalent, divalent, or trivalent organic ligand. For example, L_{402} may be selected from halogen, diketone (for example, acetylacetonate), carboxylic acid (for example, picolinate), $-\text{C}(=\text{O})$, isonitrile, $-\text{CN}$, and phosphorus (for example, phosphine or phosphite), but exemplary embodiments of the invention are not limited thereto.

In one or more exemplary embodiments, the phosphorescent dopant may be selected from, for example, Compounds PD1 to PD25 below, but exemplary embodiments of the invention are not limited thereto:

PD1

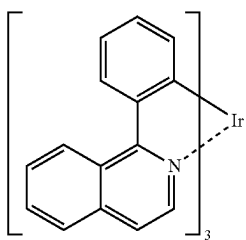
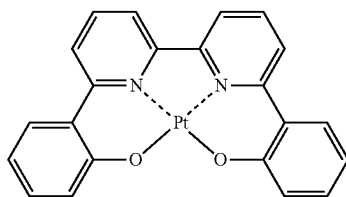
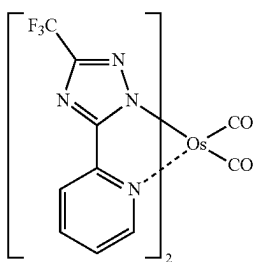
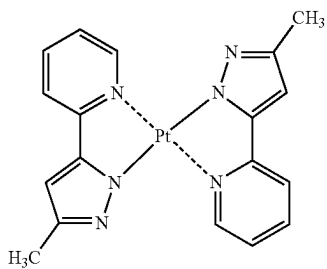
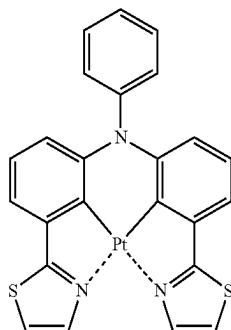
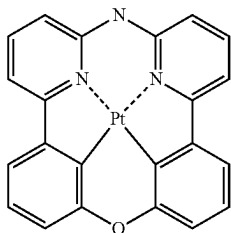


PD2



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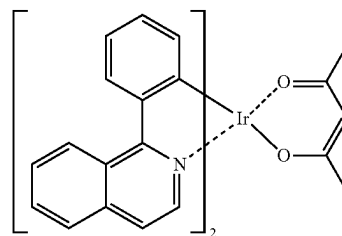
**78**

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PD3

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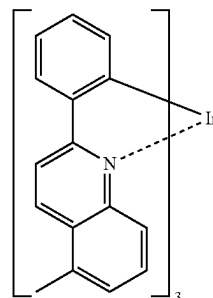


PD9

PD4

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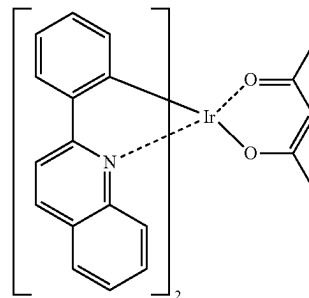
PD10

PD5

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PD11

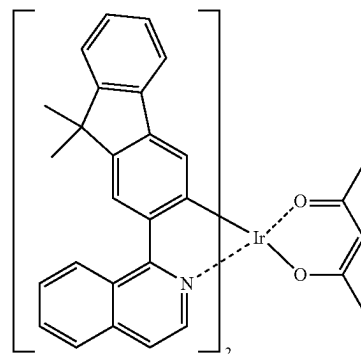
PD6

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PD7

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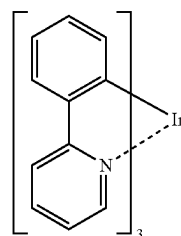
PD12

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PD8

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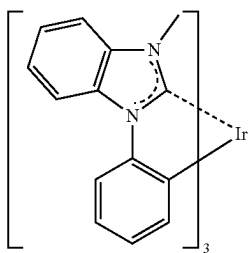
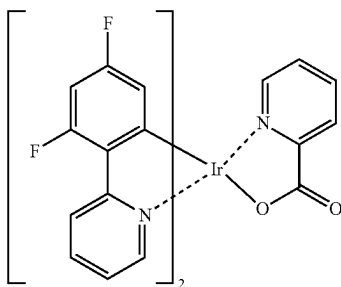
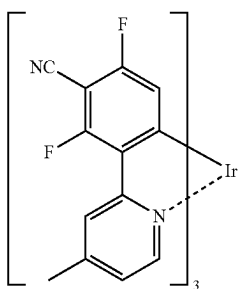
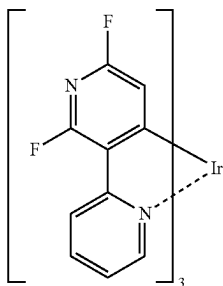
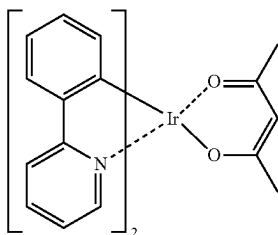
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PD13

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**80**

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PD14

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PD15

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PD16

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PD17

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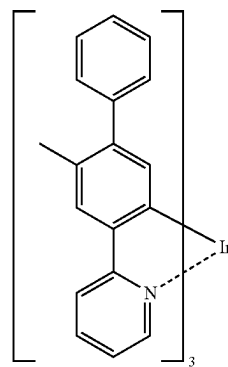
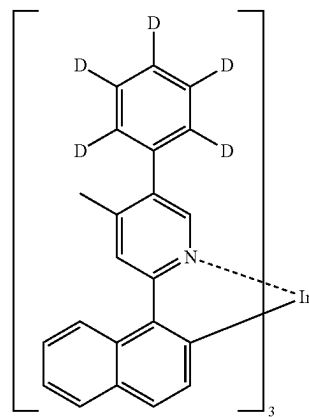
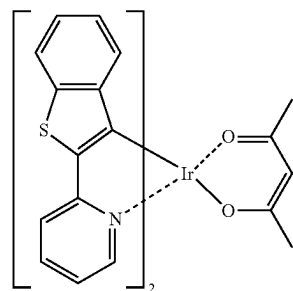
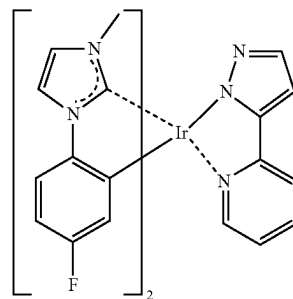
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PD18

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PD19

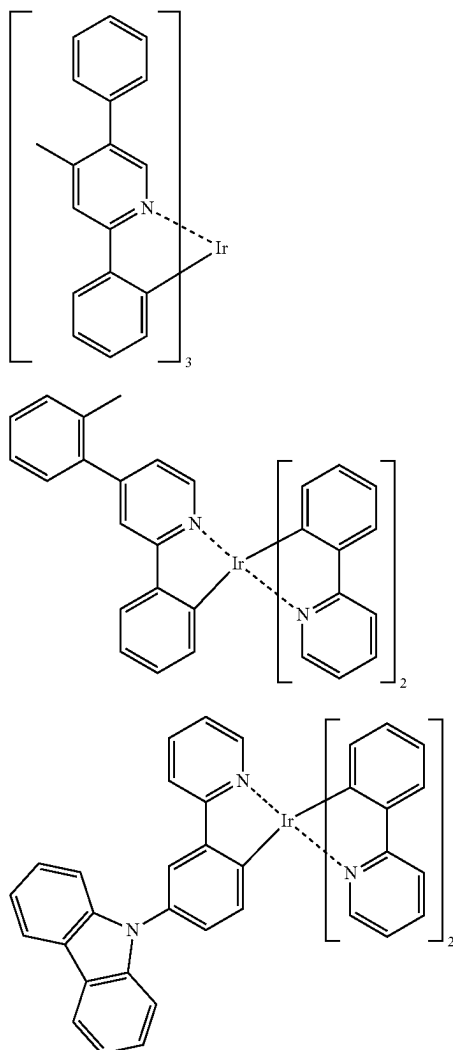
PD20

PD21

PD22

81

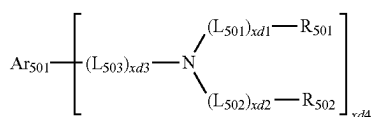
-continued



Fluorescent Dopant in Emission Layer

The fluorescent dopant may include an arylamine compound or a styrylamine compound.

The fluorescent dopant may include a compound represented by Formula 501 below:



<Formula 501>

In Formula 501,

Ar₅₀₁ may be a substituted or unsubstituted C₅-C₆₀ carbocyclic group or a substituted or unsubstituted C₁-C₆₀ heterocyclic group,

L₅₀₁ to L₅₀₃ may each independently be selected from a substituted or unsubstituted C₃-C₁₀ cycloalkylene group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkylene group, a substituted or unsubstituted C₃-C₁₀ cycloalkenylene group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenylene group, a substi-

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tuted or unsubstituted C₆-C₆₀ arylene group, a substituted or unsubstituted C₁-C₆₀ heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group,

xd1 to xd3 may each independently be an integer from 0 to 3,

R₅₀₁ and R₅₀₂ may each independently be selected from a substituted or unsubstituted C₃-C₁₀ cycloalkyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkyl group, a substituted or unsubstituted C₃-C₁₀ cycloalkenyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenyl group, a substituted or unsubstituted C₆-C₆₀ aryl group, a substituted or unsubstituted C₆-C₆₀ aryloxy group, a substituted or unsubstituted C₆-C₆₀ arylthio group, a substituted or unsubstituted C₁-C₆₀ heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, and

xd4 may be an integer from 1 to 6.

In one exemplary embodiment, Ar₅₀₁ in Formula 501 may be selected from:

a naphthalene group, a heptalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenanthracene group, and an indenophenanthrene group; and a naphthalene group, a heptalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenanthracene group, and an indenophenanthrene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₁-C₆₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group;

In one or more exemplary embodiments, L₅₀₁ to L₅₀₃ in Formula 501 may each independently be selected from:

a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthenylenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylenylene group, a thiophenylenylene group, a furanylenylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylenylene group, a benzothiophenylenylene group, a dibenzofuranylenylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group; and

a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylene group, a

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fluoranthenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylenylene group, a pentacenylene group, a thiophenylenylene group, a furanylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylene group, a benzothiophenylenylene group, a dibenzofuranylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, and a pyridinylenylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzo-
fluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group.

In one or more exemplary embodiments, R₅₀₁ and R₅₀₂ in Formula 501 may each independently be selected from:

a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzo-
fluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group; and

a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzo-
fluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzo-
fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl

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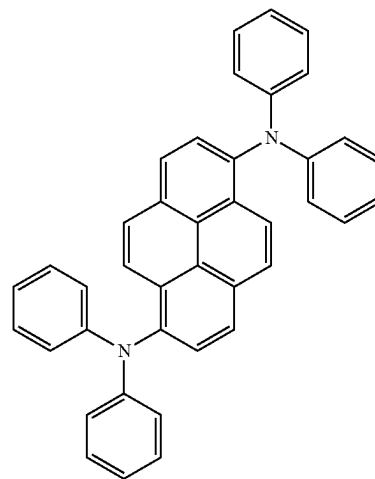
group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, and —Si(Q₃₁)(Q₃₂)(Q₃₃), and

Q₃₁ to Q₃₃ may each dependently be selected from a C₁-C₁₀ alkyl group, a C₁-C₁₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

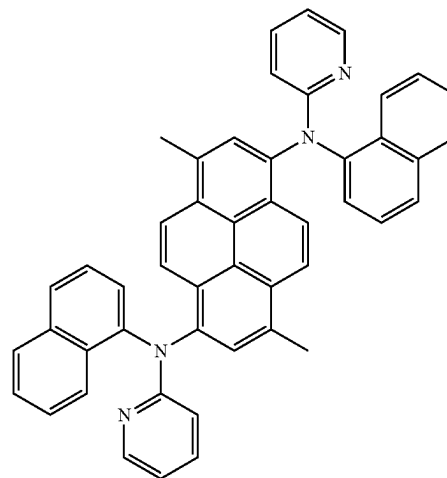
In one or more exemplary embodiments, xd4 in Formula 501 may be 2, but exemplary embodiments of the invention are not limited thereto.

For example, the fluorescent dopant may be selected from Compounds FD1 to FD22 below:

FD1

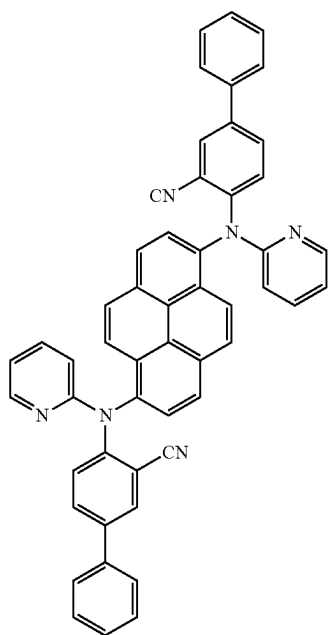
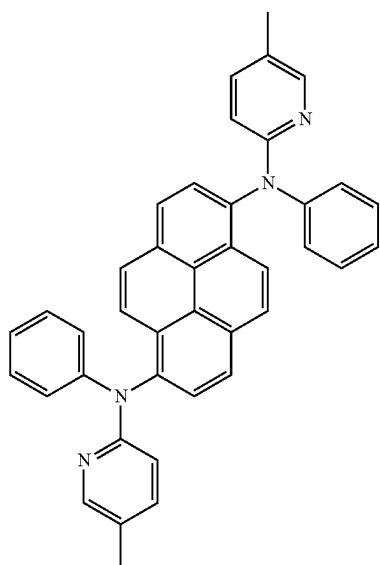


FD2



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-continued



86

-continued

FD3

FD5

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FD4

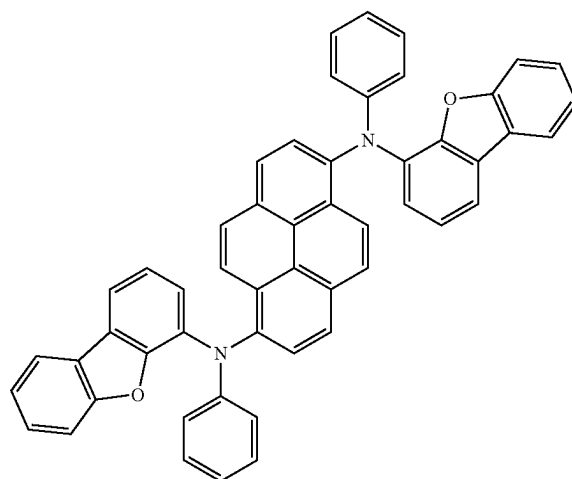
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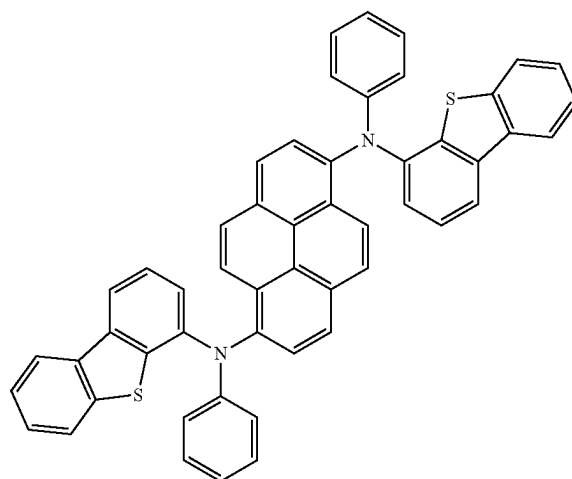
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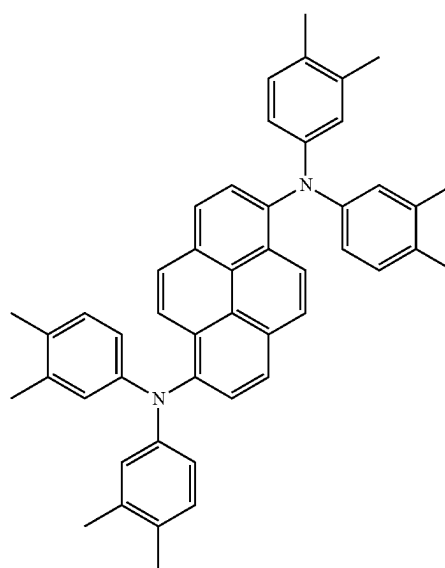
65



FD6



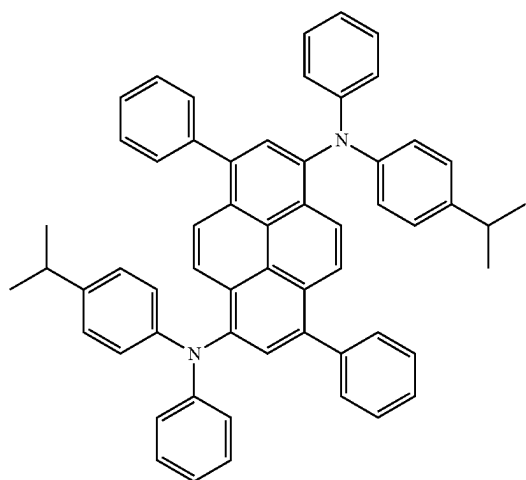
FD7



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-continued

FD8

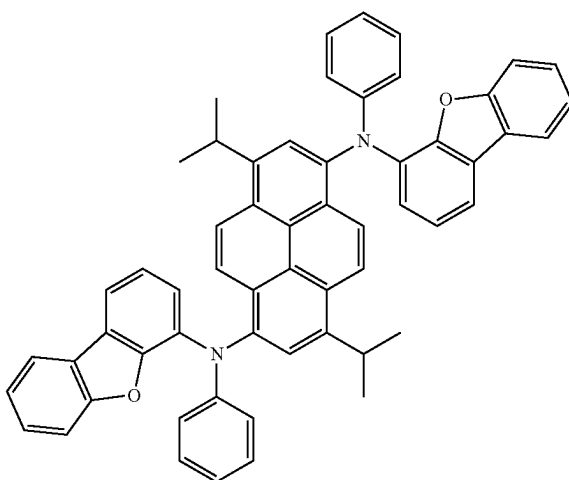


5

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FD9



20

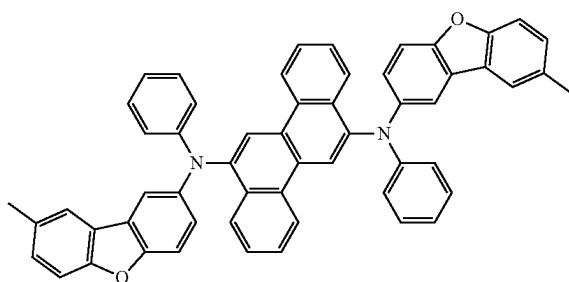
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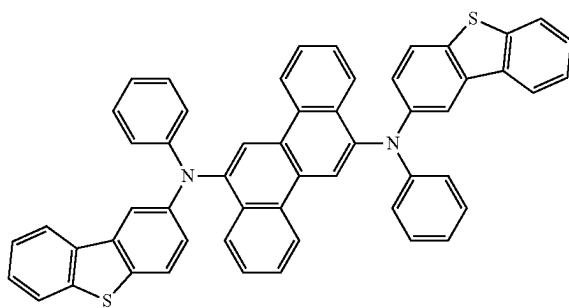
FD10



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FD11



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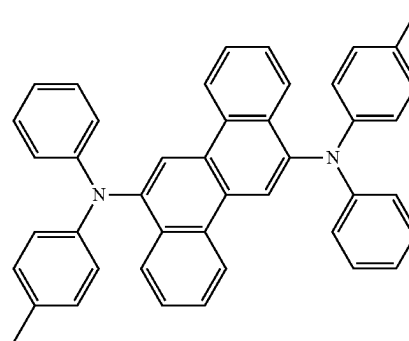
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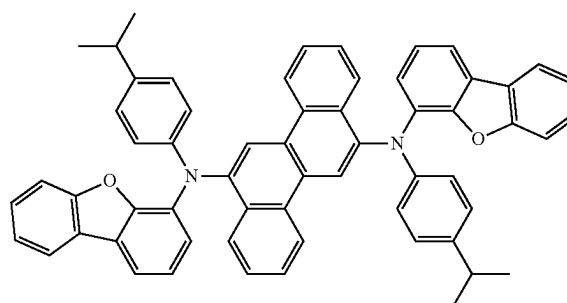
88

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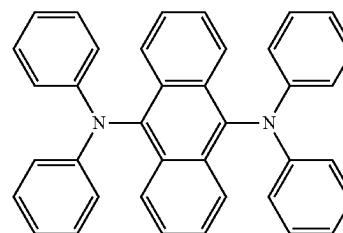
FD12



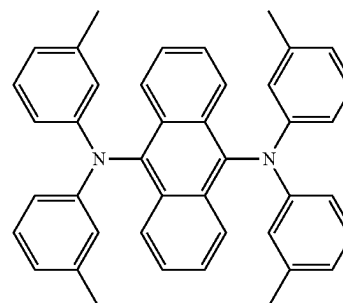
FD13



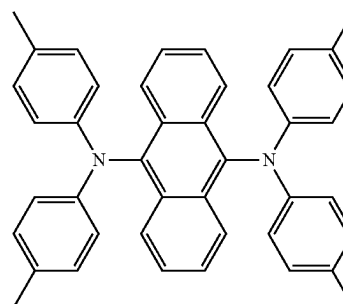
FD14



FD15

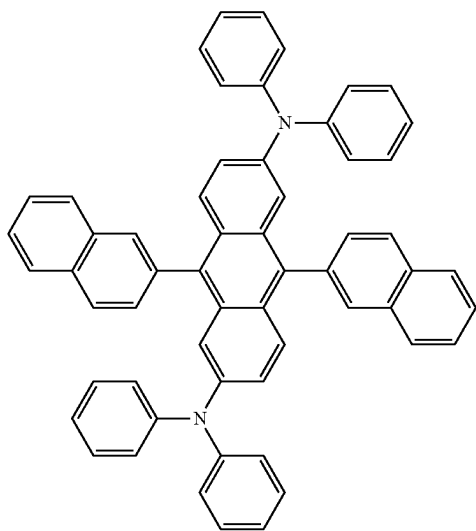
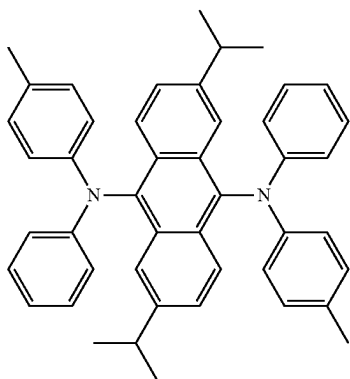
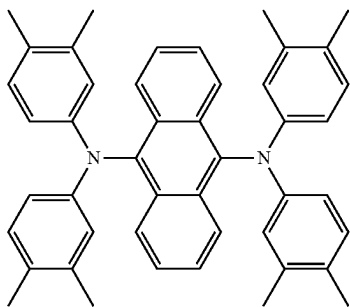


FD16



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**90**

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FD17

FD20

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FD18

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FD21

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FD19

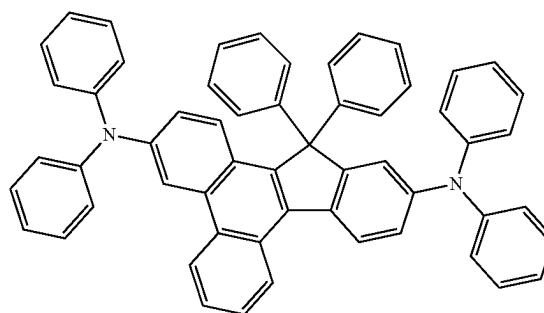
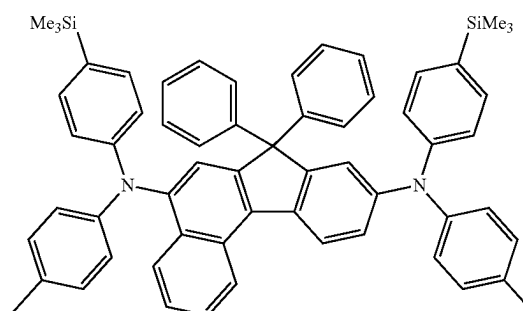
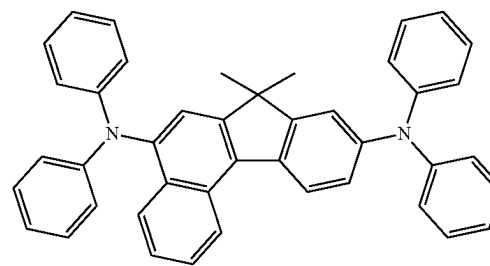
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FD22

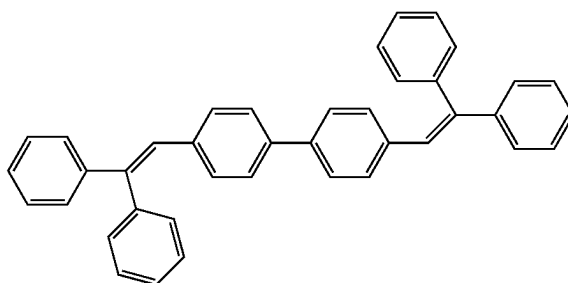
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In one or more exemplary embodiments, the fluorescent dopant may be selected from the following compounds, but exemplary embodiments of the invention are not limited thereto.

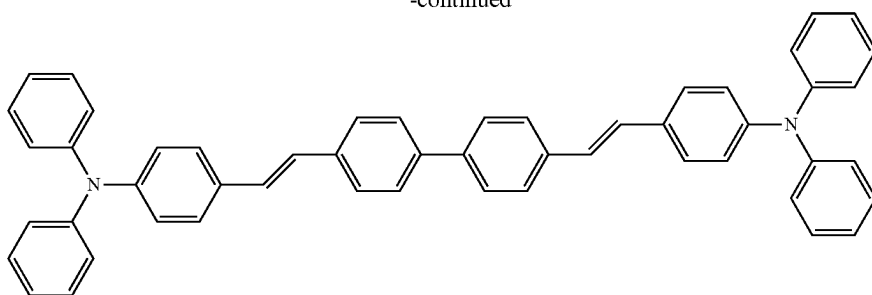


DPVBi

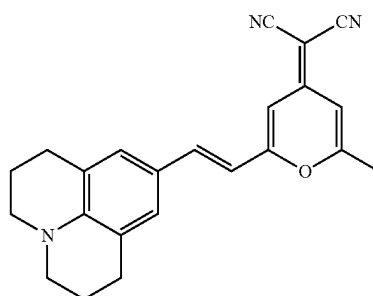
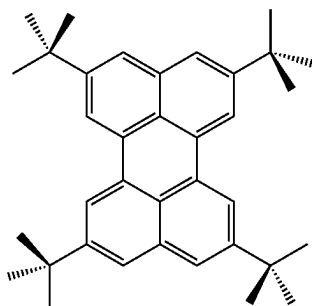
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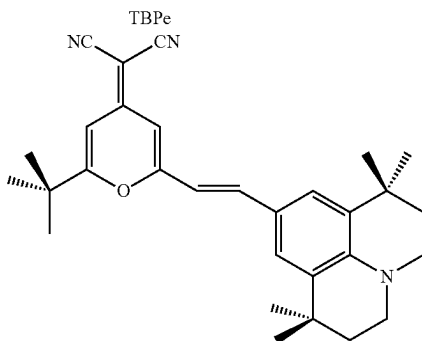
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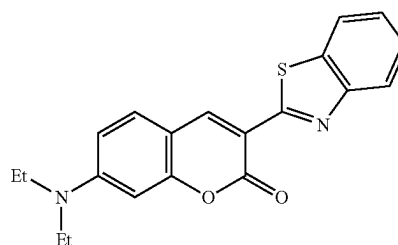
DPAVB



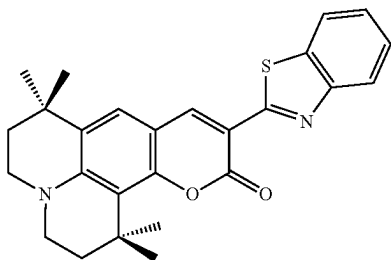
DCM



DCJTb



Coumarin 6



C545T

Electron Transport Region in Organic Layer 150

The electron transport region may have i) a single-layered structure including a single layer including a single material, ii) a single-layered structure including a single layer including a plurality of different materials, or iii) a multi-layered structure having a plurality of layers including a plurality of different materials.

The electron transport region may include at least one selected from a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, and an electron injection layer, but exemplary embodiments of the invention are not limited thereto.

For example, the electron transport region may have an electron transport layer/electron injection layer structure, a

hole blocking layer/electron transport layer/electron injection layer structure, an electron control layer/electron transport layer/electron injection layer structure, or a buffer layer/electron transport layer/electron injection layer structure, wherein for each structure, constituting layers are sequentially stacked from the emission layer. However, exemplary embodiments of the structure of the electron transport region are not limited thereto.

The electron transport region (for example, a buffer layer, a hole blocking layer, an electron control layer, or an electron transport layer in the electron transport region) may include a metal-free compound containing at least one π electron-depleted nitrogen-containing ring.

The term “ π electron-depleted nitrogen-containing ring” as used herein refers to a C_1 - C_{60} heterocyclic group having at least one $*-N=*$ moiety as a ring-forming moiety.

For example, the “ π electron-depleted nitrogen-containing ring” may be i) a 5-membered to 7-membered heteromonocyclic group having at least one $*-N=*$ moiety, ii) a heteropolycyclic group in which two or more 5-membered to 7-membered heteromonocyclic groups each having at least one $*-N=*$ moiety are condensed with each other, or iii) a heteropolycyclic group in which at least one of 5-membered to 7-membered heteromonocyclic groups, each having at least one $*-N=*$ moiety, is condensed with at least one C_5 - C_{60} carbocyclic group.

Examples of the π electron-depleted nitrogen-containing ring include an imidazole, a pyrazole, a thiazole, an isothiazole, an oxazole, an isoxazole, a pyridine, a pyrazine, a pyrimidine, a pyridazine, an indazole, a purine, a quinoline, an isoquinoline, a benzoquinoline, a phthalazine, a naphthyridine, a quinoxaline, a quinazoline, a cinnoline, a phenanthridine, an acridine, a phenanthroline, a phenazine, a benzimidazole, an isobenzothiazole, a benzoxazole, an isobenzoxazole, a triazole, a tetrazole, an oxadiazole, a triazine, a thiadiazole, an imidazopyridine, an imidazopyrimidine, and an azacarbazole, but are not limited thereto.

For example, the electron transport region may include a compound represented by Formula 601 below:



In Formula 601,

Ar_{601} may be a substituted or unsubstituted C_5 - C_{60} carbocyclic group or a substituted or unsubstituted C_1 - C_{60} heterocyclic group,

$xe11$ may be 1, 2, or 3,

L_{601} may be selected from a substituted or unsubstituted C_3 - C_{10} cycloalkylene group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkylene group, a substituted or unsubstituted C_3 - C_{10} cycloalkenylene group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkenylene group, a substituted or unsubstituted C_6 - C_{60} arylene group, a substituted or unsubstituted C_1 - C_{60} heteroarylene group, a substituted or unsubstituted divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group,

$xe1$ may be an integer from 0 to 5,

R_{601} may be selected from a substituted or unsubstituted C_3 - C_{10} cycloalkyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkyl group, a substituted or unsubstituted C_3 - C_{10} cycloalkenyl group, a substituted or unsubstituted C_1 - C_{10} heterocycloalkenyl group, a substituted or unsubstituted C_6 - C_{60} aryl group, a substituted or unsubstituted C_6 - C_{60} aryloxy group, a substituted or unsubstituted C_6 - C_{60} arylthio group, a substituted or unsubstituted C_1 - C_{60} heteroaryl group, a substituted or unsubstituted monovalent non-aromatic condensed polycyclic group, a substituted or unsubstituted monovalent non-aromatic condensed heteropolycyclic group, $-Si(Q_{601})(Q_{602})(Q_{603})$, $-C(=O)(Q_{601})$, $-S(=O)_2(Q_{601})$, and $-P(=O)(Q_{601})(Q_{602})$, Q_{601} to Q_{603} may each independently be a C_1 - C_{10} alkyl group, a C_1 - C_{10} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, or a naphthyl group, and

$xe21$ may be an integer from 1 to 5.

In one exemplary embodiment, at least one selected from $Ar_{601}(s)$ in the number of $xe11$ and $R_{601}(s)$ in the number of $xe21$ may include the π electron-depleted nitrogen-containing ring described above.

In one exemplary embodiment, ring Ar_{601} in Formula 601 may be selected from:

a benzene group, a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, a dibenzofuran group, a dibenzothienophene group, a carbazole group, an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine group, a pyrazine group, a pyrimidine group, a pyridazine group, an indazole group, a purine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a phthalazine group, a naphthyridine group, a quinoxaline group, a quinazoline group, a cinnoline group, a phenanthridine group, an acridine group, a phenanthroline group, a phenazine group, a benzimidazole group, an isobenzothiazole group, a benzoxazole group, an isobenzoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a thiadiazole group, an imidazopyridine group, an imidazopyrimidine group, and an azacarbazole group; and

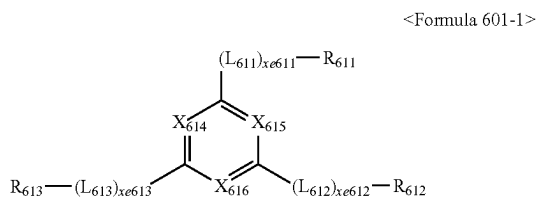
a benzene group, a naphthalene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentaphene group, an indenoanthracene group, a dibenzofuran group, a dibenzothienophene group, a carbazole group, an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine group, a pyrazine group, a pyrimidine group, a pyridazine group, an indazole group, a purine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a phthalazine group, a naphthyridine group, a quinoxaline group, a quinazoline group, a cinnoline group, a phenanthridine group, an acridine group, a phenanthroline group, a phenazine group, a benzimidazole group, an isobenzothiazole group, a benzoxazole group, an isobenzoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a thiadiazole group, an imidazopyridine group, an imidazopyrimidine group, and an azacarbazole group, each substituted with at least one selected from deuterium, $-F$, $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, $-Si(Q_{31})(Q_{32})(Q_{33})$, $-S(=O)_2(Q_{31})$, and $-P(=O)(Q_{31})(Q_{32})$, and Q_{31} to Q_{33} may each independently be selected from a C_1 - C_{10} alkyl group, a C_1 - C_{10} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

When $xe11$ in Formula 601 is 2 or more, two or more $Ar_{601}(s)$ may be linked to each other via a single bond.

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In one or more exemplary embodiments, Ar₆₀₁ in Formula 601 may be an anthracene group.

In one or more exemplary embodiments, the compound represented by Formula 601 may be represented by Formula 601-1 below:



In Formula 601-1,

X₆₁₄ may be N or C(R₆₁₄), X₆₁₅ may be N or C(R₆₁₅), X₆₁₆ may be N or C(R₆₁₆), and at least one selected from X₆₁₄ to X₆₁₆ may be N,

L₆₁₁ to L₆₁₃ may each be understood by referring to the description provided in connection with L₆₀₁, xe611 to may each be understood by referring to the description provided in connection with xe1,

R₆₁₁ to R₆₁₃ may each be understood by referring to the description provided in connection with R₆₀₁, and

R₆₁₄ to R₆₁₆ may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₁-C₆₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

In one exemplary embodiment, L₆₀₁ and L₆₁₁ to L₆₁₃ in Formulae 601 and 601-1 may each independently be selected from:

a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthrenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylene group, a pentacenylene group, a thiophenylenylene group, a furanylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylene group, a benzothiophenylenylene group, a dibenzofuranylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group; and

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a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthrenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylene group, a pentacenylene group, a thiophenylenylene group, a furanylene group, a carbazolylenylene group, an indolylenylene group, an isoindolylenylene group, a benzofuranylene group, a benzothiophenylenylene group, a dibenzofuranylene group, a dibenzothiophenylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylenylene group, a pyridinylenylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, a triazinylenylene group, a quinolinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylenylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylenylene group, a cinnolinylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylenylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylenylene group, an imidazopyrimidinylenylene group, and an azacarbazolylenylene group, but exemplary embodiments of the invention are not limited thereto.

In one or more exemplary embodiments, xe1 and xe611 to xe613 in Formulae 601 and 601-1 may each independently be 0, 1, or 2.

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In one or more exemplary embodiments, R_{601} and R_{611} to R_{613} in Formulae 601 and 601-1 may each independently be selected from:

a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthrenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group;

a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthrenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, a C_1 - C_{20} alkoxy group, a phenyl

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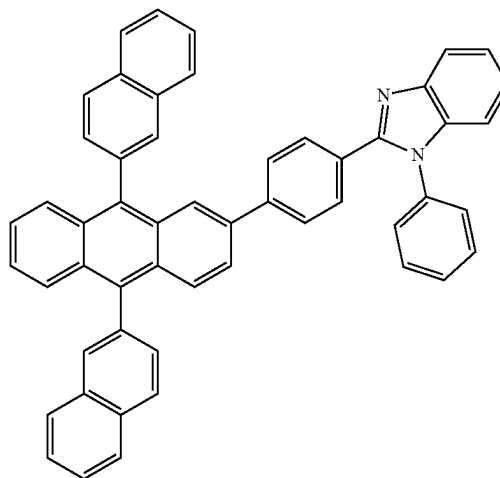
group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthrenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, and an azacarbazolyl group; and

—S(=O)₂(Q_{601}) and —P(=O)(Q_{601})(Q_{602}), and

Q_{601} and Q_{602} are the same as described above.

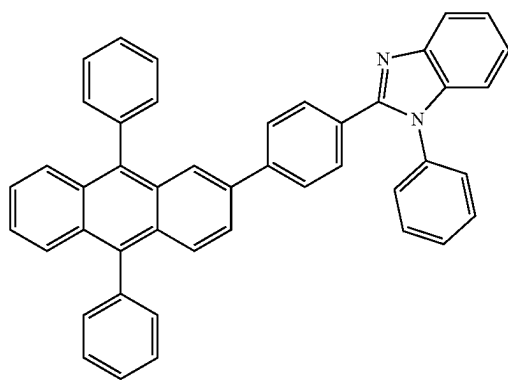
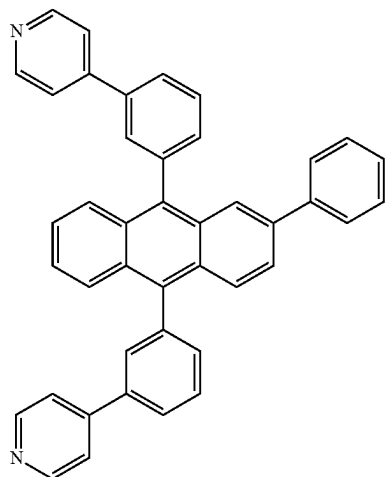
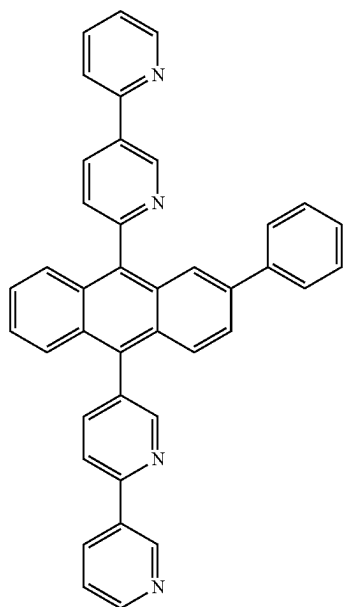
The electron transport region may include at least one compound selected from Compounds ET1 to ET36 below, but exemplary embodiments of the invention are not limited thereto:

ET1



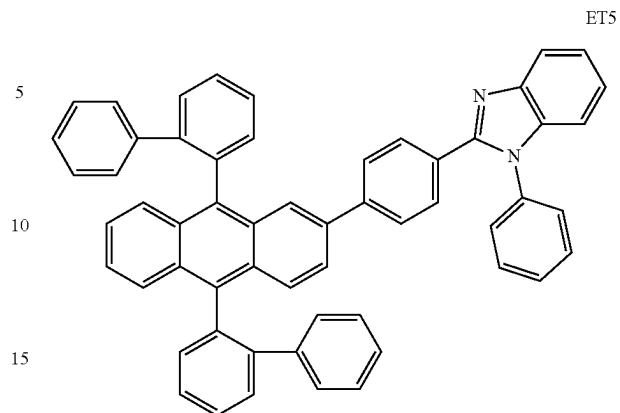
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**100**

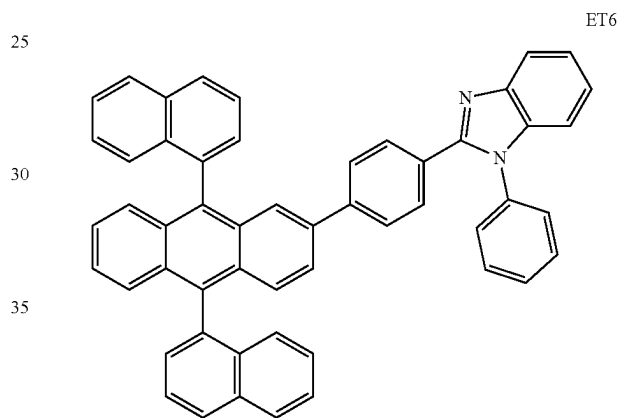
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ET2



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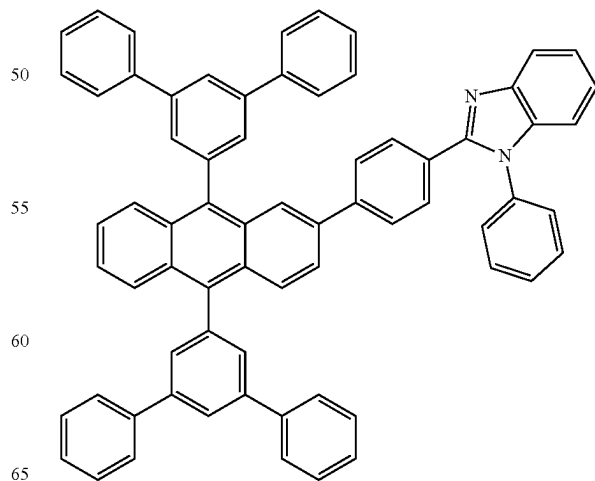
ET3



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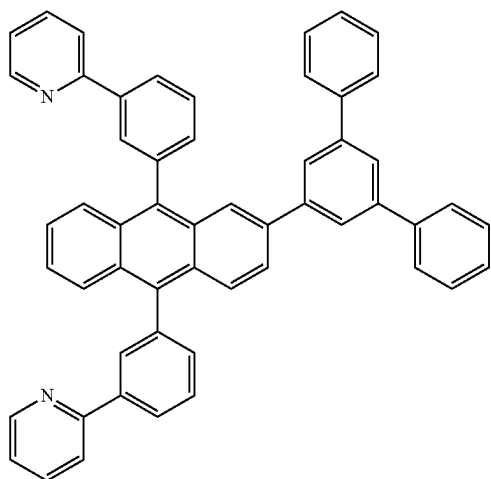
ET4



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101

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ET8

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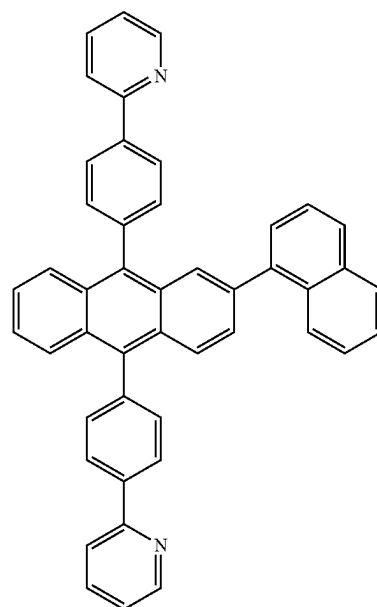
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102

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ET10

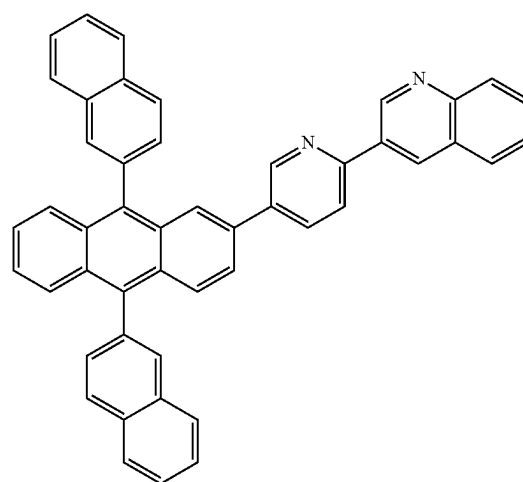
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ET9

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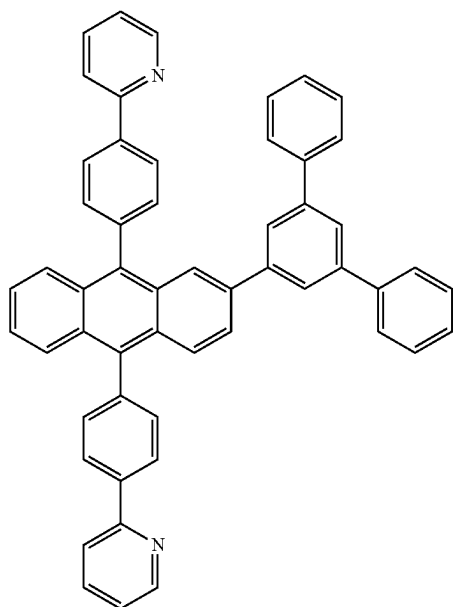
ET11

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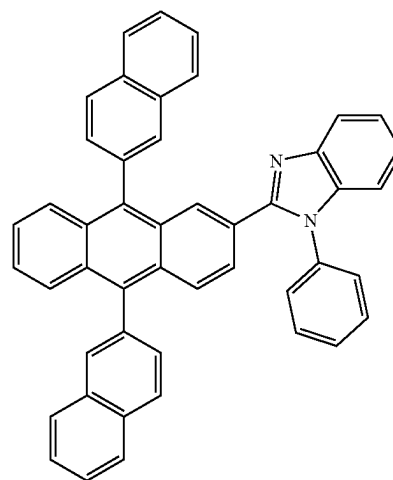
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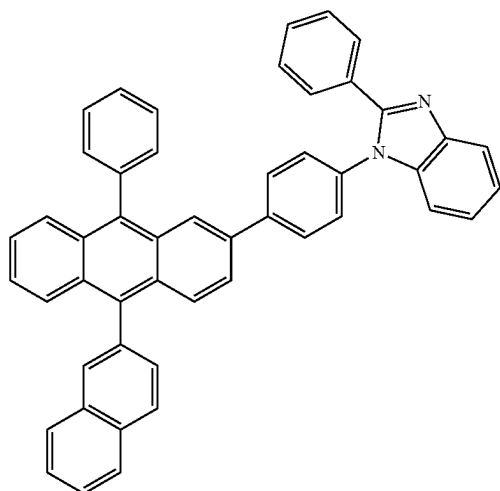
ET12



103

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ET13



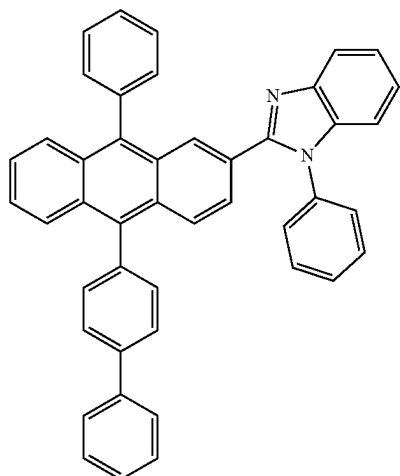
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ET14



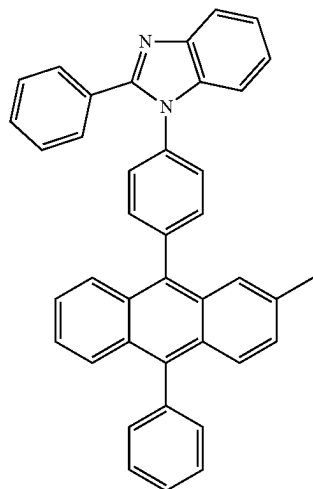
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ET15



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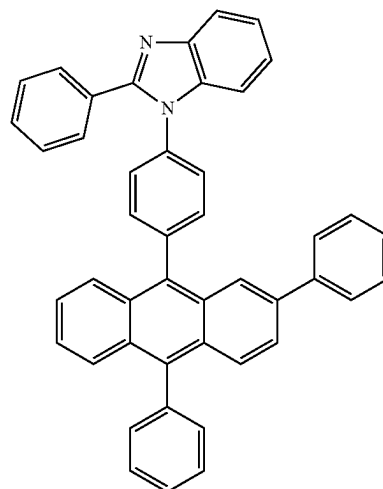
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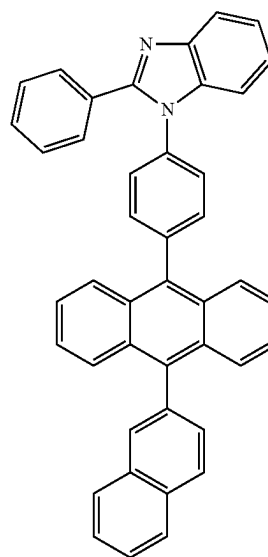
104

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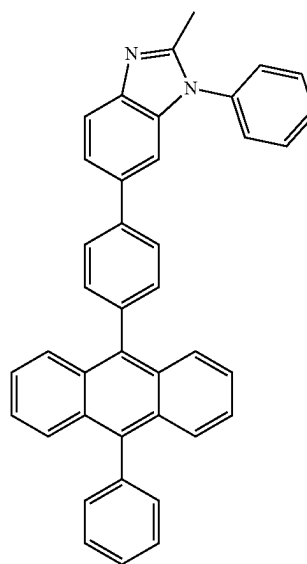
ET16



ET17

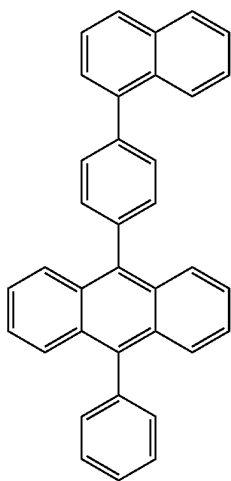
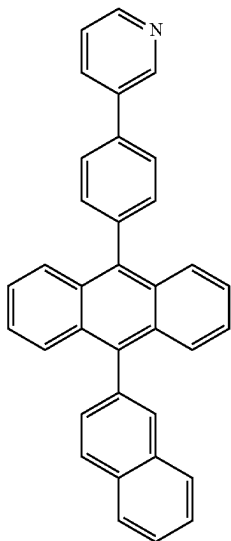
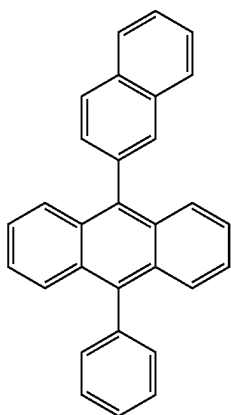


ET18



105

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**106**

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ET19

ET22

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ET20

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ET23

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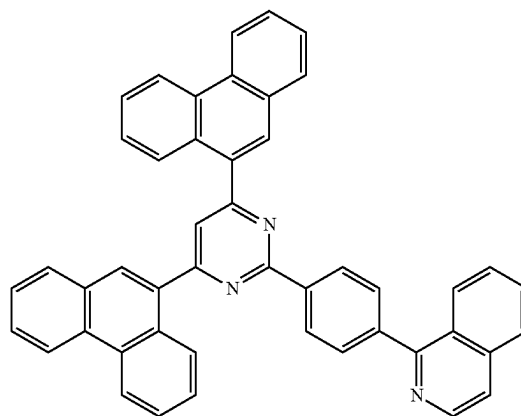
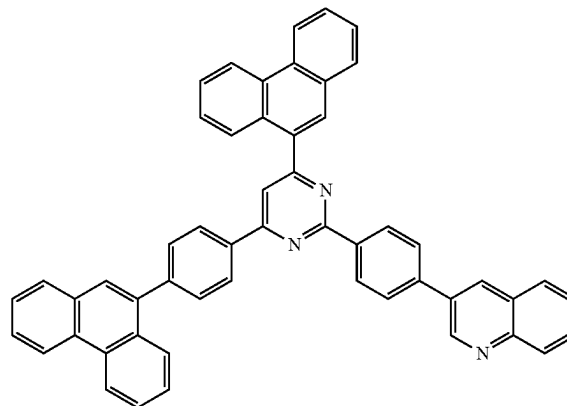
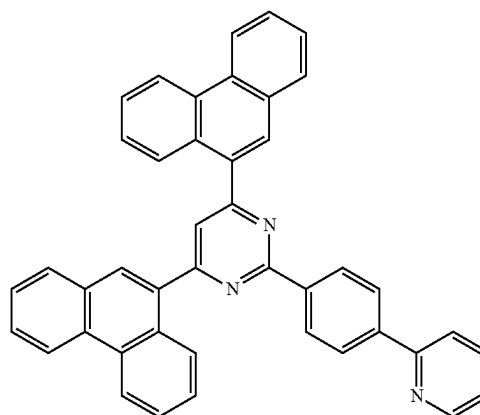
ET21

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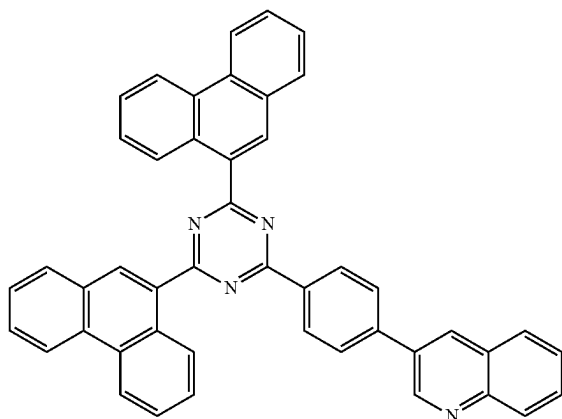
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107

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ET25

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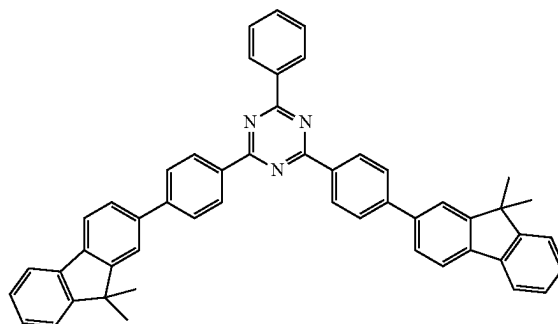
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108

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ET28

ET26

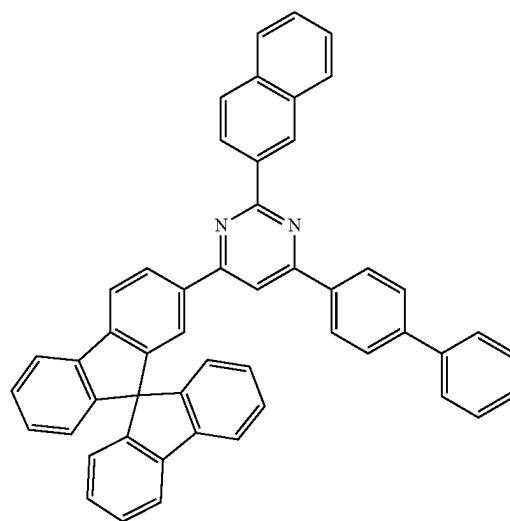
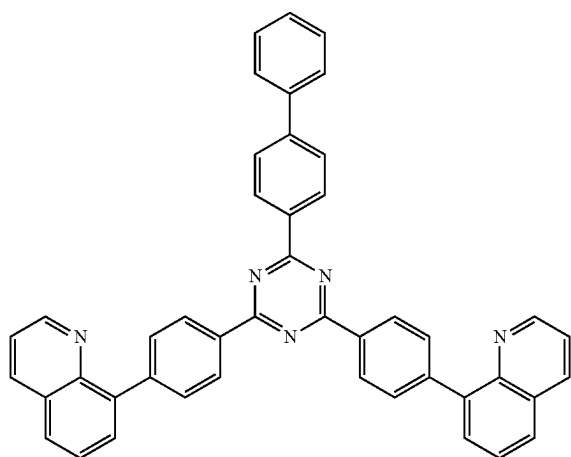
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ET29

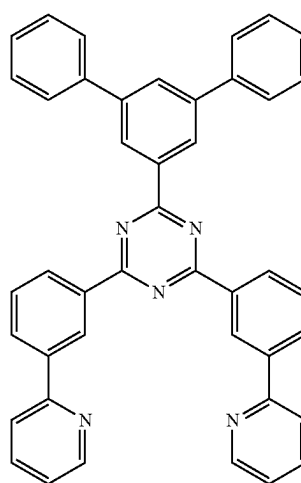
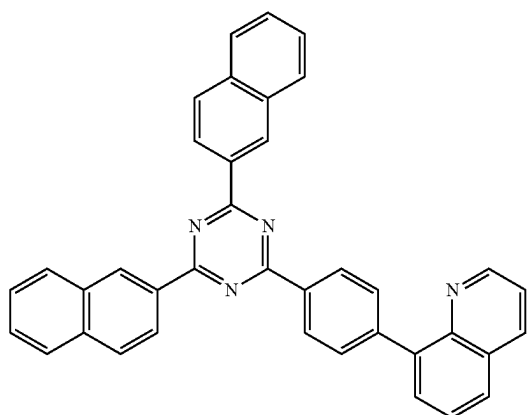
ET27

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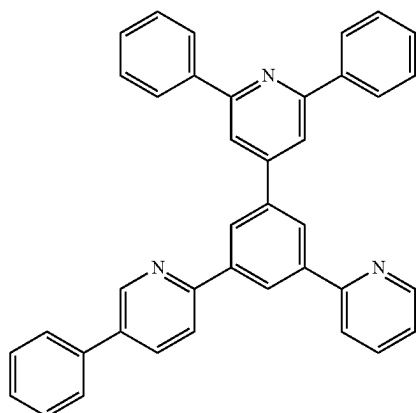
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ET30

109

-continued



ET31

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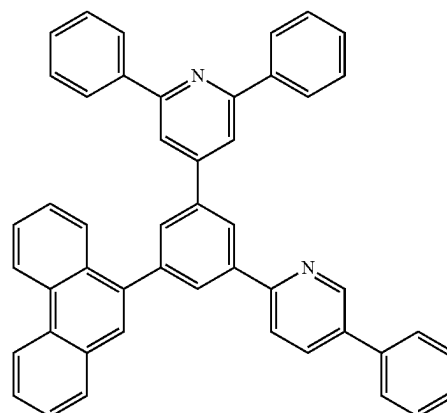
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ET34

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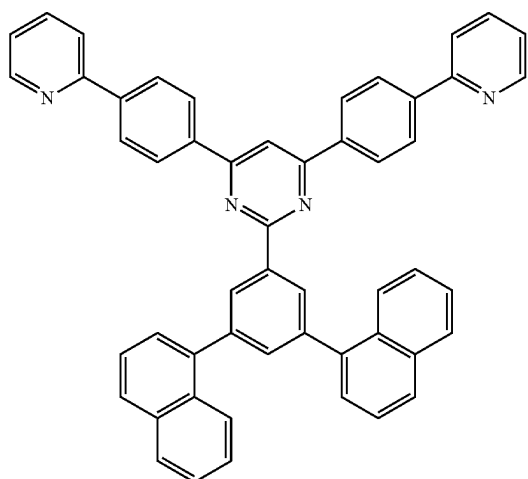
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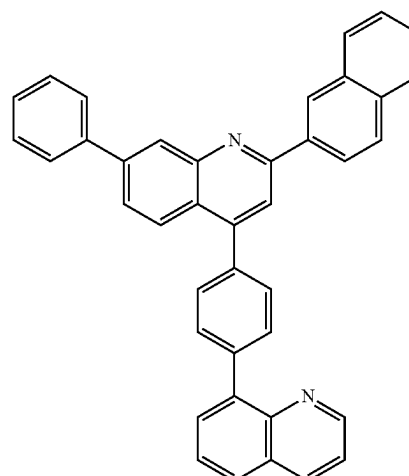
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ET33



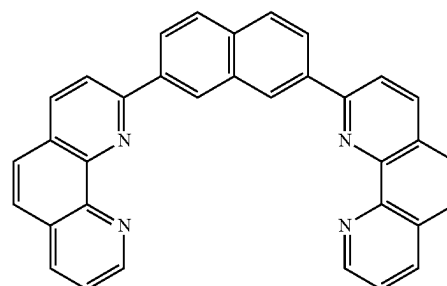
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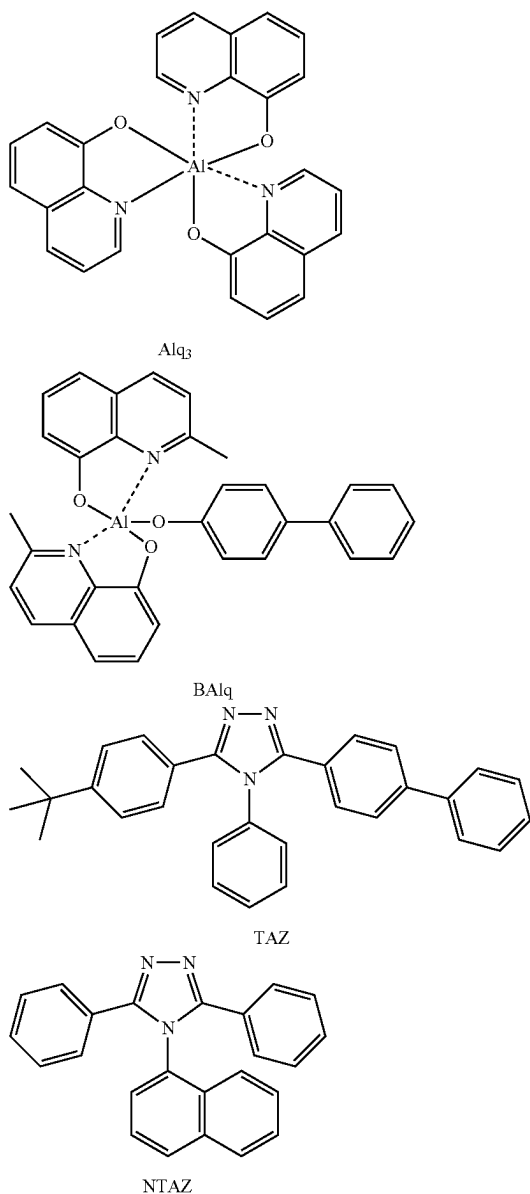
ET35

ET36



In one or more exemplary embodiments, the electron transport region may include at least one compound selected from 2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP), 4,7-diphenyl-1,10-phenanthroline (Bphen), Tris-(8-hydroxyquinoline)aluminum (Alq₃), Bis(8-hydroxy-2-methylquinoline)-(4-phenylphenoxy)aluminum (BALq), 3-(biphenyl-4-yl)-5-(4-tert-butylphenyl)-4-phenyl-1H-1,2,4-triazole (TAZ), and 4-naphthalen-1-yl-3,5-diphenyl-1,2,4-triazole (NTAZ):

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The thickness of the buffer layer, the hole blocking layer, or the electron control layer may be in a range of about 20 Å to about 1,000 Å, for example, about 30 Å to about 300 Å. When the thickness of the buffer layer, the hole blocking layer, or the electron control layer is within the range described above, excellent hole blocking characteristics or electron control characteristics may be obtained without a substantial increase in driving voltage.

The thickness of the electron transport layer may be in a range of about 100 Å to about 1,000 Å, for example, about 150 Å to about 500 Å. When the thickness of the electron transport layer is within the range described above, the electron transport layer may have satisfactory electron transport characteristics without a substantial increase in driving voltage.

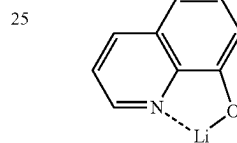
The electron transport region (for example, the electron transport layer in the electron transport region) may further include, in addition to the materials described above, a metal-containing material.

112

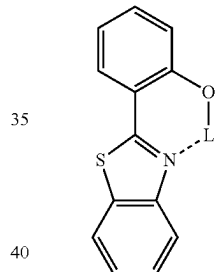
The metal-containing material may include at least one selected from an alkali metal complex and an alkaline earth-metal complex. The alkali metal complex may include a metal ion selected from a Li ion, a Na ion, a K ion, a Rb ion, and a Cs ion, and the alkaline earth-metal complex may include a metal ion selected from a Be ion, a Mg ion, a Ca ion, a Sr ion, and a Ba ion. A ligand coordinated with the metal ion of the alkali metal complex or the alkaline earth-metal complex may be selected from a hydroxy quinoline, a hydroxy isoquinoline, a hydroxy benzoquinoline, a hydroxy acridine, a hydroxy phenanthridine, a hydroxy phenyloxazole, a hydroxy phenylthiazole, a hydroxy diphenyloxadiazole, a hydroxy diphenylthiadiazole, a hydroxy phenylpyridine, a hydroxy phenylbenzimidazole, a hydroxy phenylbenzothiazole, a bipyridine, a phenanthroline, and a cyclopentadiene, but exemplary embodiments of the invention are not limited thereto.

For example, the metal-containing material may include a Li complex. The Li complex may include, for example, Compound ET-D1 (lithium quinolate, LiQ) or ET-D2 below:

ET-D1



ET-D2



The electron transport region may include an electron injection layer that facilitates electron injection from the second electrode **190**. The electron injection layer may be in direct contact with the second electrode **190**.

The electron injection layer may have i) a single-layered structure including a single layer including a single material, ii) a single-layered structure including a single layer including a plurality of different materials, or iii) a multi-layered structure having a plurality of layers including a plurality of different materials.

The electron injection layer may include an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combination thereof.

The alkali metal may be selected from Li, Na, K, Rb, and Cs. In one exemplary embodiment, the alkali metal may be Li, Na, or Cs. In one or more exemplary embodiments, the alkali metal may be Li or Cs, but exemplary embodiments of the invention are not limited thereto.

The alkaline earth metal may be selected from Mg, Ca, Sr, and Ba.

The rare earth metal may be selected from Sc, Y, Ce, Tb, Yb, and Gd.

The alkali metal compound, the alkaline earth-metal compound, and the rare earth metal compound may be selected from oxides and halides (for example, fluorides, chlorides, bromides, or iodides) of the alkali metal, the alkaline earth-metal, and the rare earth metal.

The alkali metal compound may be selected from alkali metal oxides, such as Li_2O , Cs_2O , or K_2O , and alkali metal halides, such as LiF , NaF , CsF , KF , LiI , NaI , CsI , or KI . In one exemplary embodiment, the alkali metal compound may be selected from LiF , Li_2O , NaF , LiI , NaI , CsI , and KI , but exemplary embodiments of the invention are not limited thereto.

The alkaline earth-metal compound may be selected from alkaline earth-metal oxides, such as BaO , SrO , CaO , $\text{Ba}_x\text{Sr}_{1-x}\text{O}$ ($0 < x < 1$), or $\text{Ba}_x\text{Ca}_{1-x}\text{O}$ ($0 < x < 1$). In one exemplary embodiment, the alkaline earth metal compound may be selected from BaO , SrO , and CaO , but exemplary embodiments of the invention are not limited thereto.

The rare earth metal compound may be selected from YbF_3 , ScF_3 , Sc_2O_3 , Y_2O_3 , Ce_2O_3 , GdF_3 , and TbF_3 . In one exemplary embodiment, the rare earth metal compound may be selected from YbF_3 , ScF_3 , TbF_3 , YbI_3 , ScI_3 , and TbI_3 , but exemplary embodiments of the invention are not limited thereto.

The alkali metal complex, the alkaline earth-metal complex, and the rare earth metal complex may include an ion of alkali metal, alkaline earth-metal, and rare earth metal as described above, and a ligand coordinated with a metal ion of the alkali metal complex, the alkaline earth-metal complex, or the rare earth metal complex may be selected from hydroxy quinoline, hydroxy isoquinoline, hydroxy benzoquinoline, hydroxy acridine, hydroxy phenanthridine, hydroxy phenylloxazole, hydroxy phenylthiazole, hydroxy diphenylloxadiazole, hydroxy diphenylthiadiazole, hydroxy phenylpyridine, hydroxy phenylbenzimidazole, hydroxy phenylbenzothiazole, bipyridine, phenanthroline, and cyclopentadiene, but exemplary embodiments of the invention are not limited thereto.

The electron injection layer may consist of an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combinations thereof, as described above. In one or more exemplary embodiments, the electron injection layer may further include an organic material. When the electron injection layer further includes an organic material, an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combinations thereof may be homogeneously or non-homogeneously dispersed in a matrix including the organic material.

The thickness of the electron injection layer may be in a range of about 1 Å to about 100 Å, for example, about 3 Å to about 90 Å. When the thickness of the electron injection layer is within the range described above, satisfactory electron injection characteristics may be obtained without a substantial increase in driving voltage.

Second Electrode 190

The second electrode 190 may be disposed on the organic layer 150 having such a structure. The second electrode 190 may be a cathode which is an electron injection electrode, and in this regard, a material for forming the second electrode 190 may be selected from metal, an alloy, an electri-

cally conductive compound, and a combination thereof, which have a relatively low work function.

The second electrode 190 may include at least one selected from lithium (Li), silver (Ag), magnesium (Mg), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), ITO, and IZO, but exemplary embodiments of the invention are not limited thereto. The second electrode 190 may be a transmissive electrode, a semi-transmissive electrode, or a reflective electrode.

The second 190 may have a single-layered structure, or a multi-layered structure including two or more layers.

Description of FIGS. 2 to 4

FIG. 2 is a schematic cross-sectional diagram of another exemplary embodiment of an organic light-emitting device constructed according to principles of the invention. FIG. 3 is a schematic cross-sectional diagram of yet another exemplary embodiment of an organic light-emitting device constructed according to principles of the invention. FIG. 4 is a schematic cross-sectional diagram of still another exemplary embodiment of an organic light-emitting device constructed according to principles of the invention.

An organic light-emitting device 20 of FIG. 2 includes a first capping layer 210, the first electrode 110, the organic layer 150, and the second electrode 190, which are sequentially stacked in this stated order, an organic light-emitting device 30 of FIG. 3 includes the first electrode 110, the organic layer 150, the second electrode 190, and a second capping layer 220, which are sequentially stacked in this stated order, an organic light-emitting device 40 of FIG. 4 includes the first capping layer 210, the first electrode 110, the organic layer 150, the second electrode 190, and the second capping layer 220, which are sequentially stacked in this stated order.

Regarding FIGS. 2 to 4, the first electrode 110, the organic layer 150, and the second electrode 190 may be understood by referring to the description presented in connection with FIG. 1.

In the organic layer 150 of each of the organic light-emitting devices 20 and 40, light generated in an emission layer may pass through the first electrode 110 and the first capping layer 210 toward the outside, wherein the first electrode 110 may be a semi-transmissive electrode or a transmissive electrode.

The first capping layer 210 and the second capping layer 220 may increase external luminescent efficiency according to the principle of constructive interference.

The first capping layer 210 and the second capping layer 220 may each independently be an organic capping layer including an organic material, an inorganic capping layer including an inorganic material, or a composite capping layer including an organic material and an inorganic material.

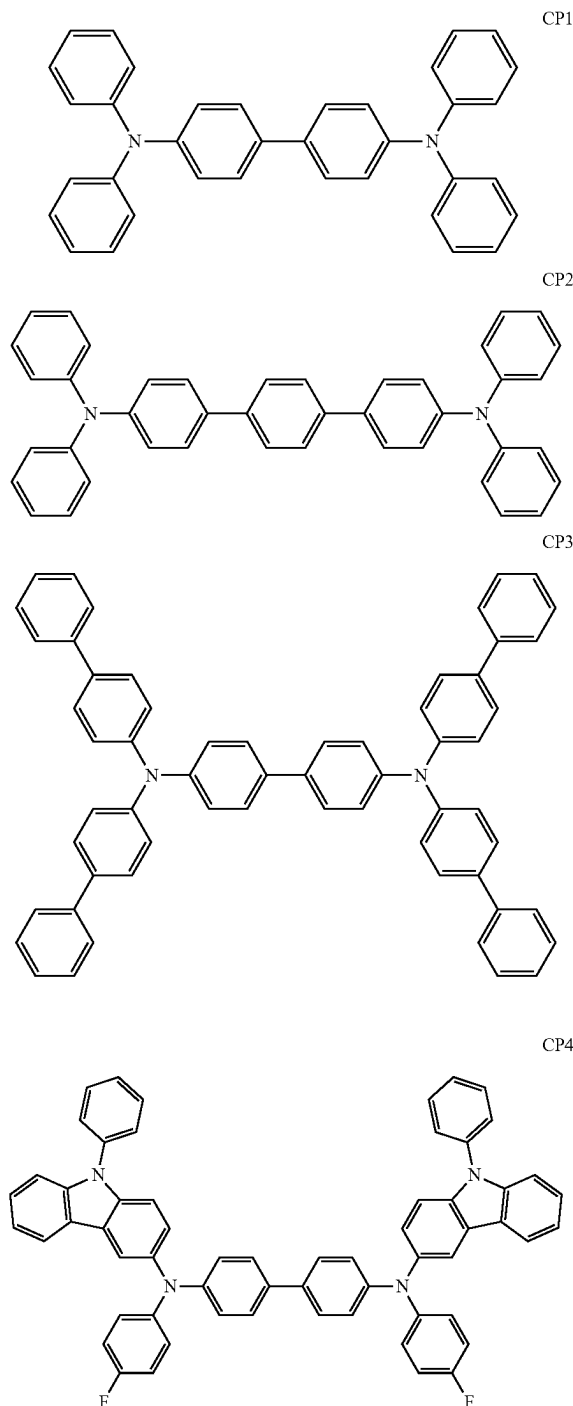
At least one of the first capping layer 210 and the second capping layer 220 may each independently include at least one material selected from carbocyclic compounds, heterocyclic compounds, amine-based compounds, porphyrine derivatives, phthalocyanine derivatives, a naphthalocyanine derivatives, alkali metal complexes, and alkaline earth-based complexes. The carbocyclic compound, the heterocyclic compound, and the amine-based compound may be optionally substituted with a substituent containing at least one element selected from O, N, S, Se, Si, F, Cl, Br, and I.

In one exemplary embodiment, at least one of the first capping layer 210 and the second capping layer 220 may each independently include an amine-based compound.

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In one exemplary embodiment, at least one of the first capping layer **210** and the second capping layer **220** may each independently include the compound represented by Formula 201 or the compound represented by Formula 202.

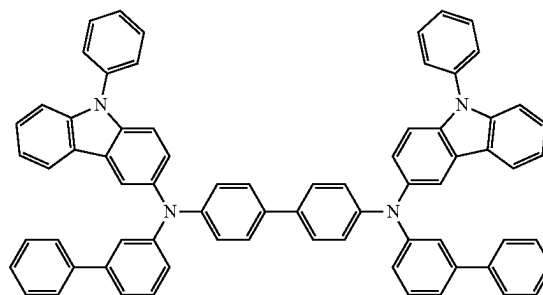
In one or more exemplary embodiments, at least one of the first capping layer **210** and the second capping layer **220** may each independently include a compound selected from Compounds HT28 to HT33 and Compounds CP1 to CP5, but exemplary embodiments of the invention are not limited thereto.



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-continued

CP5



Hereinbefore, the organic light-emitting device according to an embodiment has been described in connection with FIGS. **1** to **4**, but exemplary embodiments of the invention are not limited thereto.

Layers constituting the hole transport region, an emission layer, and layers constituting the electron transport region may be formed in a certain region by using one or more suitable methods selected from vacuum deposition, spin coating, casting, Langmuir-Blodgett (LB) deposition, ink-jet printing, laser-printing, and laser-induced thermal imaging.

When layers constituting the hole transport region, an emission layer, and layers constituting the electron transport region are formed by vacuum deposition, the deposition may be performed at a deposition temperature of about 100° C. to about 500° C., a vacuum degree of about 10⁻⁸ torr to about 10⁻³ torr, and a deposition speed of about 0.01 Å/sec to about 100 Å/sec by taking into account a material to be included in a layer to be formed, and the structure of a layer to be formed.

When layers constituting the hole transport region, an emission layer, and layers constituting the electron transport region are formed by spin coating, the spin coating may be performed at a coating speed of about 2,000 rpm to about 5,000 rpm and at a heat treatment temperature of about 80° C. to 200° C. by taking into account a material to be included in a layer to be formed, and the structure of a layer to be formed.

FIG. 5—Apparatus

FIG. **5** is a schematic diagram of an exemplary embodiment of an apparatus containing an organic light-emitting device constructed according to principles of the invention.

The organic light-emitting device may be included in various apparatuses.

Thus, another aspect of the invention provides an apparatus **300** including the organic light-emitting device **10**. Although the organic light-emitting device **10** is depicted in FIG. **5**, the one or more of the organic light-emitting devices **20**, **30**, and/or **40** may be used instead of or in addition to the organic light-emitting device **10**.

The apparatus **300** may further include, in addition to the organic light-emitting device **10**, a thin film transistor **310**. Here, the thin film transistor **310** may include a source electrode **320**, an activation layer **330**, and a drain electrode **340**, wherein the first electrode **110** of the organic light-emitting device **10** may be in electrical connection with one or both of the source electrode **320** and the drain electrode **340** of the thin-film transistor **310**.

The apparatus **300** may be, for example, a light-emitting apparatus, an authentication apparatus, or an electronic apparatus, but exemplary embodiments of the invention are not limited thereto.

The light-emitting apparatus 300 may be used as various displays, light sources, and the like.

The authentication apparatus may be, for example, a biometric authentication apparatus for authenticating an individual by using biometric information of a biometric body (for example, a finger tip, a pupil, or the like).

The authentication apparatus may further include, in addition to the organic light-emitting device, a biometric information collector.

The electronic apparatus may be applied to personal computers (for example, a mobile personal computer), mobile phones, digital cameras, electronic organizers, electronic dictionaries, electronic game machines, medical instruments (for example, electronic thermometers, sphygmomanometers, blood glucose meters, pulse measurement devices, pulse wave measurement devices, electrocardiogram (ECG) displays, ultrasonic diagnostic devices, or endoscope displays), fish finders, various measuring instruments, meters (for example, meters for a vehicle, an aircraft, and a vessel), projectors, and the like, but exemplary embodiments of the invention are not limited thereto.

General Definition of Substituents

The term “C₁-C₆₀ alkyl group” as used herein refers to a linear or branched aliphatic saturated hydrocarbon monovalent group having 1 to 60 carbon atoms, and examples thereof include a methyl group, an ethyl group, a propyl group, an isobutyl group, a sec-butyl group, a tert-butyl group, a pentyl group, an isoamyl group, and a hexyl group. The term “C₁-C₆₀ alkylene group” as used herein refers to a divalent group having a structure corresponding to the C₁-C₆₀ alkyl group.

The term “C₂-C₆₀ alkenyl group” as used herein refers to a hydrocarbon group having at least one carbon-carbon double bond in the middle or at the terminus of the C₂-C₆₀ alkyl group, and examples thereof include an ethenyl group, a propenyl group, and a butenyl group. The term “C₂-C₆₀ alkenylene group” as used herein refers to a divalent group having a structure corresponding to the C₂-C₆₀ alkenyl group.

The term “C₂-C₆₀ alkynyl group” as used herein refers to a hydrocarbon group having at least one carbon-carbon triple bond in the middle or at the terminus of the C₂-C₆₀ alkyl group, and examples thereof include an ethynyl group, and a propynyl group. The term “C₂-C₆₀ alkynylene group” as used herein refers to a divalent group having a structure corresponding to the C₂-C₆₀ alkynyl group.

The term “C₁-C₆₀ alkoxy group” as used herein refers to a monovalent group represented by —OA₁₀₁ (wherein A₁₀₁ is the C₁-C₆₀ alkyl group), and examples thereof include a methoxy group, an ethoxy group, and an isopropoxy group.

The term “C₃-C₁₀ cycloalkyl group” as used herein refers to a monovalent saturated hydrocarbon monocyclic group having 3 to 10 carbon atoms, and examples thereof include a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a cyclohexyl group, and a cycloheptyl group. The term “C₃-C₁₀ cycloalkylene group” as used herein refers to a divalent group having a structure corresponding to the C₃-C₁₀ cycloalkyl group.

The term “C₁-C₁₀ heterocycloalkyl group” as used herein refers to a monovalent monocyclic group having at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom and 1 to 10 carbon atoms, and examples thereof include a 1,2,3,4-oxatriazolidinyl group, a tetrahydrofuran group, and a tetrahydrothiophenyl group. The term “C₁-C₁₀ heterocycloalkylene group” as used herein

refers to a divalent group having a structure corresponding to the C₁-C₁₀ heterocycloalkyl group.

The term “C₃-C₁₀ cycloalkenyl group” as used herein refers to a monovalent monocyclic group that has 3 to 10 carbon atoms and at least one carbon-carbon double bond in the ring thereof and no aromaticity, and examples thereof include a cyclopentenyl group, a cyclohexenyl group, and a cycloheptenyl group. The term “C₃-C₁₀ cycloalkenylene group” as used herein refers to a divalent group having a structure corresponding to the C₃-C₁₀ cycloalkenyl group.

The term “C₁-C₁₀ heterocycloalkenyl group” as used herein refers to a monovalent monocyclic group that has at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom, 1 to 10 carbon atoms, and at least one carbon-carbon double bond in its ring. Non-limiting examples of the C₁-C₁₀ heterocycloalkenyl group include a 4,5-dihydro-1,2,3,4-oxatriazolyl group, a 2,3-dihydrofuran group, and a 2,3-dihydrothiophenyl group. The term “C₁-C₁₀ heterocycloalkenylene group” as used herein refers to a divalent group having a structure corresponding to the C₁-C₁₀ heterocycloalkenyl group.

The term “C₆-C₆₀ aryl group” as used herein refers to a monovalent group having a carbocyclic aromatic system having 6 to 60 carbon atoms, and a C₆-C₆₀ arylene group used herein refers to a divalent group having a carbocyclic aromatic system having 6 to 60 carbon atoms. Non-limiting examples of the C₆-C₆₀ aryl group include a phenyl group, a naphthyl group, an anthracenyl group, a phenanthrenyl group, a pyrenyl group, and a chrysenyl group. When the C₆-C₆₀ aryl group and the C₆-C₆₀ arylene group each include two or more rings, the rings may be fused to each other.

The term “C₁-C₆₀ heteroaryl group” as used herein refers to a monovalent group having a carbocyclic aromatic system that has at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom, in addition to 1 to 60 carbon atoms, and the term “C₁-C₆₀ heteroarylene group” as used herein refers to a divalent group having a carbocyclic aromatic system that has at least one heteroatom selected from N, O, Si, P, and S as a ring-forming atom, in addition to 1 to 60 carbon atoms. Non-limiting examples of the C₁-C₆₀ heteroaryl group include a pyridinyl group, a pyrimidinyl group, a pyrazinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, and an isoquinolinyl group. When the C₁-C₆₀ heteroaryl group and the C₁-C₆₀ heteroarylene group each include two or more rings, the rings may be fused with each other.

The term “C₆-C₆₀ aryloxy group” as used herein refers to —OA₁₀₂ (wherein A₁₀₂ is the C₆-C₆₀ aryl group), and a C₆-C₆₀ arylthio group as used herein refers to —SA₁₀₃ (wherein A₁₀₃ is the C₆-C₆₀ aryl group).

The term “monovalent non-aromatic fused polycyclic group” as used herein refers to a monovalent group (for example, having 8 to 60 carbon atoms) having two or more rings fused with each other, only carbon atoms as ring-forming atoms, and no aromaticity in its entire molecular structure. A detailed example of the monovalent non-aromatic fused polycyclic group is a fluorenyl group. The term “divalent non-aromatic fused polycyclic group” as used herein refers to a divalent group having a structure corresponding to the monovalent non-aromatic fused polycyclic group.

The term “monovalent non-aromatic fused heteropolycyclic group” as used herein refers to a monovalent group (for example, having 1 to 60 carbon atoms) having two or more rings fused with each other, at least one heteroatom selected from N, O, Si, P, and S, other than carbon atoms, as a ring-forming atom, and no aromaticity in its entire molecu-

lar structure. A detailed example of the monovalent non-aromatic fused heteropolycyclic group is a 1,2,3,4-tetrahydroquinolinyl group. The term “divalent non-aromatic fused heteropolycyclic group” as used herein refers to a divalent group having a structure corresponding to the monovalent non-aromatic fused heteropolycyclic group.

The term “C₅-C₆₀ carbocyclic group” as used herein refers to a monocyclic or polycyclic group having 5 to 60 carbon atoms in which a ring-forming atom is a carbon atom only. The term “C₅-C₆₀ carbocyclic group” as used herein refers to an aromatic carbocyclic group or a non-aromatic carbocyclic group. The C₅-C₆₀ carbocyclic group may be a ring, such as benzene, a monovalent group, such as a phenyl group, or a divalent group, such as a phenylene group. In one or more exemplary embodiments, depending on the number of substituents connected to the C₅-C₆₀ carbocyclic group, the C₅-C₆₀ carbocyclic group may be a trivalent group or a quadrivalent group.

The term “C₁-C₆₀ heterocyclic group” as used herein refers to a group having a structure corresponding to the C₅-C₆₀ carbocyclic group, except that as a ring-forming atom, at least one heteroatom selected from N, O, Si, P, and S is used in addition to carbon (the number of carbon atoms may be in a range of 1 to 60).

The terms “hydrogen” and “deuterium” refers to their respective atoms and corresponding radicals.

In the exemplary embodiments, at least one substituent of the substituted C₅-C₆₀ carbocyclic group, the substituted C₁-C₆₀ heterocyclic group, the substituted C₃-C₁₀ cycloalkylene group, the substituted C₁-C₁₀ heterocycloalkylene group, the substituted C₃-C₁₀ cycloalkenylene group, the substituted C₁-C₁₀ heterocycloalkenylene group, the substituted C₆-C₆₀ arylene group, the substituted C₁-C₆₀ heteroarylene group, the substituted divalent non-aromatic fused polycyclic group, the substituted divalent non-aromatic fused heteropolycyclic group, the substituted C₁-C₆₀ alkyl group, the substituted C₂-C₆₀ alkenyl group, the substituted C₂-C₆₀ alkynyl group, the substituted C₁-C₆₀ alkoxy group, the substituted C₃-C₁₀ cycloalkyl group, the substituted C₁-C₁₀ heterocycloalkyl group, the substituted C₃-C₁₀ cycloalkenyl group, the substituted C₁-C₁₀ heterocycloalkenyl group, the substituted C₆-C₆₀ aryl group, the substituted C₆-C₆₀ aryloxy group, the substituted C₆-C₆₀ arylthio group, the substituted C₁-C₆₀ heteroaryl group, the substituted monovalent non-aromatic fused polycyclic group, and the substituted monovalent non-aromatic fused heteropolycyclic group may be selected from:

deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, and a C₁-C₆₀ alkoxy group;

a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, and a C₁-C₆₀ alkoxy group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic fused polycyclic group, a monovalent non-aromatic fused heteropolycyclic group, —Si(Q₁₁)(Q₁₂)(Q₁₃), —N(Q₁₁)(Q₁₂), —B(Q₁₁)(Q₁₂), —C(=O)(Q₁₁), —S(=O)₂(Q₁₁), and —P(=O)(Q₁₁)(Q₁₂);

a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic fused polycyclic group, and a monovalent non-aromatic fused heteropolycyclic group;

a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic fused polycyclic group, and a monovalent non-aromatic fused heteropolycyclic group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic fused polycyclic group, a monovalent non-aromatic fused heteropolycyclic group, —Si(Q₂₁)(Q₂₂)(Q₂₃), —N(Q₂₁)(Q₂₂), —B(Q₂₁)(Q₂₂), —C(=O)(Q₂₁), —S(=O)₂(Q₂₁), and —P(=O)(Q₂₁)(Q₂₂); and —Si(Q₃₁)(Q₃₂)(Q₃₃), —N(Q₃₁)(Q₃₂), —B(Q₃₁)(Q₃₂), —C(=O)(Q₃₁), —S(=O)₂(Q₃₁), and —P(=O)(Q₃₁)(Q₃₂), and

Q₁₁ to Q₁₃, Q₂₁ to Q₂₃ and Q₃₁ to Q₃₃ may each independently be selected from hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₁-C₆₀ heteroaryl group, a monovalent non-aromatic fused polycyclic group, a monovalent non-aromatic fused heteropolycyclic group, a biphenyl group, and a terphenyl group.

The term “Ph” as used herein refers to a phenyl group, the term “Me” as used herein refers to a methyl group, the term “Et” as used herein refers to an ethyl group, the term “ter-Bu” or “Bu” as used herein refers to a tert-butyl group, and the term “OMe” as used herein refers to a methoxy group.

The term “biphenyl group” as used herein refers to “a phenyl group substituted with a phenyl group. In other words, the “biphenyl group” is a substituted phenyl group having a C₆-C₆₀ aryl group as a substituent.

The term “terphenyl group” as used herein refers to “a biphenyl group substituted with a phenyl group. In other words, the “terphenyl group” is a phenyl group having, as a substituent, a C₆-C₆₀ aryl group substituted with a C₆-C₆₀ aryl group.

* and * as used herein, unless defined otherwise, each refer to a binding site to a neighboring atom in a corresponding formula.

Hereinafter, a compound according to exemplary embodiments and an organic light-emitting device according to exemplary embodiments will be described in detail with reference to Synthesis Examples and Examples. The word-

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ing "B was used instead of A" used in describing Synthesis Examples refers to that an identical molar equivalent of B was used in place of A.

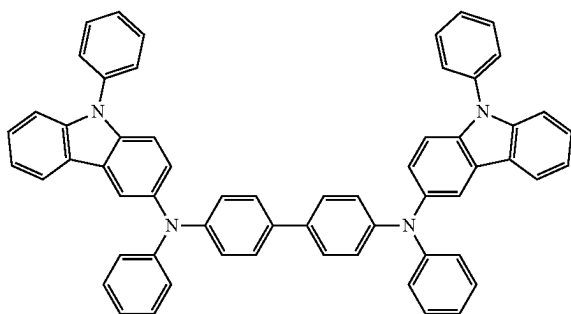
EXAMPLES

Example 1

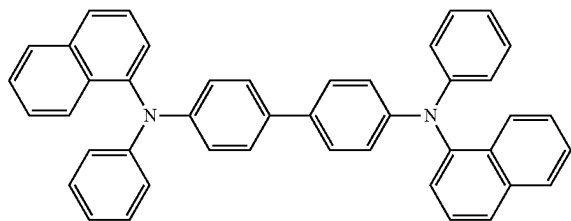
As a substrate and an anode, a glass substrate on which ITO (Corning) $15 \Omega/\text{cm}^2$ (120 nm) was formed was cut to the size of 50 mm×50 mm×0.5 mm, sonicated with acetone, isopropyl alcohol, and pure water, each for 5 minutes, and then, cleaned by exposure to ultraviolet rays and ozone for 10 minutes. Then, the glass substrate was provided to a vacuum deposition apparatus.

Compound HTL1 was vacuum deposited on the ITO anode to form a hole injection layer having the thickness of 600 Å, and 4,4'-bis[N-(1-naphthyl)-N-phenylamino]biphenyl (hereinafter, referred to as NPB) was deposited on the hole injection layer to form a hole transport layer having the thickness of 600 Å.

Compounds HT2-1 and ET2-1 and bis[2-(2-pyridinyl-N) phenyl-C](acetylacetonato)iridium(III) ($\text{Ir}(\text{mppy})_2(\text{acac})$) (8 weight %) which is a phosphorescent dopant were co-deposited on the hole transport layer to form an emission layer having the thickness of 400 Å. Subsequently, Compounds ETL1 and LiQ were mixed (at a ratio of 50:50) and deposited on the emission layer to form an electron transport layer having the thickness of 360 Å, and LiQ was deposited on the electron transport layer to form an electron injection layer having the thickness of 10 Å. Then, Mg and Ag were co-deposited (at a 10 weight %) on the electron injection layer to form a cathode having the thickness of 130 Å, thereby completing the manufacture of an organic light-emitting device.



<HTL1>

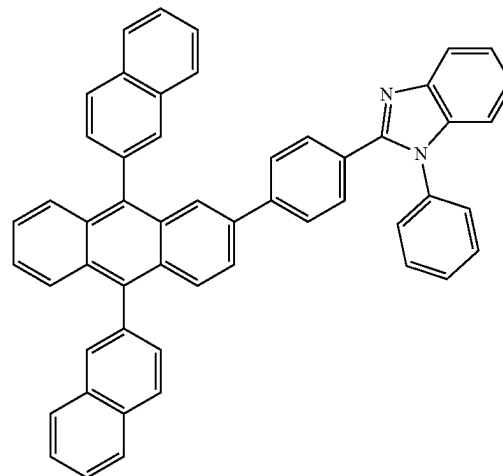


<NPB>

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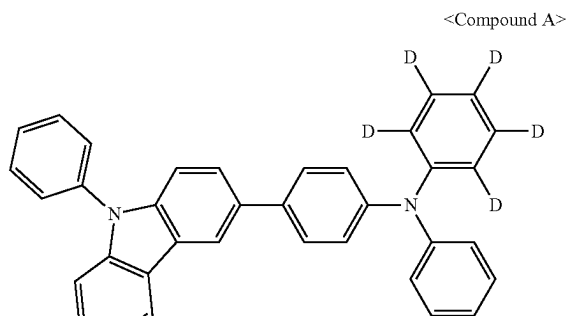
101



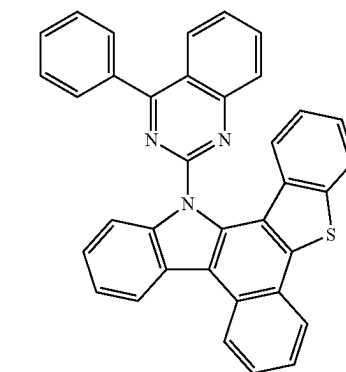
<ETL1>

Examples 2 to 16 and Comparative Examples 1 to 10

Organic light-emitting devices were manufactured in the same manner as in Example 1, except that an emission layer was formed by using compounds listed in Tables 1 and 2 were used. Structures of Compounds A, B, ET1-1, ET1-2, ET3-1, and ET3-2 used in Comparative Examples 1 to 10 are as follows:



<Compound A>

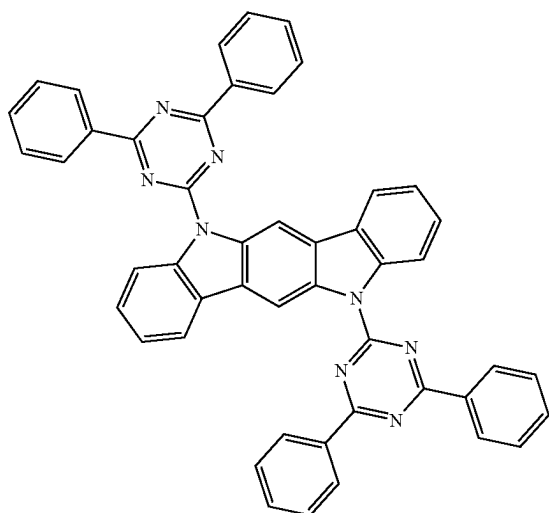


<Compound B>

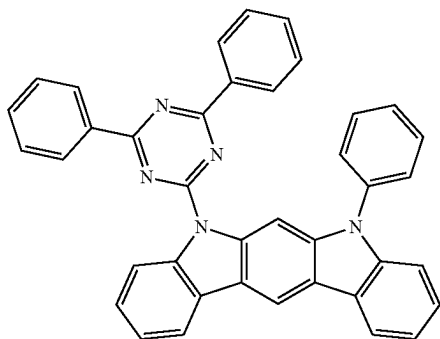
123

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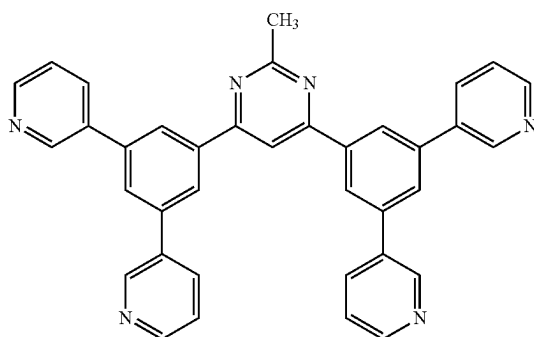
<ET1-1>



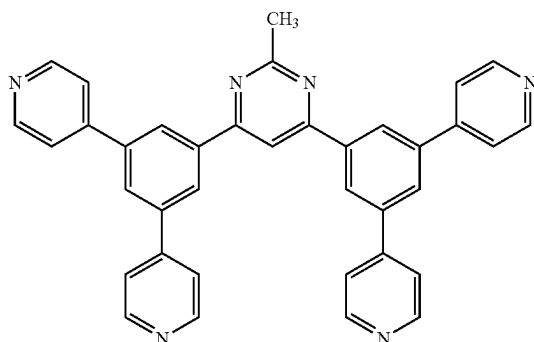
<ET1-2>



<ET3-1>



<ET3-2>



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Synthesis methods of the compounds used in the Examples may be recognized by one of ordinary skill in the art with reference to known techniques in the art.

Evaluation Example 1

The efficiency and lifespan of the organic light-emitting devices manufactured according to Examples 1 to 8 and Comparative Examples 1 to 5 were measured by using Keithley SMU 236 and a luminance meter PR650 at 600 nit, and results thereof are shown in Table 1.

In addition, the efficiency and lifespan of the organic light-emitting devices manufactured according to Examples 9 to 16 and Comparative Examples 6 to 10 were measured by using Keithley SMU 236 and a luminance meter PR650 at 600 nit, and results thereof are shown in Table 2.

In Tables 1 and 2, T97 indicates an amount of time to reach a luminance of 97% of the initial luminance.

TABLE 1

	First host	Second host	Efficiency (cd/A)	Color coordinate CIE_x	Lifespan (T97) (relative value)
Example 1	HT2-1	ET2-1	162	0.250	115%
Example 2	HT2-1	ET2-2	161	0.250	120%
Example 3	HT2-2	ET2-1	163	0.250	118%
Example 4	HT2-2	ET2-2	161	0.250	120%
Example 5	HT1-1	ET2-1	162	0.250	110%
Example 6	HT1-1	ET2-2	160	0.250	108%
Example 7	HT1-2	ET2-1	162	0.250	115%
Example 8	HT1-2	ET2-2	161	0.250	111%
Comparative Example 1	Compound A	Compound B	150	0.250	51%
Comparative Example 2	HT2-1	—	135	0.250	48%
Comparative Example 3	—	ET2-2	127	0.250	30%
Comparative Example 4	HT1-1	ET1-1	157	0.250	100%
Comparative Example 5	HT1-2	ET1-2	157	0.250	98%

TABLE 2

	First host	Second host	Efficiency (cd/A)	Color coordinate CIE_x	Lifespan (T97) (relative value)
Example 9	HT2-1	ET4-1	165	0.250	111%
Example 10	HT2-1	ET4-2	167	0.250	110%
Example 11	HT2-2	ET4-1	164	0.250	113%
Example 12	HT2-2	ET4-2	167	0.250	109%
Example 13	HT1-1	ET4-1	165	0.250	110%
Example 14	HT1-1	ET4-2	168	0.250	106%
Example 15	HT1-2	ET4-1	167	0.250	108%
Example 16	HT1-2	ET4-2	165	0.250	105%
Comparative Example 6	Compound A	Compound B	150	0.250	74%
Comparative Example 7	HT2-1	—	135	0.250	65%
Comparative Example 8	—	ET4-1	115	0.250	52%
Comparative Example 9	HT1-1	ET3-1	160	0.250	100%
Comparative Example 10	HT1-2	ET3-2	162	0.250	98%

Referring to Tables 1 and 2, the organic light-emitting devices of Examples 1 to 16 had unexpected and surprising results in terms of high efficiency and excellent lifespan

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characteristics, as compared with those of the organic light-emitting devices of Comparative Examples 1 to 10.

Accordingly, organic light-emitting device constructed according to the principles and exemplary embodiments of the invention have high efficiency and long lifespan.

Although certain exemplary embodiments and implementations have been described herein, other exemplary embodiments and modifications will be apparent from this description. Accordingly, the inventive concepts are not limited to such exemplary embodiments, but rather to the broader scope of the appended claims and various obvious modifications and equivalent arrangements as would be apparent to a person of ordinary skill in the art.

What is claimed is:

1. An organic light-emitting device comprising:

a first electrode;

a second electrode facing the first electrode;

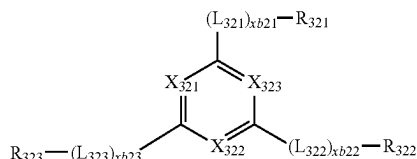
an organic layer disposed between the first electrode and the second electrode and includes an emission layer; and

a first compound and a second compound;

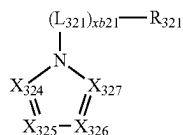
wherein the first compound comprises a hole transport compound not including an electron transport moiety, and

the second compound comprises an electron transport compound including an electron transport moiety or a bipolar compound including an electron transport moiety, wherein the second compound includes deuterium, wherein the emission layer comprises the first compound and the second compound, and the second compound is a compound of Formulae 302-1 to 302-4:

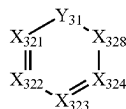
<Formula 302-1>



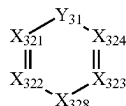
<Formula 302-2>



<Formula 302-3>



<Formula 302-4>



wherein:

X₃₂₁ to X₃₂₇, independently of one another, are N or C[(L₃₂₄)_{xb24}-R₃₂₄],

Y₃₁ is O or S,

X₃₂₈ is N[(L₃₂₅)_{xb25}-R₃₂₅] or C[(L₃₂₅)_{xb25}-R₃₂₅],

xb21 to xb26, independently of one another, are 0, 1, 2, 3, 4, or 5,

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L₃₂₁ to L₃₂₆ are each independently selected from:

a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthrenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylene group, a hexacenylene group, a pentacenylene group, a thiophenylene group, a furanylene group, a carbazolylenylene group, an indolylene group, an isoindolylene group, a benzofuranylene group, a benzothiophenylene group, a dibenzofuranylene group, a dibenzothiophenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylene group, a pyridinylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylene group, a pyrimidinylene group, a pyridazinylene group, a triazinylene group, a quinolinylene group, an isoquinolinylene group, a benzoquinolinylene group, a phthalazinylene group, a naphthyridinylene group, a quinoxalinylene group, a quinazolinylene group, a cinnolinylene group, a phenanthridinylene group, an acridinylene group, a phenanthrolinylene group, a phenazinylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylene group, an imidazopyrimidinylene group, and an azacarbazolylenylene group; and

a phenylene group, a naphthylene group, a fluorenylene group, a spiro-bifluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthrenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a perylenylene group, a pentaphenylene group, a hexacenylene group, a pentacenylene group, a thiophenylene group, a furanylene group, a carbazolylenylene group, an indolylene group, an isoindolylene group, a benzofuranylene group, a benzothiophenylene group, a dibenzofuranylene group, a dibenzothiophenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a dibenzosilolylene group, a pyridinylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a thiadiazolylenylene group, an oxadiazolylenylene group, a pyrazinylene group, a pyrimidinylene group, a pyridazinylene group, a triazinylene group, a quinolinylene group, an isoquinolinylene group, a benzoquinolinylene group, a phthalazinylene group, a naphthyridinylene group, a quinoxalinylene group, a quinazolinylene group, a cinnolinylene group, a phenanthridinylene group, an acridinylene group, a phenanthrolinylene group, a phenazinylene group, a benzimidazolylenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an imidazopyridinylene group, an imidazopyrimidinylene group, and an azacarbazolylenylene group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a

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biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a hexacacenyl group, a pentacenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a thiadiazolyl group, an oxadiazolyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an azacarbazolyl group, $-\text{Si}(\text{Q}_{31})(\text{Q}_{32})(\text{Q}_{33})$, $-\text{N}(\text{Q}_{31})(\text{Q}_{32})$, $-\text{B}(\text{Q}_{31})(\text{Q}_{32})$, $-\text{C}(=\text{O})(\text{Q}_{31})$, $-\text{S}(=\text{O})_2(\text{Q}_{31})$, and $-\text{P}(=\text{O})(\text{Q}_{31})(\text{Q}_{32})$, R_{321} to R_{326} are each independently selected from a substituted or unsubstituted C_1 - C_{60} alkyl group, a substituted or unsubstituted C_3 - C_{10} cycloalkyl group, a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, and a C₁-C₆₀ alkyl group, a C₃-C₁₀ cycloalkyl group, a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group; and

a C₁-C₆₀ alkyl group, a C₃-C₁₀ cycloalkyl group, a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl

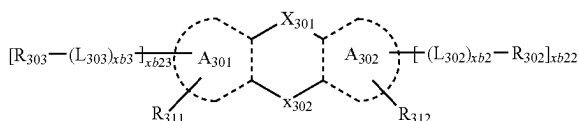
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group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, and a pyridinyl group, each substituted with at least one selected from deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a phenyl group substituted with a C₁-C₁₀ alkyl group, a phenyl group substituted with —F, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a thiophenyl group, a furanyl group, a carbazolyl group, an indolyl group, an isoindolyl group, a benzofuranyl group, a benzothiophenyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a dibenzosilolyl group, a pyridinyl group, $-\text{Si}(\text{Q}_{31})(\text{Q}_{32})(\text{Q}_{33})$, and $-\text{N}(\text{Q}_{31})(\text{Q}_{32})$, at least one of R_{321} to R_{326} , independently of one another, comprises a C₃-C₁₀ cycloalkyl group substituted with deuterium, two or more substituents among R_{321} to R_{326} are, independently, optionally linked to each other to form a substituted or unsubstituted C₁-C₆₀ carbocyclic group or a substituted or unsubstituted C₁-C₆₀ heterocyclic group, Q_{31} to Q_{33} are each independently selected from a C₁-C₁₀ alkyl group, a C₁-C₁₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, and a naphthyl group.

2. The organic light-emitting device of claim 1, wherein the electron transport moiety comprises at least one of a cyano group, a phosphine oxide group, a sulfone oxide group, a sulfonate group, and a Ir electron-depleted nitrogen-containing ring.

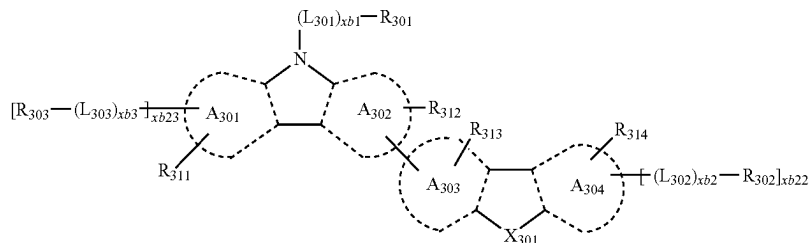
3. The organic light-emitting device of claim 1, wherein the first compound is a compound of Formulae 301-1 to 301-5:

<Formula 301-1>

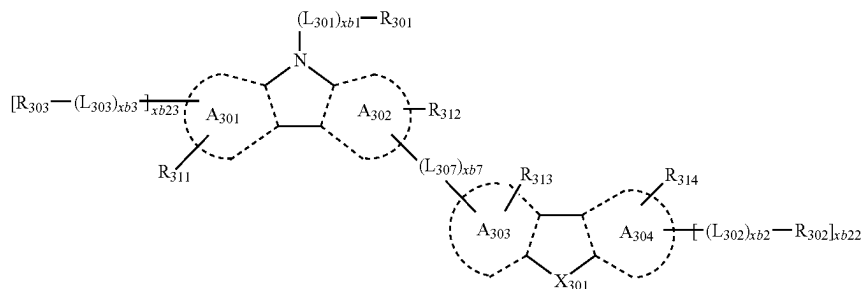


-continued

<Formula 301-2>

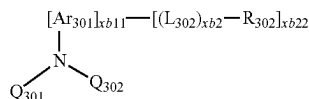
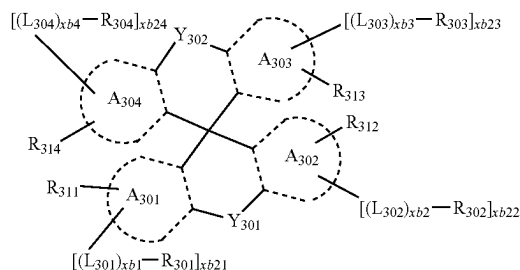


<Formula 301-3>



<Formula 301-4>

<Formula 301-5>



wherein, in Formulae 301-1 to 301-5,

A₃₀₁ to A₃₀₄, independently of one another, are a benzene group, a naphthalene group, a phenanthrene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, an indene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, an indole group, a carbazole group, a benzocarbazole group, a dibenzocarbazole group, a furan group, a benzofuran group, a dibenzofuran group, a naphthofuran group, a benzonaphthofuran group, a dinaphthofuran group, a thiophene group, a benzothiophene group, a dibenzothiophene group, a naphthothiophene group, a benzonaphthothiophene group, or a dinaphthothiophene group,

X₃₀₁ is O, S, N-[(L₃₀₅)_{xb5}-R₃₀₅], C [(L₃₀₅)_{xb5}-R₃₀₅], [(L₃₀₆)_{xb6}-R₃₀₆], or Si[(L₃₀₅)_{xb5}-R₃₀₅][(L₃₀₆)_{xb6}-R₃₀₆], X₃₀₂, Y₃₀₁, and Y₃₀₂, independently of one another, are a single bond, O, S, N-[(L₃₀₅)_{xb5}-R₃₀₅], C[(L₃₀₅)_{xb5}-R₃₀₅], [(L₃₀₆)_{xb6}-R₃₀₆], Si[(L₃₀₅)_{xb5}-R₃₀₅][(L₃₀₆)_{xb6}-R₃₀₆], or S(=O)₂,

xb11 is 1, 2, or 3,

xb22 and xb23, independently of one another, are 0, 1, or 2,

xb21 and xb24, independently of one another, are 1, 2, 3, 4, or 5,

L₃₀₁ to L₃₀₇, independently of one another, are a substituted or unsubstituted C₃-C₁₀ cycloalkylene group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkylene group, a substituted or unsubstituted C₃-C₁₀ cycloalkenylene group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenylene group, a substituted or unsubstituted C₆-C₆₀ arylene group, a substituted or unsubstituted C₁-C₆₀ heteroarylene group, a substituted or unsubstituted divalent non-aromatic fused polycyclic group, or a substituted or unsubstituted divalent non-aromatic fused heteropolycyclic group,

xb1 to xb6, independently of one another, are 0, 1, 2, 3, 4, or 5,

xb7 is 1, 2, 3, 4, or 5,

R₃₀₁ to R₃₀₆, independently of one another, are deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group,

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a nitro group, an amidino group, a hydrazino group, a hydrazono group, a substituted or unsubstituted C₁-C₆₀ alkyl group, a substituted or unsubstituted C₂-C₆₀ alkenyl group, a substituted or unsubstituted C₂-C₆₀ alkynyl group, a substituted or unsubstituted C₁-C₆₀ alkoxy group, a substituted or unsubstituted C₃-C₁₀ cycloalkyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkyl group, a substituted or unsubstituted C₃-C₁₀ cycloalkenyl group, a substituted or unsubstituted C₁-C₁₀ heterocycloalkenyl group, a substituted or unsubstituted C₆-C₆₀ aryl group, a substituted or unsubstituted C₆-C₆₀ aryloxy group, a substituted or unsubstituted C₆-C₆₀ arylthio group, a substituted or unsubstituted C₁-C₆₀ heteroaryl group, a substituted or unsubstituted monovalent non-aromatic fused polycyclic group, a substituted or unsubstituted monovalent non-aromatic fused heteropolycyclic group, —Si(Q₃₀₁)(Q₃₀₂)(Q₃₀₃), —N(Q₃₀₁)(Q₃₀₂), —B(Q₃₀₁)(Q₃₀₂), —C(=O)(Q₃₀₁), —S(=O)₂(Q₃₀₁), or —P(=O)(Q₃₀₁)(Q₃₀₂),

R₃₁₁ to R₃₁₄, independently of one another, are hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group —Si(Q₃₀₁)(Q₃₀₂)(Q₃₀₃), —N(Q₃₀₁)(Q₃₀₂), —B(Q₃₀₁)(Q₃₀₂), —C(=O)(Q₃₀₁), —S(=O)₂(Q₃₀₁), or —P(=O)(Q₃₀₁)(Q₃₀₂), and

Q₃₀₁ to Q₃₀₃, independently of one another, are hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, a C₃-C₁₀ cycloalkyl group, a C₁-C₁₀ heterocycloalkyl group, a C₃-C₁₀ cycloalkenyl group, a C₁-C₁₀ heterocycloalkenyl group, a C₆-C₆₀ aryl group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryl group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, a monovalent non-aromatic fused polycyclic group, a monovalent non-aromatic fused heteropolycyclic group, a biphenyl group, or a terphenyl group.

4. The organic light-emitting device of claim 1, wherein the first compound is a fluorene-containing compound, a carbazole-containing compound, a diarylamine compound, a triarylamine compound, a dibenzofuran-containing compound, a dibenzothiophene-containing compound, or a dibenzosilole-containing compound.

5. The organic light-emitting device of claim 1, wherein the first compound comprises deuterium.

6. The organic light-emitting device of claim 3, wherein R₃₀₁ to R₃₀₆, and R₃₁₁ to R₃₁₄, independently of one another, are deuterium, a C₁-C₆₀ alkyl group substituted with deuterium, a C₃-C₁₀ cycloalkyl group substituted with deuterium, a C₆-C₆₀ aryl group substituted with deuterium, or a C₁-C₆₀ heteroaryl group substituted with deuterium.

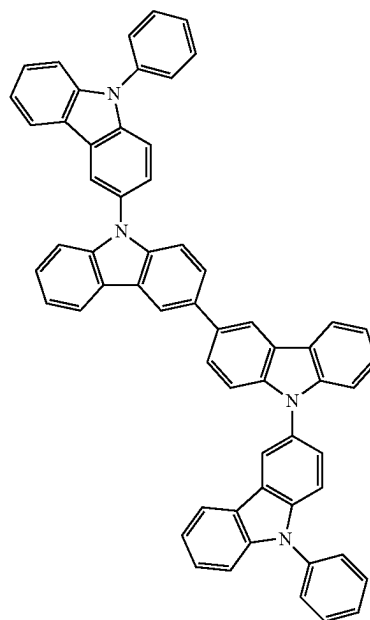
7. The organic light-emitting device of claim 1, wherein the second compound is a pyridine-containing compound, a pyrimidine-containing compound, a triazine-containing

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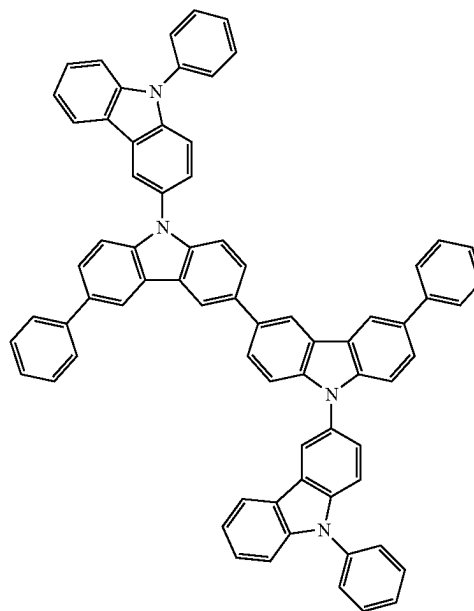
compound, a triazole-containing compound, an imidazole-containing compound, or an oxazine-containing compound.

8. The organic light-emitting device of claim 1, wherein the first compound is a compound of HT1-1, HT1-2, HT2-1, or HT2-2:

<HT1-1>

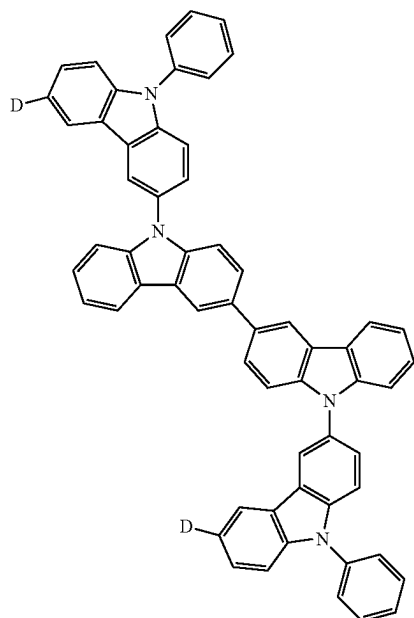


<HT1-2>



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-continued



<HT2-1>

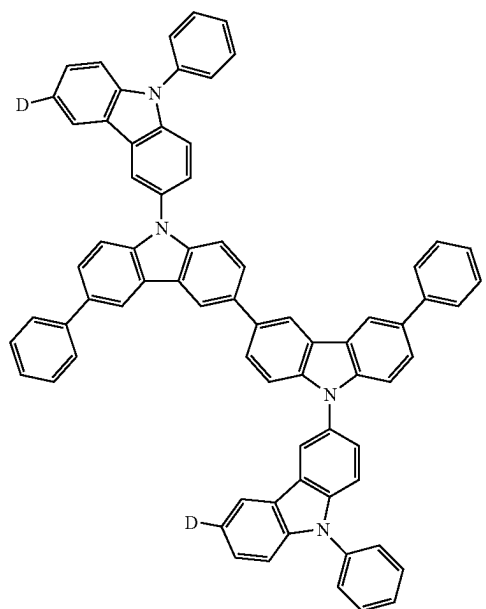
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<HT2-2>



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9. An organic light-emitting device comprising:
 a first electrode;
 a second electrode facing the first electrode;
 an organic layer disposed between the first electrode and the second electrode and includes an emission layer;
 and
 a first compound and a second compound;
 wherein the first compound comprises a hole transport compound not including an electron transport moiety,
 and
 the second compound comprises an electron transport compound including an electron transport moiety or a bipolar compound including an electron transport moiety, wherein the second compound includes deuterium,
 wherein the emission layer comprises the first compound and the second compound, and

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wherein the second compound is a compound of ET4-1 or ET4-2:

<ET4-1>

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<ET4-2>

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10. The organic light-emitting device of claim 1, wherein the emission layer comprises a first emission layer and a second emission layer,

the first emission layer comprises a first host and a second host, and the second emission layer comprises a third host and a fourth host,

wherein the first host in the first emission layer and the third host in the second emission layer each include the first compound, and the first host and the third host are same or different from each other,

the second host in the first emission layer and the fourth host in the second emission layer each include the second compound, the second compound included in the second host comprises the electron transport compound, the second compound included in the fourth host comprises the bipolar compound,

the first host and the second host each satisfy Equation 1 below, and the third host and the fourth host each satisfy Equation 2 below:

$$E_{HOMO1} - E_{HOMO2} \leq 0.2 \text{ eV} \quad \text{<Equation 1>}$$

$$E_{HOMO3} - E_{HOMO4} \geq 0.3 \text{ eV} \quad \text{<Equation 2>}$$

wherein, in Equations 1 and 2,

E_{HOMO1} indicates a highest occupied molecular orbital ("HOMO") energy level of the first host,

E_{HOMO2} indicates a HOMO energy level of the second host,

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E_{HOMO3} indicates a HOMO energy level of the third host,
and

E_{HOMO4} indicates a HOMO energy level of the fourth
host.

11. The organic light-emitting device of claim 1, wherein
the emission layer emits green light having a maximum
emission wavelength in a range of about 490 nm to about
560 nm.

12. The organic light-emitting device of claim 1, wherein
the emission layer further comprises a dopant, and the
dopant comprises a phosphorescent dopant or a fluorescent
dopant.

13. The organic light-emitting device of claim 1, wherein:
the first electrode is an anode,

the second electrode is a cathode,

the organic layer, further comprises at least one of a hole
transport region disposed between the first electrode
and the emission layer and at least one of an electron
transport region disposed between the emission layer
and the second electrode,

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the hole transport region comprises a hole injection layer,
a hole transport layer, an emission auxiliary layer, an
electron blocking layer, or any combination thereof,
and

the electron transport region comprises a hole blocking
layer, an electron transport layer, an electron injection
layer, or any combination thereof.

14. The organic light-emitting device of claim 13, wherein
the hole transport region comprises an amine-based com-
pound.

15. The organic light-emitting device of claim 13, wherein
the electron transport region comprises a metal-containing
material.

16. An apparatus comprising:

the organic light-emitting device of claim 1, wherein
the first electrode of the organic light-emitting device is in
electrical contact with one of a source electrode and a
drain electrode of a thin-film transistor.

* * * * *